

USING MACHINE LEARNING TO PREDICT HIP FRACTURE RISK FROM
DUAL-ENERGY X-RAY ABSORPTIOMETRY IMAGES AND HEALTH
FACTORS

USING MACHINE LEARNING TO PREDICT HIP FRACTURE RISK FROM
DUAL-ENERGY X-RAY ABSORPTIOMETRY IMAGES AND HEALTH
FACTORS

By TAYLOR KRAMER, B.Eng.BME

A Thesis Submitted to the School of Graduate Studies in Partial Fulfilment of the Requirements
for the Degree Master of Applied Science

McMaster University © Copyright by Taylor Kramer, January 2026

McMaster University MASTER OF APPLIED SCIENCE (2026) Hamilton, Ontario (Biomedical Engineering)

TITLE: Using Machine Learning to Predict Hip Fracture Risk Using Dual-Energy X-Ray Absorptiometry Images and Health Factors

AUTHOR: Taylor Kramer, B.Eng.BME (McMaster University)

SUPERVISOR: Cheryl E. Quenneville, B.Sc., M.E.Sc., Ph.D.

NUMBER OF PAGES: xvii, 127

Lay Abstract

Osteoporosis, a disease associated with aging, is a major public health concern causing a substantial increase in morbidity and mortality in older adults. Fragility fractures are often the first clinical manifestation of this disease, with hip fractures in particular being especially detrimental to both individual health outcomes and healthcare systems. Current diagnostic tools have limitations in predicting fracture risk for older adults. Recent advances in machine learning and image processing techniques have shown promise in improving fracture risk prediction for older adults beyond what is currently achievable with standard assessment methods.

The purpose of this work was to use dual-energy x-ray absorptiometry images and clinical health data to develop machine learning models capable of predicting fracture risk. Four predictive models were developed and evaluated to compare performance across different modeling strategies and to assess the added value of integrating imaging features with clinical risk factors. The results of this work demonstrated the potential of integrating imaging and clinical data to enhance fracture risk prediction, while also highlighting important considerations for clinical utility.

Abstract

Osteoporosis is a highly prevalent skeletal disease that greatly increases the risk of fragility fractures and affects millions of older adults worldwide. Hip fractures are especially dangerous, often resulting in long-term disability, increased need for institutional care, and increased healthcare costs. Current methods for diagnosing osteoporosis and subsequent fracture risk, such as bone mineral density (BMD) T-score from dual-energy x-ray absorptiometry (DXA) imaging and the Fracture Risk Assessment Tool (FRAX) have limitations in their ability to accurately predict fracture risk. Developments in machine learning and image processing tools have shown promise for improved fracture risk prediction. This work builds on these advances by integrating DXA imaging data with longitudinal clinical information to develop and evaluate predictive models aimed at improving fracture risk prediction.

To evaluate the independent predictive ability of DXA images, a simple feed-forward neural network trained on DXA images reduced using principal component analysis (PCA) was developed. This model demonstrated that images alone contain predictive information for fracture risk; however, performance outcomes improved when clinical risk factors (CRFs) were incorporated in a multimodal feed-forward neural network approach. SHapley Additive exPlanations analysis revealed that imaging features contributed most strongly to the model's predictions while age and body mass index (BMI) had comparatively smaller contributions. Both models achieved performance metrics within the range of values typically reported for FRAX-based assessments. Next, deep learning and transfer learning models were developed to assess how model performance was impacted by the addition of spatial pattern recognition within the images. Deep learning achieved moderate predictive ability, with performance similar to FRAX, while transfer learning models struggled to learn meaningful patterns from the available data.

The purpose of this work was to improve fracture risk prediction by integrating DXA imaging data with clinical health information. The results of this work can inform the development of clinically relevant fracture risk prediction tools that support early intervention for at-risk individuals.

Acknowledgments

Thank you to my supervisor, Dr. Cheryl Quenneville, for your mentorship and support throughout graduate school. Having the opportunity to learn from you and work under your guidance has been a great privilege, and I feel very fortunate to have gotten to pursue my degree in the Injury Biomechanics Laboratory. Your positivity and encouragement, along with your ability to clearly work through challenges and pivot when needed, were invaluable throughout the course of this project. Thank you for your kindness and for supporting me throughout the highs and lows of graduate studies!

To the members of the McMaster Injury Biomechanics Lab: Carson, Ary, Emelyn, Aiden, Morgan, Edward, Aidan, Erin, Maryam, Ali, Cole, and Julia, thank you for making my graduate school experience so rewarding! I'm grateful to have been part of a supportive and fun lab environment, and I'm going to miss seeing you all regularly, playing the geography games with Carson, and hearing about what's going on with everyone's projects. Thank you to all my friends for the laughs, support, and for making Hamilton a home away from home for the past seven years. A special thank you to the DeMelo family for welcoming me in and making me feel like part of the family as I finished my studies.

I would also like to acknowledge McMaster University, the McMaster Institute for Research on Aging, and the NSERC-funded sMAP CREATE program for the funding that made this work possible. Thank you to Amgen for providing the Injury Biomechanics Lab with the data that were used to perform this work. Thank you as well to Dr. Jonathan (Rick) Adachi and Dr. Katherine Zukotynski for your valuable feedback and support in my project direction. I would also like to thank Glen Walencik, the DXA technologist of the Hamilton Osteoporosis Clinic, for teaching me how the DXA machines worked when I first started my studies in 2024.

Finally, to my mom, who always believed in my abilities and was my biggest cheerleader, I know you would have been so proud of me for completing this work. To my dad, for always supporting me, encouraging me, coming to visit, and picking up the phone when I called, thank you for the immense role that you have played in both my education and my life.

Table of Contents

Chapter 1 – Introduction.....	1
1.1 Motivation	1
1.2 Hip Joint Anatomy	3
1.3 Osteoporosis and Hip Fractures	4
1.3.1 Clinical Diagnosis and Treatment.....	5
1.3.2 Osteoporotic Hip Fractures	6
1.3.3 Risk Factors and Existing Risk Assessment Tools.....	7
1.3.4 Longitudinal Studies	10
1.4 Dual Energy X-Ray Absorptiometry (DXA)	11
1.5 Machine Learning and Artificial Intelligence	12
1.5.1 Machine Learning Techniques	12
1.5.2 Transfer Learning.....	15
1.5.3 Interpretability Techniques	15
1.5.4 Pre-Processing Techniques for Machine Learning	16
1.6 Methods for Evaluating Model Performance	17
1.7 Study Objectives	18
1.8 Thesis Overview.....	19
Chapter 2 – Study Data and Pre-processing.....	20
2.1 Introduction	20
2.2 Study Data.....	22
2.2.1 Inclusion Criteria	22
2.3 Pre-Processing Techniques.....	24
2.3.1 Image Pre-Processing Techniques	24
2.3.2 Clinical Risk Factor Pre-Processing Techniques	28
2.4 Study Demographics and Outcome Classification.....	31
2.5 Conclusion.....	36
Chapter 3 – Design and Implementation of a Feed-Forward Neural Network for Binary Fracture Classification from DXA Images	37
3.1 Introduction	37
3.2 Methodology	39
3.2.1 Train/Validate/Test Split.....	39
3.2.2 Principal Component Analysis.....	40

3.2.3 Selection of the Loss Function.....	44
3.2.4 Image-Only Algorithm.....	48
3.3 Results	53
3.4 Discussion	56
3.4.1 Relevance, Strengths, and Limitations.....	59
3.5 Conclusions	61
Chapter 4 – Design and Implementation of a Feed-Forward Neural Network for Binary Fracture Classification from DXA Images and Clinical Risk Factors	62
4.1 Introduction	62
4.2 Methodology	63
4.2.1 Model Architecture	64
4.2.2 Model Training.....	66
4.2.3 Model Validation.....	67
4.3 Results	69
4.4 Discussion	74
4.4.1 Relevance, Strengths, and Limitations.....	76
4.5 Conclusions	79
Chapter 5 – Deep Learning and Transfer Learning for Binary Fracture Classification from DXA Images and Clinical Risk Factors	80
5.1 Introduction	80
5.2 Methodology	82
5.2.1 Deep Learning Model	82
5.2.2 Transfer Learning Model	86
5.3 Results	87
5.3.1 Deep Learning Model	87
5.3.2 Transfer Learning Model	92
5.4 Discussion	94
5.4.1 Deep Learning Model	94
5.4.2 Transfer Learning Model	96
5.5 Conclusions	98
Chapter 6 – General Discussion and Conclusions	99
6.1 Summary	99
6.2 Limitations and Strengths	105
6.3 Future Directions.....	108
6.4 Significance.....	109

References	111
Appendix A – Glossary of Anatomical Terms.....	125

List of Figures

Figure 1.1: Anterior View of the Pelvis and Proximal Femur	4
Figure 1.2: Healthy and Osteoporotic Bone Tissue	5
Figure 1.3: Classification of Hip Fractures.....	7
Figure 2.1: Lunar vs. Hologic DXA Images of the Proximal Femur.....	23
Figure 2.2: Study Data	24
Figure 2.3: Complete DICOM Report	25
Figure 2.4: Complete DICOM Report	27
Figure 2.5: Correlation Heatmap of Clinical Variables	30
Figure 2.6: Normal Distribution Assessment for Age, BMI, and Total Hip BMD	32
Figure 2.7: Fracture History at Baseline Assessment	33
Figure 2.8: Study Fracture Outcomes	35
Figure 3.1: Basic Architecture of a Simple Neural Network	38
Figure 3.2: Cumulative Explained Variance Plot.....	41
Figure 3.3: Principal Component Images	42
Figure 3.4: Anatomical Detail at Various PCA Reductions	43
Figure 3.5: Training and Validation Loss Comparison	46
Figure 3.6: Focal Loss vs. Ground-Truth Probability	48
Figure 3.7: Architecture Design.....	49
Figure 3.8: Training and Validation Loss.....	53
Figure 3.9: Validation AUC	53
Figure 3.10: Confusion Matrix	54
Figure 3.11: Confusion Matrices for Thresholds in Range 0.40-0.54 (Misclassification Percentage in Brackets)	55
Figure 3.12: ROC Curve	56
Figure 4.1: Architecture Design.....	66
Figure 4.2: Training and Validation Loss.....	69
Figure 4.3: Validation AUC	70
Figure 4.4: Confusion Matrix	70
Figure 4.5: Confusion Matrices for Thresholds in Range 0.40-0.48 (Misclassification Percentage in Brackets)	71
Figure 4.6: ROC Curve	72
Figure 4.7: SHAP Analysis	73
Figure 4.8: Branch Specific SHAP Analysis	74
Figure 5.1: Deep Learning Architecture Overview.....	84
Figure 5.2: Training and Validation Loss.....	88
Figure 5.3: Validation AUC	88
Figure 5.4: Confusion Matrix	89
Figure 5.5: ROC Curve	90
Figure 5.6: SHAP Analysis	91
Figure 5.7: Image-Level SHAP Analysis.....	92
Figure 5.8: Training and Validation Loss.....	93
Figure 5.9: Validation AUC	93
Figure 5.10: Confusion Matrix	94

List of Tables

Table 2.1: Clinical Risk Factors Included in Analysis.....	29
Table 3.1: Fracture Distribution in Train/Validate/Test Splits	40
Table 3.2: Principal Component Analysis Groups to be Tested in Logistic Regression.....	42
Table 3.3: Area Under the Curve Performance in Logistic Regression	44
Table 3.4: Tuned Model Parameters	52
Table 3.5: Model Performance Metrics.....	55
Table 4.1: Global Training Parameters	68
Table 4.2: Model Performance Metrics.....	71
Table 5.1: Global Training Parameters	85
Table 5.2: Model Performance Metrics.....	89
Table 5.3: Model Performance Metrics.....	94
Table 6.1: Comparison of Risk Prediction Tools	103

List of Abbreviations and Symbols

\$	Dollar
%	Percent
α	Alpha
β	Beta
γ	Gamma
p	Probability
n	Number
μSv	Micro Sievert
g	Gram
cm	Centimeter
1D	One-dimensional
2D	Two-dimensional
AdamW	Adaptive Moment Estimation with Weight Decay
AI	Artificial Intelligence
AUC	Area Under the Receiver Operating Characteristic Curve
Backpropagation	Backward Propagation of Errors
BCE	Binary Cross-Entropy
BMD	Bone Mineral Density
BMI	Body Mass Index
CaMos	Canadian Multicentre Osteoporosis Study
CAROC	Canadian Association of Radiologists and Osteoporosis Canada

CNN	Convolutional Neural Network
CRF	Clinical Risk Factor
DICOM	Digital Imaging and Communications in Medicine
DL	Deep Learning
DXA	Dual-Energy X-Ray Absorptiometry
FFNN	Feed-Forward Neural Network
FRAX	Fracture Risk Assessment Tool
FREEDOM	Fracture REduction Evaluation of Denosumab in Osteoporosis Every 6 Months
Grad-CAM	Gradient-weighted Class Activation Mapping
JPG	Joint Photographic Experts Group
KFCV	K-Fold Cross Validation
LR	Learning Rate
ML	Machine Learning
MLP	Multi Layer Perceptron
MOF	Major Osteoporotic Fracture
MSE	Mean Squared Error
NLM	Non-Local Means
NN	Neural Network
NumPy	Numerical Python
PCA	Principal Component Analysis
PNG	Portable Network Graphics
Q-Q	Quantile-Quantile

ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SD	Standard Deviation
SHAP	SHapley Additive exPlanations
TBS	Trabecular Bone Score
TF	TensorFlow
TL	Transfer Learning
wBCE	Weighted Binary Cross-Entropy
WHO	World Health Organization

Declaration of Academic Achievement

The following is a declaration that I, Taylor Kramer, completed the research outlined in this thesis and recognizes the contributions of Dr. Cheryl Quenneville. I contributed to the study design and was responsible for the model development, validation, evaluation, data analysis, and writing of the manuscript. Dr. Cheryl Quenneville assisted with the study design and review of the manuscript.

Chapter 1 – Introduction

***Overview:** Osteoporotic hip fractures are especially detrimental to patient quality of life, and one of the leading causes of mortality in older adults. This disease is diagnosed through dual-energy x-ray absorptiometry (DXA) imaging, where a patient's bone mineral density (BMD) is calculated and translated to a T-score that is interpreted to determine their risk of osteoporotic fracture. This diagnostic approach ignores many factors that contribute to bone strength, and fragility fractures often occur in individuals with BMD T-score values in the normal or osteopenic range. This introductory chapter provides an overview of hip joint anatomy, risk factors and treatment of the disease, and principles of machine learning and statistical analysis that were used to quantify fracture risk. This chapter concludes with study objectives and a thesis overview.¹*

1.1 Motivation

Osteoporosis is a major public health concern. This disease is characterized by a loss in bone mass that causes bones to become porous and thin, making the skeleton more susceptible to fracture. The first symptom of osteoporosis is often a fragility fracture, a non-traumatic fracture caused by low-energy impact, making it challenging to diagnose the disease before bone health has largely deteriorated [1]. Osteoporosis affects over 2.3 million Canadians, and it is estimated that at least one in three women and one in five men will suffer an osteoporotic fracture in their lifetime [2]. Over 80% of all fractures in individuals over the age of 50 are caused by osteoporosis,

¹ A glossary of frequently used anatomical terms is included in Appendix A.

and no single cause for the disease has been identified [2]. Hip fractures represent a serious clinical condition, with one-year mortality rates of 22% in women and 33% in men [3].

In 2010/2011, osteoporosis-attributed fractures in Canada resulted in substantial healthcare costs. There were approximately 64,884 admissions to acute care and 216,753 emergency or same-day surgery visits, leading to 983,074 days of inpatient hospitalization and \$1.5 billion in associated acute care costs [4]. Approximately 3 million physician visits were recorded, with an estimated cost of \$253 million; the cost of prescription drugs for treatment incurred another \$346 million. Post-acute care costs were also significant, including \$274 million for home care, \$1.03 billion for long-term care, \$334 million for rehabilitation, and \$434 million for complex continuing care. Productivity losses for patients and caregivers contributed another \$304 million. Overall, the total national cost of osteoporosis-related fractures reached \$4.6 billion, a figure that has likely risen by 2026 as a result of the country's aging population. The direct medical cost of a hip fracture alone is estimated to be \$61,540 within just the first year following injury [4]. This represents a serious economic burden to Canadians and justifies the need for preventive tools that can more accurately identify individuals at high risk of osteoporotic fracture and guide timely clinical intervention.

Currently, bone mineral density (BMD) is the clinical standard used to diagnose osteoporosis. While it has been proven that a lower BMD does contribute to an increased risk of fracture [5], many individuals experience fragility fractures despite having hip or vertebral BMD values in the normal or osteopenic range [6]. The Fracture Risk Assessment Tool (FRAX), developed by the World Health Organization (WHO) Centre for Metabolic Bone Diseases, was developed to estimate an individual's ten-year risk of major osteoporotic fracture (vertebral, hip, forearm, proximal humerus), and is a broadly used fracture risk assessment tool [6]. Despite its

wide use, the tool has many limitations. For example, FRAX does not consider dose-dependent relationships, ignores variables known to affect bone health (*e.g.*, physical activity and vitamin D deficiency), and does not account for medication use [7]. The successive edition to this tool, FRAXplus, is currently available in beta form, and has made attempts to account for several of these limitations [8].

Advances in artificial intelligence and machine learning offer new opportunities to improve fracture risk prediction. DXA images have been used to develop statistical shape and appearance models with a demonstrated strong ability to predict fracture risk [9], [10]. This suggests that neural network-based approaches could similarly learn complex patterns from DXA scans, enabling automated, accurate fracture risk assessment that can be integrated into existing clinical workflows, without increasing the cost of assessment. This research focuses on leveraging neural network development and interpretability techniques to improve fracture risk prediction.

1.2 Hip Joint Anatomy

The femur is the longest, strongest, and heaviest bone in the body. The proximal end of the femur contains the femoral neck, the greater trochanter, and the lesser trochanter (Figure 1.1a). The femoral neck connects the femoral head to the femoral shaft, and the greater and lesser trochanters are bony projections that serve as attachment points for tendons and muscles [11]. The hip joint is a synovial ball-and-socket joint that is formed by an articulation between the femoral head (the spherical portion of the proximal end of the femur) and the pelvic acetabulum (the cup-like depression located on the lateral aspect of the pelvis) (**Error! Reference source not found.**b) [11]. Responsible for facilitating movement, maintaining balance, and supporting the weight of the upper body, the hip joint is a critical structure within the musculoskeletal system. The joint's articulating surfaces are lined with articular cartilage that provides a smooth surface and helps

reduce friction during movement [11]. The synovial cavity is located between the articulating surfaces and encased by the articular capsule, which strengthens the joint to prevent dislocation and secretes synovial fluid for lubrication [11].

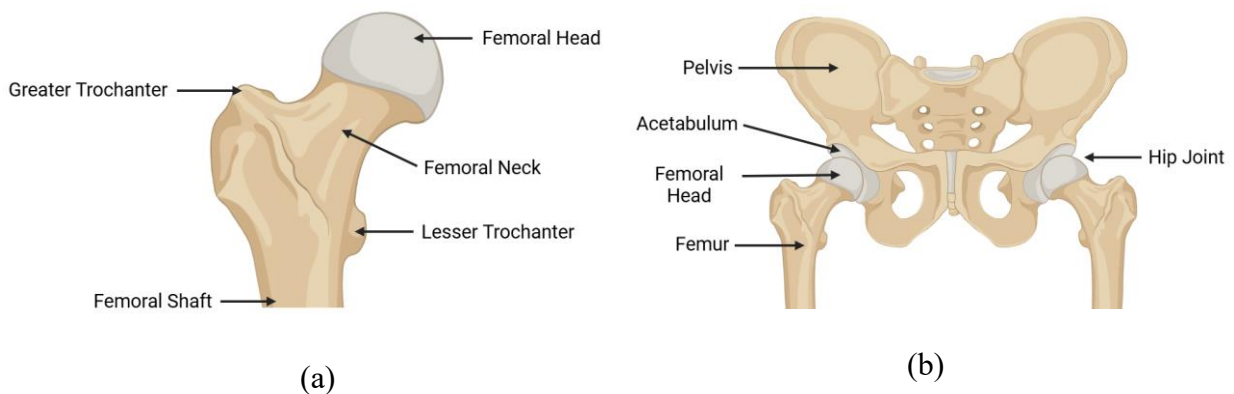


Figure 1.1: Anterior View of the Pelvis and Proximal Femur

(a) Isolated view of the proximal aspect of the femur. (b) Anterior view of the pelvic region including the hip joint (Created with Biorender.com).

1.3 Osteoporosis and Hip Fractures

Osteoporosis is a skeletal disease that is characterized by reduced BMD and the microarchitectural deterioration of bone tissue (Figure 1.2) [6]. This disease compromises bone strength and increases the risk of fragility fractures, which occur from low-energy trauma, such as a fall from standing height [12]. Osteoporosis is prevalent among Canada's aging population, affecting approximately 10% of Canadians above the age of 40 [13]. The disease is asymptomatic, and usually only comes to clinical attention after a fragility fracture has occurred [12].

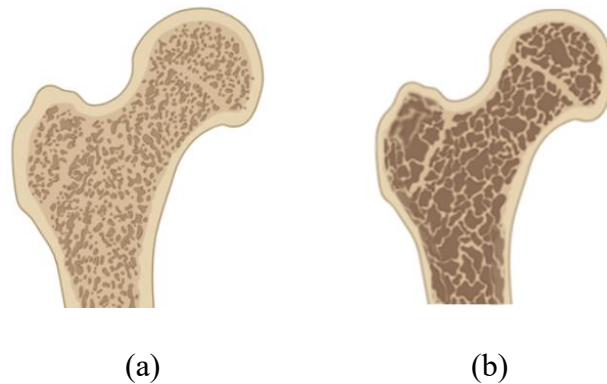


Figure 1.2: Healthy and Osteoporotic Bone Tissue

Proximal end of the femur with magnification of internal (a) healthy bone and (b) osteoporotic bone. With osteoporosis, the cortical and trabecular bone become more porous and weaker, making bones more fragile and susceptible to fractures (Created with Biorender.com).

The human skeleton consists of cortical and trabecular bone. Cortical bone represents the compact outer layer of bone tissue that provides strength and is essential in supporting the body's weight. The diaphysis (shaft) of long bones consists mainly of cortical bone [14]. Trabecular bone (also known as cancellous bone) is the spongy, porous bone that is found within the vertebrae and at the epiphyses (ends) of the long bones. Its lattice-like structure allows for improved mechanical stress attenuation, and it houses fat and bone marrow [15]. Osteoporosis affects both cortical and trabecular bone, with aging and menopause driving increased bone loss that leads to cortical and trabecular thinning [16].

1.3.1 Clinical Diagnosis and Treatment

Osteoporosis is diagnosed through areal BMD measurement, expressed in grams of mineral per square centimeter (g/cm^2), using DXA imaging [12]. An individual's BMD measurement is expressed as T-score that quantifies the number of standard deviations (SDs) their specific BMD deviates from the average of a young, healthy population, adjusted for sex [12]. The WHO classifies osteoporosis as a BMD T-score that is 2.5 SDs below the mean, osteopenia (low bone

mass) as a BMD T-score between -2.5 and -1.0 SDs, and normal bone density as a T-score greater than -1.0 [12].

Treatment of osteoporosis aims to reduce fracture risk by improving bone strength and preventing further bone loss. Managing this disease typically involves a combination of lifestyle measures and pharmacologic therapy. Medications include antiresorptive agents, such as bisphosphonates (*e.g.*, alendronate, risedronate, zoledronic acid) and denosumab, which block osteoclast activity. Adequate vitamin intake is essential for maintaining bone health, and sufficient serum levels of vitamin D have been associated with a reduced risk of fracture [17]. Dietary interventions, such as adequate protein and calcium intake, vitamin D supplementation, and limited caffeine (no more than 4 cups per day) /alcohol (no more than 2 units per day) consumption can reduce osteoporotic fracture risk [18]. Remaining physically active, particularly in weight-bearing and resistance exercises, and refraining from smoking can also improve outcomes in patients with osteoporosis [18]. Hip protectors have also been recommended for osteoporosis management, particularly in institutionalized individuals with a high fall risk [18], [19]. Evidence supporting the use of hip protectors outside institutional settings is limited, and their effectiveness is often hindered by challenges with patient compliance [18]. The selection of treatment is guided by the patient's fracture risk, overall health status, and response or tolerance to therapy.

1.3.2 Osteoporotic Hip Fractures

Hip fractures are generally classified into two main types: intracapsular fractures, which occur within the joint capsule such as a femoral neck fracture, and extracapsular fractures, which occur outside the joint capsule, such as an intertrochanteric or a subtrochanteric fracture (Figure 1.3) [20]. Femoral neck fractures are most common [11], [21]. As humans age, bone porosity can rise by 46% and microcracks can accumulate in the cortical bone, altering the material properties

that determine the overall bone strength [20]. With the progression of osteoporosis, bone stiffness, strength, and fracture toughness all decrease, increasing susceptibility to low-energy fractures [20]. Additionally, the cortical bone of the femoral neck, which provides essential structural support, becomes thinner due to endosteal resorption. Although some periosteal expansion occurs to help preserve the moment of inertia, this compensatory process is insufficient, leading to overall weakening of the region and an increased risk of fracture [16].

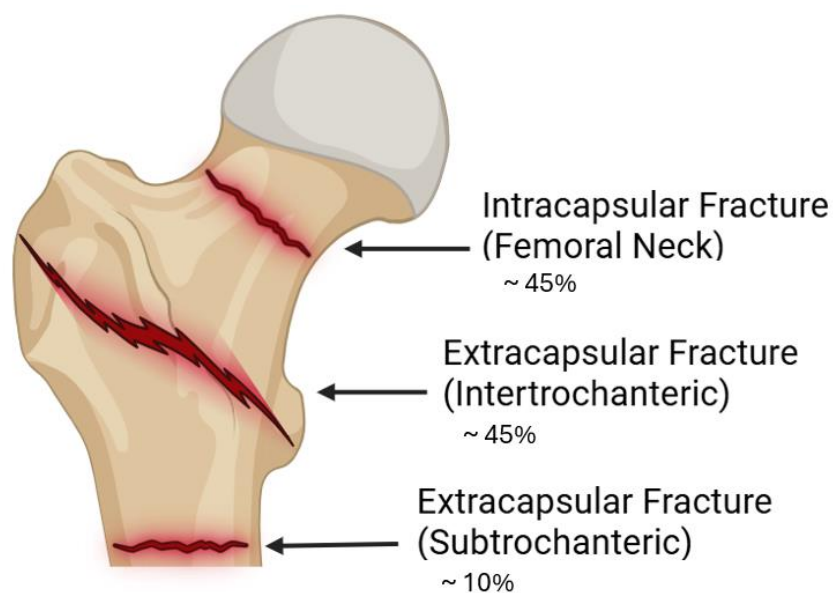


Figure 1.3: Classification of Hip Fractures

Anterior view of the proximal femur highlighting intracapsular (femoral neck) and extracapsular (intertrochanteric, subtrochanteric) fracture localization. Relative frequencies are indicated from a study of Americans over age 65 [21] (Created with Biorender.com).

1.3.3 Risk Factors and Existing Risk Assessment Tools

Osteoporosis is a complex condition influenced by many factors beyond bone density, and current diagnostic methods often miss those at risk. To better identify vulnerable individuals, assessments need to investigate further than BMD measurements alone. Additional factors that have been known to contribute to fracture risk include age, sex, body mass index (BMI), previous

fracture history, family history, tobacco and alcohol use, chronic inflammatory diseases, and use of glucocorticoids [5]. These factors disrupt bone remodeling, resulting in a weakened structure with reduced stress resistance and greater fracture risk. Women have an especially increased risk of developing osteoporosis [22]. While Type II osteoporosis affects both men and women, Type I osteoporosis is driven by the decline in estrogen levels following menopause, which accelerates bone resorption and leads to a rapid loss in trabecular bone [22].

Given there are many factors outside of BMD that contribute to an individual's overall fracture risk, additional approaches combining BMD with Clinical Risk Factors (CRFs) have been developed. Three widely used tools include FRAX, the updated model FRAXplus, and the Canadian Association of Radiologists and Osteoporosis Canada (CAROC) tool.

The FRAX algorithm was developed and validated using data from multiple meta-analyses and prospective cohort studies, encompassing individuals with diverse geographic and demographic backgrounds [6]. This free web-based tool estimates an individual's 10-year probability of sustaining a major osteoporotic fracture (MOF) through a series of self-reported questions based on CRFs [6]. Designed for individuals aged 40–90, the CRFs included in this tool were selected based on their association with fracture risk [6]. FRAX can provide risk estimates with or without a BMD value, making it more accessible to regions where DXA assessments are unavailable [6]. FRAX recommends intervention when the 10-year fracture risk exceeds 3% for hip fracture or 20% for MOF [23]. It has been suggested that these thresholds were selected based on cost-effectiveness, not clinical significance, and that the 3% threshold for hip fracture is likely too low [24]. Thresholds also vary among countries, reflecting differing healthcare priorities and resources. Selecting the appropriate threshold for clinical intervention is complex, while lowering the threshold to encourage earlier treatment could reduce long-term fracture-related healthcare

costs and prevent more fractures, it also raises the risk of overmedication and unnecessary exposure to treatment side effects [24].

While FRAX represents a widely available tool, it has some limitations. The tool is not comprehensive for all CRFs, and many are discretized to yes/no answers [7]. The FRAX tool does not consider dose-dependent relationships (*e.g.*, number of prior fractures, magnitude of alcohol consumption, etc.), and it assumes an average value for individuals reporting glucocorticoid exposure [6]. The algorithm does not account for smoking history and models current smokers as a homogeneous group with an assumed uniform daily consumption, failing to differentiate former smokers from those who have never smoked [6]. Overall, while FRAX effectively excludes individuals unlikely to sustain a fracture, it struggles to accurately identify those at risk [23].

The FRAXplus tool was introduced in beta version in 2023 and was designed to correct the limitations of the FRAX model. It builds on the traditional model by allowing adjustments for recency of osteoporotic fracture, level of exposure to glucocorticoids, duration of type 2 diabetes, fall history, and supplementary bone quality indicators such as lumbar spine BMD, trabecular bone score (TBS), and hip axis length [8]. Applying these changes at the population level has the potential to shift individuals across treatment thresholds, such as from high risk to very high risk, which could alter clinical decisions. It is important to note that as of 2024 the accuracy of using multiple simultaneous adjustments in FRAXplus has not been fully validated, and results should be interpreted cautiously [8].

Another risk assessment tool includes the CAROC tool. Similar to FRAX, CAROC is an assessment algorithm designed to predict the 10-year fracture risk for major osteoporotic fracture. CAROC stratifies fracture risk into three clusters: low (<10%), moderate (10-20%), and high (>20%) [25]. It was developed using cohorts from the Canadian Multicentre Osteoporosis Study

(CaMos) and the Manitoba Bone Mineral Density Registry [26]. The predictors included in this algorithm are sex, age, BMD, fracture history, and glucocorticoid use [26]. In postmenopausal women, fracture risk estimates derived from BMD reports using CAROC have been reported to differ from those of FRAX [27]. This discrepancy can lead to inconsistent treatment decisions, highlighting the need for a standardized and accurate assessment process.

1.3.4 Longitudinal Studies

Longitudinal studies play a vital role in human health research, as they provide the opportunity to observe changes in patient health over time. By following populations for extended periods of time, researchers can make predictions, evaluate treatments, and understand how patient health variables evolve over time. Longitudinal studies are essential for osteoporosis research because tracking patients over time helps identify fracture risk factors, evaluate how treatments impact bone health, and guide the development of reliable risk assessment tools. Both the Amgen (Amgen, Thousand Oaks, CA, USA) Fracture REduction Evaluation of Denosumab in Osteoporosis Every 6 Months (FREEDOM) and CaMos studies have provided long-term data on older adults to evaluate the progression of osteoporosis [28], [29].

The Amgen FREEDOM trial was a randomized, double-blind, placebo-controlled, international multi-centre study investigating postmenopausal women with a clinical osteoporosis diagnosis [19]. In this Phase 3 clinical trial, 7808 women between the ages of 60 and 90 were randomized to receive denosumab or placebo every 6 months for 36 months [30]. Hip DXA scans were collected at baseline and at months 12, 24, and 36, and the primary endpoint was the reduction in new vertebral fractures [30].

The Canadian Multi-Centre Osteoporosis Study was a population based, prospective, non-interventional, observational study focused on the incidence and prevalence of osteoporosis and

fractures among Canadians [31]. A total of 6,539 women and 2,884 men aged 25 years and older were followed for at least 10 years, with extended follow-up continuing for up to 16 years [31]. DXA scans of the hip and lumbar spine were obtained at baseline, and participants aged 40–60 years at study entry underwent additional scans at 3 and 5 years of follow-up [29]. The CaMos study includes important data on fractures, bone mineral density, demographic characteristics, and more [32].

1.4 Dual Energy X-Ray Absorptiometry (DXA)

Medical imaging techniques are important tools in both research and clinical practice. Numerous medical imaging modalities exist, each offering unique diagnostic capabilities by allowing for the visualization of internal anatomical structures. Some examples include standard X-rays, Computer Tomography, Magnetic Resonance Imaging, and ultrasound. In assessing osteoporosis and bone health, DXA imaging is the standard of care.

Dual-energy x-ray absorptiometry is a two-dimensional medical imaging technique used to assess bone porosity. The system works by measuring the signal attenuation through the body from two independent X-ray beams. The patient's bone density is then calculated by measuring the difference between the signal attenuation of the two beams. Standard DXA assessment for BMD typically focuses on two main anatomical locations: the hip and the lumbar spine. Apart from BMD assessments, DXA technology is also used to measure factors such as body fat percentage, fat/lean mass, and total body composition [33].

Unlike some other imaging modalities, DXA imaging exposes the patient to minimal radiation, reducing the likelihood of harmful side effects from radiation exposure. The effective radiation emission from a DXA scan ranges from 1-10 μSv (microsieverts). A typical chest X-ray exposes individuals to around 100 μSv of radiation, while flight crews can receive an average

annual dose of approximately 3 mSv [34]. Due to the relatively low radiation dose, patients can undergo multiple DXA scans and remain below the exposure of a single conventional X-ray scan, allowing for more frequent patient assessments.

The two major DXA manufacturers are Hologic Inc (Hologic Inc., Marlborough, MA, USA) and GE Healthcare (GE HealthCare, Chicago, IL ,USA). Both manufacturers' systems use similar DXA technology and can assess the same variables, but BMD values may differ slightly due to variations in proprietary calibration methods, the use of different algorithms, and the selection in region of interest [33]. To minimize machine-to-machine error, attempts have been made to develop calibration equations between the machines, but this has not yet gained widespread adoption [33].

1.5 Machine Learning and Artificial Intelligence

1.5.1 Machine Learning Techniques

Artificial intelligence (AI) is the ability of machines to execute operations that normally require human intelligence, such as problem solving and decision making. Artificial intelligence systems do this by using different algorithm structures to analyze large volumes of data and learn patterns that enable them to then make predictions, recognize trends, and perform tasks autonomously. Machine learning (ML) is a subset of artificial intelligence in which systems improve their performance on a task over time by learning from data. Unlike broader AI, which encompasses any machine capability that mimics human intelligence, ML specifically focuses on algorithms that identify patterns in datasets to make predictions or decisions without being explicitly programmed for each task. Deep learning (DL) is a subset of machine learning that employs multi-layered neural networks to automatically learn complex patterns from large datasets. Unlike traditional ML, which often requires manual feature selection, DL can

automatically extract features from raw data, enabling tasks such as image recognition, natural language processing, and predictive modeling [35].

Designing and deploying a machine learning model typically follows a standard workflow. Raw data are first collected and pre-processed to address missing values, normalize scales, and extract relevant features. The processed data are then divided into training, validation, and test sets, which are used to train the model, tune hyperparameters, and evaluate performance, respectively. Once trained, the model can make predictions or classify new, unseen data [35].

Supervised machine learning is a process by which an algorithm can make predictions on novel data after learning and optimizing itself from a specific training set that has been labeled (for example, medical images can be labeled as “fractured” or “non-fractured” from follow-up data). Supervised ML algorithms include logistic regression, decision trees, random forests, support vector machines, k-nearest neighbours, neural networks, and convolutional neural networks. While these algorithms all use labeled data to learn patterns and make predictions, they differ in how they learn decision boundaries and handle complexity or nonlinearity. In neural networks, training often involves minimizing a loss function, which measures the difference between the model’s predictions and the true outcomes. This is achieved through the backward propagation of errors (backpropagation), a process that calculates how changes in each model parameter affect the loss and updates the parameters accordingly to improve predictions. The test set is kept separate and is used only to evaluate the model’s final performance.

Unsupervised machine learning is a technique where ML algorithms analyze unlabeled datasets. In this, the algorithms are designed to uncover hidden patterns in the dataset without the need for human intervention. Typical unsupervised learning approaches include clustering (*e.g.*, k-means) and dimensionality reduction (*e.g.*, principal component analysis). Unlike supervised

learning, these different unsupervised approaches differ by their underlying objectives. Clustering optimizes similarity-based grouping of data points, while dimensionality reduction preserves variance by projecting multi-dimensional data into a lower-dimensional space, creating latent variables.

Traditional machine learning algorithms such as gradient boosting, random forest, decision tree, and logistic regression have been applied to predict fracture risk from FRAX parameters [36]. Their performance was evaluated using the area under the receiver operating characteristic (ROC) curve, a measure of a model's ability to distinguish between classes, with reported values ranging from 0.63 to 0.66, indicating reasonable predictive ability [36]. This demonstrates that there is potential for machine learning algorithms to predict osteoporotic fracture risk, with room for improvement. Another study applied DXA scan measurements from a cohort of 4,722 women to 24 statistical models, achieving a test area under the curve (AUC) of 0.91 [37]. While this metric indicates high predictive ability, this study only had 293 participants experience a fracture, representing only a 6.6% fracture distribution in the cohort.

When applying ML tools to an imbalanced dataset, AUC can be misleading. While it measures a model's ability to rank predictions, it does not account for class prevalence or the real-world consequences of false negatives and false positives [38]. In rare-event settings like fracture prediction, a high AUC can mask the fact that many true positive cases are missed, limiting clinical utility. The F2-score, which emphasizes recall over precision, provides a more meaningful evaluation by prioritizing the detection of positive cases while still penalizing false positives. This makes F2 a practical metric that can be used in parallel with AUC to assess model performance, ensuring predictions align with clinical priorities. Although many studies have incorporated tabular

BMD variables and medical images into their algorithms, only few have investigated the use of DXA in their predictive models [39], [40].

1.5.2 Transfer Learning

Transfer learning (TL) is a deep learning approach where a model developed for a particular task is reused as the starting point for a model on a second task. Instead of training a neural network from scratch, which requires large amounts of labeled data, transfer learning leverages pre-trained models, allowing knowledge learned from large benchmark datasets to be applied to new, often smaller, datasets [41]. This is particularly useful in medical imaging, where annotated datasets are frequently limited. In transfer learning, the early layers of a neural network, which typically extract low-level features such as edges and textures, are retained, while the later layers are fine-tuned or replaced to suit the target task. This approach accelerates convergence, improves generalization, and often achieves high performance even on modest datasets. Transfer learning approaches have been used to classify radiographic images and detect fractures, with a high degree of success [42].

1.5.3 Interpretability Techniques

Model interpretability is important for ML applications in medicine as it can help clinicians understand and trust model predictions. Gradient-weighted Class Activation Mapping (Grad-CAM) is a visualization technique that highlights the regions of an input image that are most influential in the model's decision making. This tool has also been used in detecting and visualizing hip fractures from pelvic radiographs [43]. In previous work, Grad-CAM has localized the fracture site in 95.9% of test images, demonstrating that the model accurately identified and classified hip fractures [43]. In images without fractures, Grad-CAM did not highlight the hip as a high-risk region, indicating that the model focused on clinically relevant features rather than the hip

indiscriminately [43]. SHapley Additive exPlanations (SHAP) is another interpretability framework that explains machine learning model predictions by assigning a contribution value to each input feature based on its impact on the output. This property makes SHAP particularly useful for assessing feature importance, as attribution values can be aggregated across samples to rank and compare the relative influence of different image regions or input features on model predictions [44].

1.5.4 Pre-Processing Techniques for Machine Learning

In machine learning studies that involve image-based data, it is critical to ensure that the images are pre-processed to eliminate noise, standardize image dimensions, and normalize pixel values. Min-max normalization is a common pre-processing technique that is performed to scale pixel values between [0, 1]. By ensuring all values fall within a specific range, the model is more likely to improve model convergence. Normalizing image data ensures that all features contribute equally to the model's learning process [45], [46], limiting a single feature's ability to disproportionately influence the model's output.

It is also important to pre-process CRFs, as these features often have drastically different ranges. For instance, patient age will likely fall between zero and 90 years, whereas BMD values will typically fall between 0.5-1.5 g/cm². Without proper scaling, features with larger numeric ranges can disproportionately influence model training, potentially overshadowing clinically relevant variables with smaller magnitudes. Standardization techniques are commonly applied to ensure that all features contribute proportionally to the learning process.

Principal component analysis is a dimensionality reduction technique that is used to simplify data while retaining its variance. This technique is especially useful with image-based neural networks, where each pixel represents one node in the input layer of the network. Analysis of high-resolution

images with neural networks can result in models becoming overfit due to the large number of pixels present, while also requiring high-level computational resources to train. Principal component analysis allows for pixel quantities to be reduced, without potentially losing important radiographic information, reducing computational power and minimizing overfitting [47].

Principal component analysis works by projecting the image input space onto a smaller set of new independent, orthogonal, and uncorrelated vectors known as principal components. These components are ranked by the amount of variance that they capture from the original dataset, allowing for a more compact representation while retaining key data patterns and relationships [47]. Selecting more principal components to include in the model will capture additional variance; however, most of the data's variance is typically captured by the first few components. This technique largely reduces the feature space, which can lead to improved results [47]. Different numbers of principal components could be retained across multiple datasets to assess how the results change with varying amounts of included components, as demonstrated in previous work [48].

1.6 Methods for Evaluating Model Performance

The receiver operating curve is a widely used tool for evaluating the performance of binary classification models. It plots the true positive rate (sensitivity) against the false positive rate ($1 - \text{specificity}$) across different decision thresholds, providing a comprehensive view of a model's ability to discriminate between classes. The AUC quantifies the overall performance of the classifier: an AUC of 1.0 indicates perfect discrimination, while an AUC of 0.5 corresponds to performance no better than random chance.

In classification tasks, particularly in medical applications, it is crucial to carefully balance the trade-off between correctly identifying positive cases and minimizing false predictions.

Precision measures the proportion of predicted positive cases that are truly positive, while recall (sensitivity) measures the proportion of actual positive cases correctly identified by the model. The $F\beta$ -score provides a single metric that combines precision and recall, allowing the relative importance of each to be adjusted depending on the clinical context. The F1-score gives equal weight to precision and recall, making it suitable when a balance between false positives and false negatives is desired. The F0.5-score emphasizes precision, which is useful when minimizing false positives is critical, whereas the F2-score emphasizes recall, prioritizing the identification of true positives. In medical diagnostics, prioritizing recall and higher $F\beta$ -scores that favor true positive detection is meaningful, as failing to detect a condition can have serious consequences for patient care and treatment outcomes [36], [49].

1.7 Study Objectives

The overall goal of this research was to evaluate the effectiveness of neural network and deep learning methods to predict osteoporotic fracture risk from DXA images and CRFs in postmenopausal women.

The specific objectives of this thesis were to:

1. Develop a machine learning model based solely on DXA images to quantify an individual's risk of sustaining an osteoporotic hip fracture within a three-year period;
2. Develop and evaluate a multi-modal machine learning model integrating DXA images with clinical risk factors to quantify an individual's risk of sustaining an osteoporotic hip fracture within a three-year period;
3. Investigate alternative deep learning methods for enhancing risk prediction and interpretation.

1.8 Thesis Overview

This thesis examines the various approaches used to classify fracture risk in postmenopausal women diagnosed with osteoporosis. Chapter 2 provides an overview of the study dataset and describes the pre-processing steps applied prior to training the machine learning models. Chapter 3 details the development of a feed-forward neural network trained exclusively on DXA images. Chapter 4 evaluates a multi-modal architecture, demonstrating how the integration of clinical risk factors enhances the performance of the image-only model. Chapter 5 explores initial experiments with deep learning and transfer learning approaches. Finally, Chapter 6 summarizes the main findings of this work and outlines directions for future research.

Chapter 2 – Study Data and Pre-processing

Overview: *This chapter introduces the study data that were used to train, validate, and test the machine learning (ML) models developed in this thesis work. The study data consist of dual energy X-ray absorptiometry (DXA) images and clinical risk factors (CRFs) obtained during the Amgen Fracture REduction Evaluation of Denosumab in Osteoporosis Every 6 Months (FREEDOM) Trials, a Phase 3 clinical trial aimed at evaluating the clinical efficacy and safety of denosumab (Prolia[®]) for treating postmenopausal osteoporosis. This chapter presents the inclusion criteria, variable distributions, and pre-processing techniques applied to prepare the data for integration into machine learning pipelines.*

2.1 Introduction

Longitudinal studies are essential in health research because they allow researchers to examine how patients respond to environmental or pharmacological factors over time. By repeatedly collecting data from the same individuals, these studies make it possible to observe changes in health outcomes, track disease progression, identify long-term trends, and explore cause-and-effect relationships. Clinical trials are a form of longitudinal study, as they follow participants over time to evaluate the outcome of a medical intervention. These studies are conducted in phases. Earlier phase trials (phase 0, 1, and 2) focus largely on safety, effectiveness, and side effects. Phase 3 studies focus on confirming efficacy, monitoring adverse reactions, and comparing the intervention to standard treatments in larger populations.

The Amgen (Amgen, Thousand Oaks, CA, USA) Fracture REduction Evaluation of Denosumab in Osteoporosis Every 6 Months (FREEDOM) clinical trial was a Phase 3 clinical trial that enrolled 7,808 postmenopausal women aged 60-90 years with an osteoporosis diagnosis [28]. Participants enrolled in the trial were randomized (1:1) to receive placebo or 60 mg subcutaneous denosumab every six months for three years [28]. Bone mineral density (BMD) measurements at the proximal femur and lumbar spine were obtained for all participants at baseline and at 12-month follow-up visits over three years [28]. The dual-energy x-ray absorptiometry (DXA) scans were performed using either Hologic (Hologic Inc., Marlborough, MA, USA) or Lunar (GE HealthCare, Chicago, IL, USA) machine types. Additional clinical variables relevant to fracture risk such as age, body mass index (BMI), fracture history, race, and treatment status were also recorded at baseline assessment. At follow-up, any fractures occurring since baseline were recorded for 15 anatomical locations. At the end of the three-year study, denosumab was found to reduce new vertebral fractures (VFs) by 68%, hip fractures by 40%, and non-vertebral fractures by 20%, demonstrating that the monoclonal antibody was effective in mitigating fracture risk caused by osteoporosis [28].

In machine learning (ML) studies, it is important to correctly pre-process data before beginning model development. This typically involves handling missing values, normalizing or standardizing features, and encoding categorical variables into a machine-readable format [50]. When selecting which variables to include, it is important to consider how they relate to one another and the quantity of available data. Models that include a larger number of variables require sufficient data to capture meaningful relationships and avoid overfitting, which occurs when the ML model memorizes the training data rather than learning the underlying relationships. This is often evident when the model shows a very low training loss but a much higher validation loss.

The Amgen FREEDOM trials collected both structured and unstructured data types. The unstructured data consisted of DXA images collected at baseline, while the structured data included a set of clinical risk factors. Because these data types differ in format and content, distinct pre-processing approaches were required; the methods used are described in this chapter.

2.2 Study Data

2.2.1 Inclusion Criteria

Data from the Amgen FREEDOM trials were provided to the McMaster Injury Biomechanics Laboratory to develop fracture risk prediction models. Of the 7,808 participants enrolled in the FREEDOM study, 1,330 discontinued, and 6,478 completed the study. Only participants who had consented to future research use and were covered under applicable country regulations could have their data included in this work. In addition, some scans were acquired using an older proprietary DXA system that did not support Digital Imaging and Communications in Medicine (DICOM) conversion; participants who received their BMD assessments on these scanner types were excluded. As such, the dataset available for the present work was for a cohort of 1,600 participants, and consisted of DXA images obtained at baseline, 12-, 24-, and 36-month assessments, as well as clinical variables collected at baseline and 36-month follow-up. Participants were further excluded if they were missing a readable baseline DXA scan, had incomplete CRF data, or were under 50 years of age at baseline.

To avoid inter-manufacturer variability, only DXA scans from one system type were included. Given most scans were acquired on Lunar machines (864 vs. 633) and these images showed higher contrast with less prominent bounding box lines (Figure 2.1), only Lunar scans were used for model development. It should be noted that multiple DXA machine models exist within each manufacturer; however, detailed information on the specific Lunar machine models

used at individual study centers was not available, so the variability between machine types could not be considered. Fracture status at follow-up was a required variable, as it was used to label the unstructured data and convert it into a semi-structured format. Since DXA images are not suitable for fracture diagnosis, follow-up data were used to identify participants who experienced a fracture after baseline and those who did not. After applying the inclusion criteria, the final cohort consisted of 806 participants (Figure 2.2).

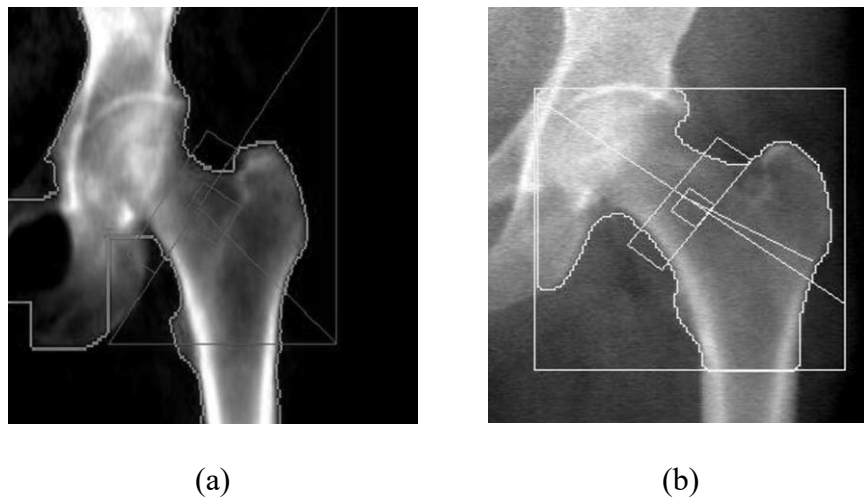
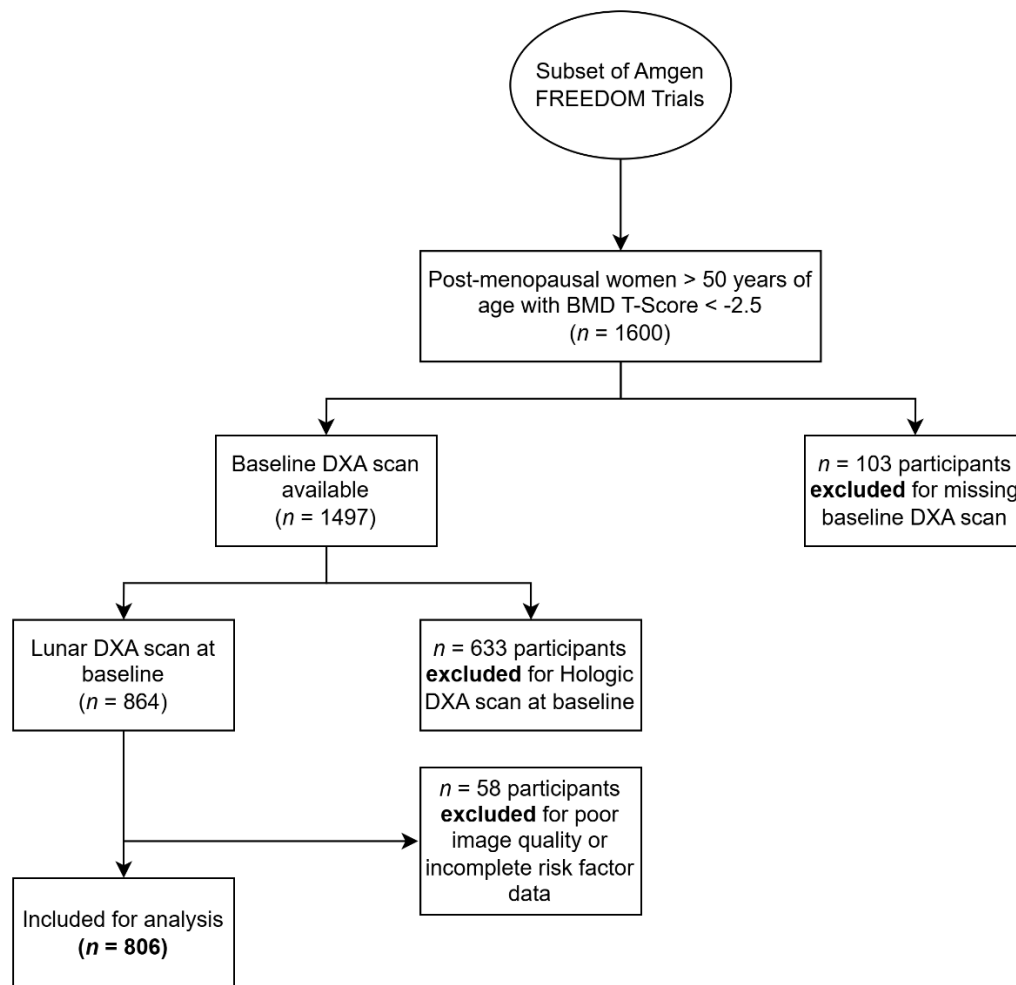


Figure 2.1: Lunar vs. Hologic DXA Images of the Proximal Femur

(a) Proximal aspect of the femur collected through Lunar DXA imaging. (b) Proximal aspect of the femur collected through Hologic DXA imaging.

**Figure 2.2: Study Data**

Flowchart illustrating participant inclusion and exclusion, with justification for excluded cases.

2.3 Pre-Processing Techniques

Various pre-processing techniques were included for the image and tabular data. Pre-processing techniques were consistent for all models developed throughout the study.

2.3.1 Image Pre-Processing Techniques

The images were received as complete DICOM reports (Figure 2.3). This is the standardized format for handling, storing, and transferring medical images. These files contain anatomical image data and metadata that includes information on the patient, study, and image

acquisition. The anatomical region of interest (ROI) needed to be isolated from the full DXA acquisition. The anatomical region of interest (ROI) needed to be isolated from the full DXA DICOM report (Figure 2.3) before algorithm development to ensure that the model only learned from the image input and not external noise.

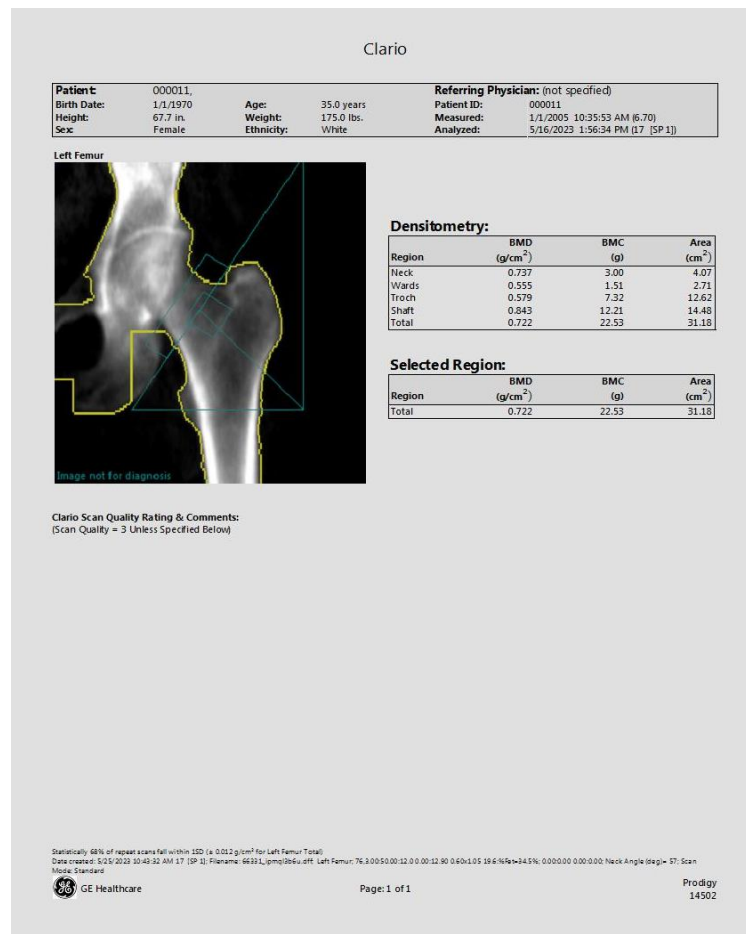


Figure 2.3: Complete DICOM Report
Lunar DICOM report of hip DXA imaging.

All data were anonymized to ensure patient privacy. While DICOM images are an excellent solution for efficient data transfer, other file formats such as joint photographic experts group (JPG) or portable network graphics (PNG) are more suitable for machine learning and image processing tasks [51]. The specified file types differ in their compression methods. Files in the JPG format use lossy compression, which reduces file size by discarding some image data, whereas

files in the PNG format use lossless compression, which reduces file size without discarding any image data. Previous research has shown that JPG compression does not negatively impact image-based model performance and, in some cases, can even reduce noise or improve classification accuracy in ML tasks [51], [52]. Although PNG files retain all image details, JPG was used in this study because it offered a balance between image quality, storage size, and processing efficiency. In addition, because large population-based datasets such as the Canadian Longitudinal Study on Aging (CLSA) commonly use JPG files, this format was prioritized for development.

The goal of image pre-processing was to convert all DICOM images to JPG, isolate the region of interest from the complete DXA report, and then perform standard image pre-processing steps, (*e.g.*, normalization and resizing) to prepare the images for model development. Once the images were pre-processed, they were tagged with fracture status, which identified participants who sustained a fracture during the follow-up period versus those who did not. This labeling process effectively converted the imaging data from an unstructured format into a semi-structured dataset, allowing for more efficient data handling, analysis, and integration with clinical variables. To isolate the ROI from each DXA image, images were cropped using a custom MATLAB (MathWorks, Natick, MA, USA) script. This approach avoided the use of screenshots, which could introduce additional compression and pixel distortion. By extracting the ROI directly from the original image data, pixel integrity was preserved and all subsequent image-based analyses were performed on high-quality, unaltered data. Following cropping, images were reviewed, and any right hip images were mirrored using a custom MATLAB script to ensure consistent anatomical orientation with the left hip.

Once the images were isolated from the DXA report they were center cropped to have a size of 224×224 pixels for consistency. Standardizing image size also prevented distortions during

[illegible]

(a) Complete patient report from DICOM image. (b) Final processed output.

2.3.2 Clinical Risk Factor Pre-Processing Techniques

All clinical risk factors included in this study were obtained at baseline assessment. The FREEDOM study recorded previous fracture history into three variables: number of non-vertebral fractures, number of vertebral fractures, and number of prevalent vertebral fractures (fractures identified at baseline via radiographic assessment). Previous fracture incidence is highly predictive of future fracture [5], so all prior fracture events were included. Treatment was also considered, as antiresorptive drugs are known to influence fracture risk [28]. Finally, age and BMI were included as key demographic and physiological risk factors [5]. Table 2.1 highlights the included CRFs, their initial range, pre-processing method, and range after pre-processing.

Table 2.1: Clinical Risk Factors Included in Analysis

Clinical risk factors used for machine learning model development, including corresponding pre-processing methods and value ranges before and after pre-processing, for the cohort of 806 individuals.

CRF	Range in Study Dataset	Average ($\pm$ Standard Deviation)	Pre-processing Step	Range After Pre-processing
Age (Years)	[60, 89]	72.0 (5.0)	Standardized	[-2.35, 3.35]
Body Mass Index (kg/m ²)	[15.05, 41.33]	25.7 (4.0)	Standardized	[-2.66, 3.88]
# Previous Non-Vertebral Fractures	[0, 6]	0.64 (0.94)	Standardized	[-0.68, -0.55]
# Previous Vertebral Fractures	[0, 7]	0.34 (0.82)	Standardized	[-0.42, 8.15]
# Prevalent Vertebral Fractures at Baseline	[0, 5]	0.45 (0.80)	Standardized	[-0.56, 5.68]
Total Hip DXA BMD (g/cm ²)	[0.50, 1.04]	0.76 (0.09)	Standardized	[-2.63, 3.90]
Treatment	Denosumab (n = 398) Placebo (n = 408)	N/A	Binarized	Placebo = 0 Denosumab = 1

Standardization was performed according to Eq 2.1:

$$x' = \frac{x - \mu}{\sigma} \quad \text{Eq. 2.1}$$

Each new value (x') was calculated by subtracting the variable mean (μ) from the sample value (x) and dividing by the standard deviation (σ). It is important to implement standardization for continuous variables in ML applications to ensure that the range is comparable among predictors.

In ML model development, it is important to not include predictors that are highly linearly correlated. Multi-collinearity can distort variable importance and reduce model stability, potentially harming predictive performance. To examine whether the included variables were

linearly independent, a correlation heatmap was generated (Figure 2.5). Variables with pairwise correlations exceeding ± 0.7 were not both included in model development [53].

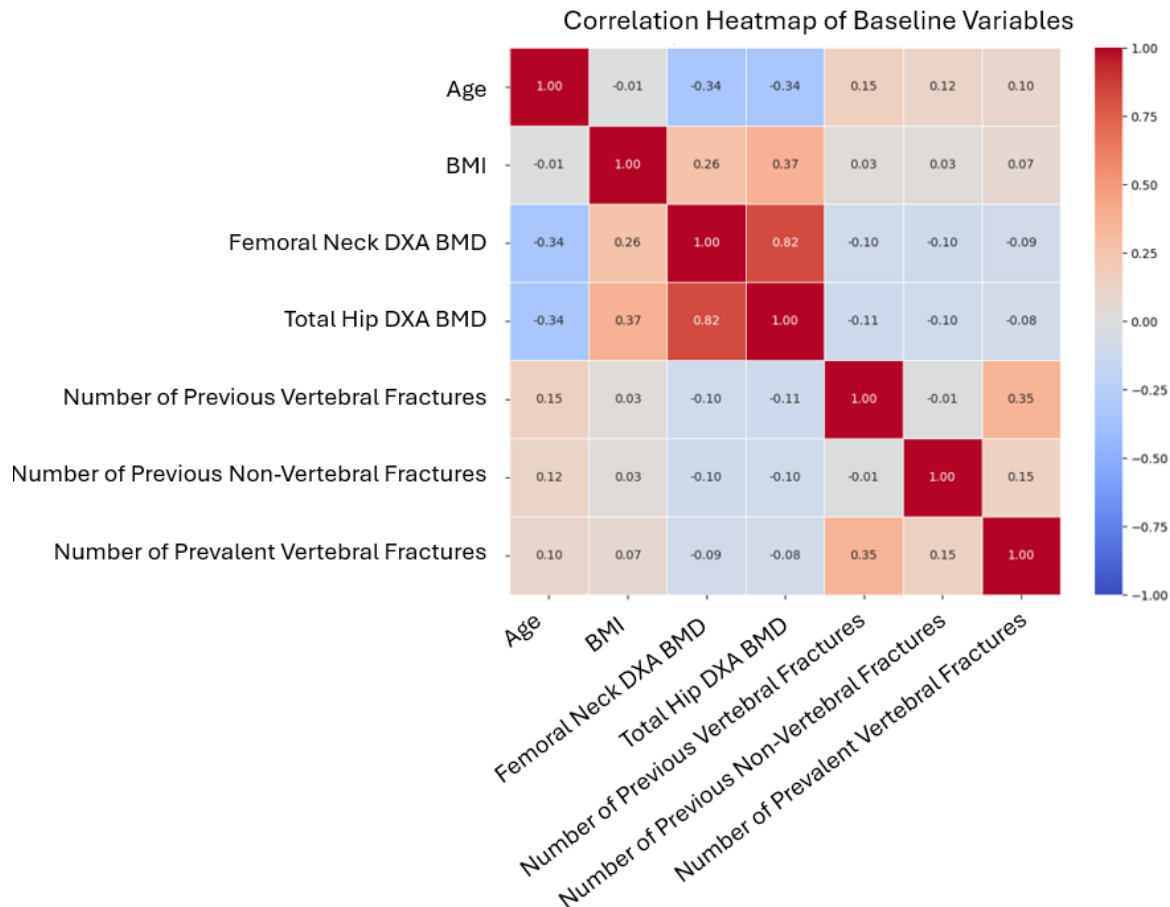


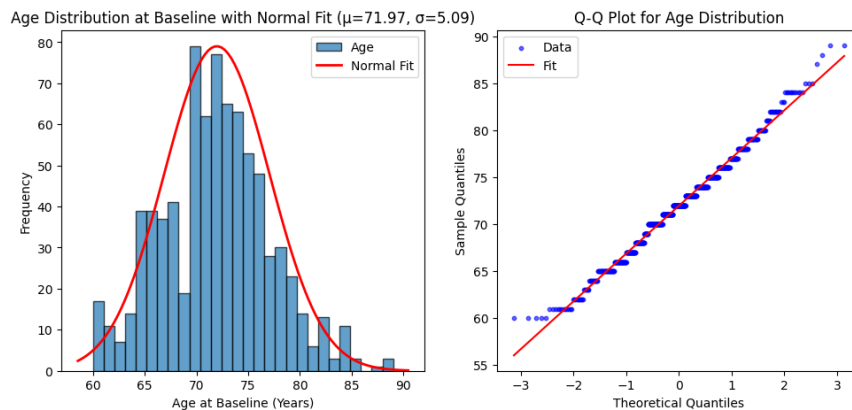
Figure 2.5: Correlation Heatmap of Clinical Variables

Correlation heatmap identifying the linear relationship between known fracture risk factors.

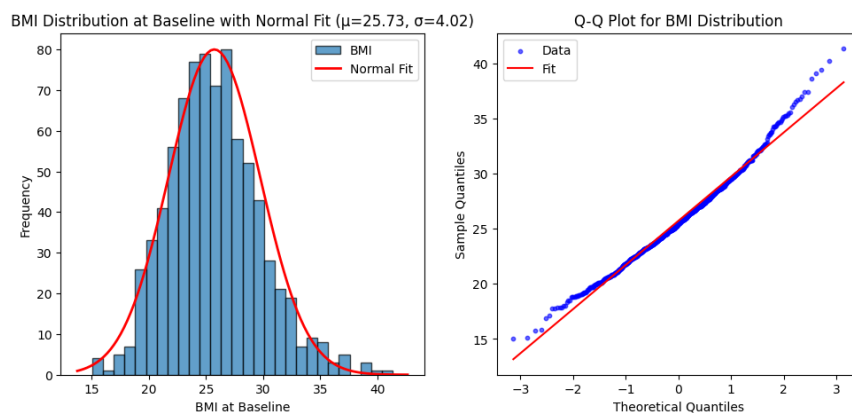
The only correlation that exceeded the established threshold was total hip DXA BMD with femoral neck DXA BMD, with a positive correlation of 0.82. Between the two, total hip DXA BMD was considered more meaningful, as it may better capture information relevant to extracapsular fractures. For this reason, femoral neck DXA BMD was excluded from the analysis.

2.4 Study Demographics and Outcome Classification

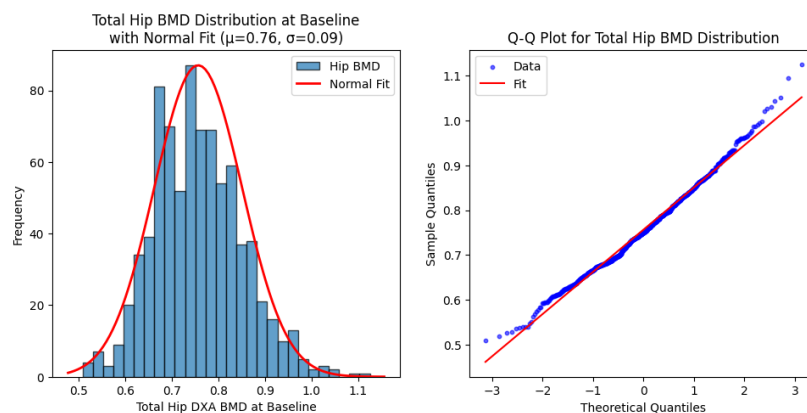
Prior to model development, all variables were assessed for normality. This was done using Quantile-Quantile (Q-Q) plots, which compare the distribution of each variable to a theoretical normal distribution to visually identify deviations from normality. Histograms were also developed, with the probability density function overlaid to visually assess how closely each variable followed a normal distribution. This analysis was done for age, BMI, and total hip BMD (Figure 2.6). No significant deviations from the line of best fit were observed, indicating that the data were approximately normally distributed.



(a)



(b)



(c)

Figure 2.6: Normal Distribution Assessment for Age, BMI, and Total Hip BMD
Age distribution, (b) BMI distribution (c), Total Hip BMD distribution.

Fracture history at baseline was also investigated (Figure 2.7). 172 participants had a history of VF, 339 participants had a history of non-VF, and 245 participants had a prevalent VF at baseline assessment. Prevalent VFs were defined at baseline as vertebral bodies with a semi-quantitative grade of ≥ 1 (≥ 20 -25% vertebral height reduction) on a spine radiograph, which is indicative of a previous fracture [54].

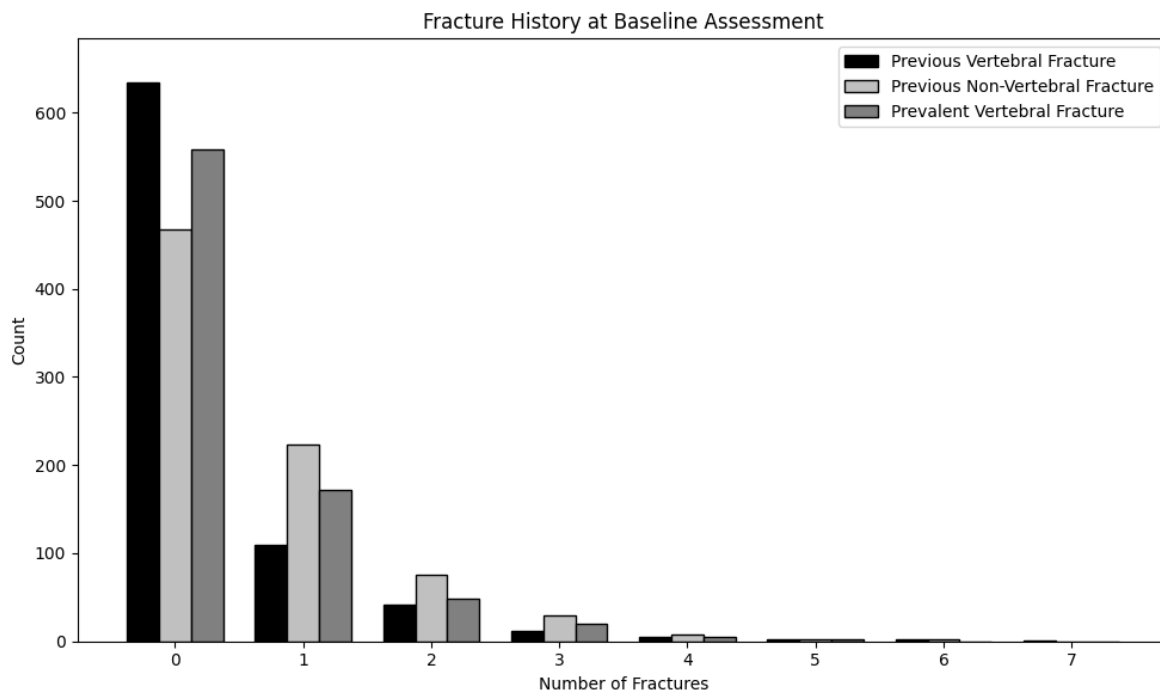


Figure 2.7: Fracture History at Baseline Assessment

At study entry, 21% of participants had a history of vertebral fracture, 42% had a history of non-vertebral fracture, and 31% had a prevalent vertebral fracture identified at baseline assessment. Additionally, 8% of participants had experienced more than one vertebral fracture, 14% had experienced more than one non-vertebral fracture, and 9% had more than one prevalent vertebral fracture.

The primary outcome of interest for this study was the incidence of hip fracture over a three-year follow-up period. In the cohort of 806 participants, 33 individuals (4.1%) sustained a hip fracture. When training machine learning models, it is important to consider the effect of class imbalance. This phenomenon occurs when one outcome class is largely under-represented compared to the other, leading the model to favor the majority class and potentially overlook

patterns associated with the minority class [55]. This can result in biased predictions, poor sensitivity to clinically meaningful outcomes, and misleading performance metrics [56]. Class imbalance is common in medical datasets because many clinically important events are relatively rare. In this study, class imbalance was further amplified by the fact that 398 participants were receiving denosumab, a treatment proven to reduce fracture risk, resulting in even fewer fracture cases for the model to learn from [28], [54]. To help mitigate the impact of class imbalance, the positive fracture class was expanded to include hip fractures, major osteoporotic fractures (MOFs), and new vertebral fractures (VFs). The working premise was that fractures at these sites reasonably reflect underlying skeletal weakness, relevant to hip strength, which was appropriate given that this study focused on hip DXA scans.

Broadening the outcome increased the number of fracture cases available for model training, an approach that has been used in previous machine-learning studies of fracture risk assessment [44]. 173 participants in the cohort experienced a new VF (21.5%), and 119 participants experienced an MOF (14.8%). The Amgen FREEDOM trials defined MOFs as any clinical vertebral, hip, forearm, or humerus fracture [57]. Clinical vertebral fractures were defined as any new VF that presented with pain or symptoms that were suggestive of fracture, with fractures due to high-energy trauma or pathological causes excluded. New vertebral fractures were defined as vertebral fractures identified on scheduled follow-up radiographs that showed an increase of at least one Genant grade compared with a prior grade of 0, 1, or 2 at vertebral levels T4 through L4. Because new vertebral fractures include both symptomatic and asymptomatic events, their total count exceeds that of major osteoporotic fractures. New vertebral fractures were therefore used in this study, as even asymptomatic fractures reflect underlying skeletal fragility and provide valuable information for model learning. Hip fracture was defined as any fracture occurring at the femoral

neck, intertrochanteric, or subtrochanteric region; however the specific fracture site was not available for analysis [57] (Figure 2.8).

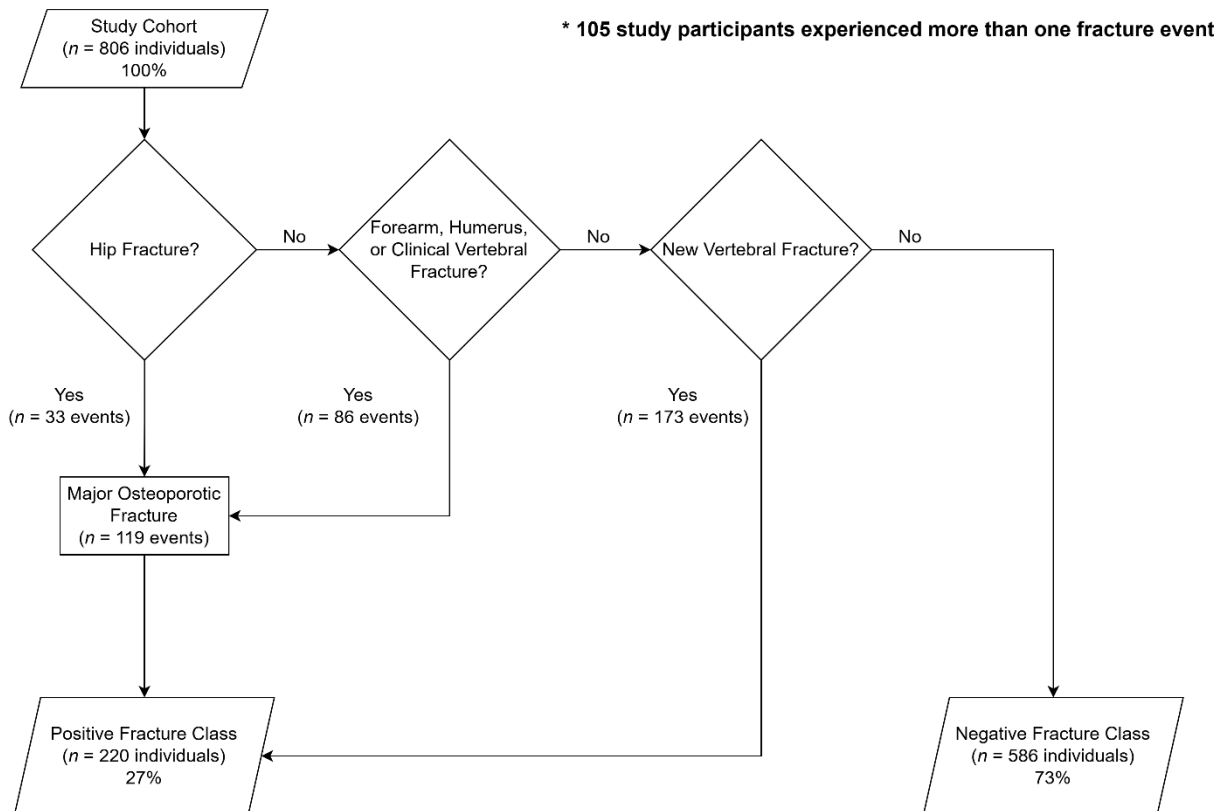


Figure 2.8: Study Fracture Outcomes

Fracture outcome distribution in the cohort of 806 participants that were used for all model development in this thesis work. A total of 220 participants experienced at least one fracture at follow-up, including 105 individuals who sustained more than one fracture event.

Within the study cohort, 33 hip fracture events, 119 major osteoporotic fracture events, and 173 new vertebral fracture events were recorded. To be included in the positive fracture class, participants only had to experience one fracture in any of the aforementioned fracture groups. Participants with no fracture at any of these locations were included in the negative fracture class. The positive fracture class included 220 individuals because although the total number of fracture events in the cohort was 325, certain participants experienced a fracture at more than one location

(*e.g.*, new vertebral fracture and forearm fracture resulting in new VF classification and MOF classification). The number of individuals included in the positive fracture class represented the number of unique individuals who experienced a fracture at one of the three selected locations.

2.5 Conclusion

This chapter described the study data, inclusion criteria, pre-processing methods, demographic characteristics, and outcome classification of the cohort used in this work. These elements provide the basis for the predictive modeling approach presented in Chapter 3. Understanding the data and pre-processing steps used to develop the ML models is important for interpreting their performance. The next chapter introduces a machine learning framework for predicting future fracture risk, which integrates processed imaging features with clinical data to assess their combined predictive ability.

Chapter 3 – Design and Implementation of a Feed-Forward Neural Network for Binary Fracture Classification from DXA Images

***Overview:** This chapter introduces a traditional machine learning approach to classifying fracture risk from dual energy x-ray absorptiometry (DXA) images. The proposed algorithm is a feed-forward neural network (FFNN) for binary classification, aimed at classifying study participants as either high- or low-risk for fracture. It provides a foundation on which future tools for predicting fracture risk can be built. This chapter describes the process used to train, tune, and test the model. It also discusses the techniques used to evaluate model performance.*

3.1 Introduction

Traditional machine learning models have been used in many medical applications, offering valuable tools for risk assessment and clinical decision support [37], [44], [58], [59]. These methods, which include algorithms such as logistic regression, support vector machines, decision trees, and neural network (NN) architectures, have played a role in translating complex clinical data into meaningful predictions.

The feed-forward neural network (FFNN) is a fundamental structure within the broader family of neural network models [55]. It is composed of layers of interconnected nodes, or “neurons,” through which information moves from the input layer, through one or more hidden layers, to the output layer (Figure 3.1) [55]. Each neuron applies a weighted transformation to the

incoming data and passes the result through an activation function, allowing the network to capture patterns and relationships that may not be apparent through simpler analytical methods [55].

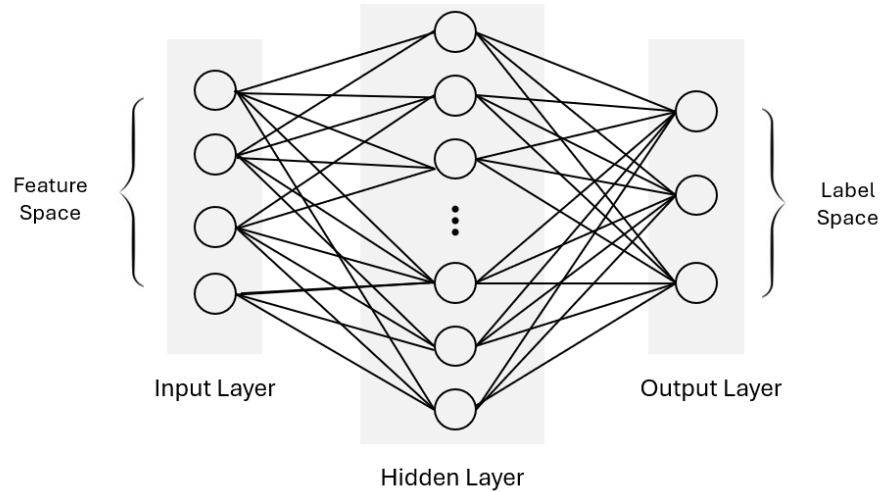


Figure 3.1: Basic Architecture of a Simple Neural Network

Diagram of a simple neural network containing an input layer with four nodes, one hidden layer, and one output layer with three nodes.

Principal component analysis (PCA) is an unsupervised machine learning (ML) approach aimed at reducing a model's feature space [55]. By projecting the original variables onto a new set of principal components, PCA can capture the dominant patterns of variance in a dataset. In this study, PCA was applied to the dual-energy x-ray absorptiometry (DXA) images to reduce their dimensionality and allow the model to better understand the underlying relationships within the dataset. This unsupervised approach is also useful when limited data are available. When image data are used to train ML models, each individual pixel will represent one input to the feature space. This means that a single 224×224 pixel image would generate 50,176 input values, requiring an equal number of input neurons. When building a predictive model, the input space (*i.e.*, the number of features or input dimensions) should be smaller than the cohort size because a model needs substantially more observations than features to learn meaningful, generalizable patterns [55]. If the number of inputs approaches or exceeds the number of samples, the model can

fit the data too closely, capturing noise over signal. This phenomenon, known as overfitting, leads to poor performance on new, unseen data [55].

Machine learning has been applied to fracture risk prediction using clinical and administrative health data [36], [37], [44], [45], [60], [61]; however, its direct application to raw DXA images has not been widely studied [39], [40]. Although many studies have developed models that outperform the Fracture Risk Assessment Tool (FRAX) and bone mineral density (BMD) T-score predictions, their ability to discriminate between individuals at high and low risk of fragility fracture remains only moderate [36], [44], [45], [60]. This study aimed to assess whether DXA images can provide predictive value for fracture risk prediction. The objective was to develop a machine learning model and evaluate its performance when trained on PCA-reduced DXA images. The results of this study will be used to examine how the level of PCA reduction affects model performance, assess the ability of DXA images to predict hip fracture risk, and inform the development of future predictive models.

3.2 Methodology

3.2.1 Train/Validate/Test Split

A 70/20/10 train/validate/test split was used in this study. This allocated 564 participants to the training group, 169 participants to the validation group, and 73 participants to the testing group. With limited datasets, dedicating a larger portion of samples to the training set is essential to ensure the model has sufficient data to learn meaningful patterns [55]. The validation set provides a separate group for tuning hyperparameters and decision thresholds, helping to optimize performance while preventing information leakage into the held-out test set used for the final, unbiased evaluation.

When the data were split, they were stratified based on incidence of major osteoporotic fracture (MOF) (Table 3.1). Major osteoporotic fracture was used to stratify the data instead of hip fracture because it provided more event cases while still capturing clinically relevant events. The standard approach in stratification is that the fracture classes are balanced across all splits, so that no subset contains disproportionately difficult cases [56]. This ensures a fair evaluation of the model's performance and prevents any single split from artificially inflating or deflating the results. The test set had the highest fracture prevalence (32.9%), compared with 27.8% in the validation set and 26.4% in the training set.

Table 3.1: Fracture Distribution in Train/Validate/Test Splits

Proportion of each fracture outcome in the train/validate/test splits. Efforts were made to balance the minority class in each split to ensure that model training and evaluation was fair.

Split	Hip Fracture (%)	New Vertebral Fracture (%)	MOF (%)	Total Unique Fracture Count (%)
Train	4.1	21.1	14.7	26.4
Validate	3.6	20.1	14.8	27.8
Test	5.5	27.4	15.1	32.9

3.2.2 Principal Component Analysis

Rather than retaining the full image resolution, principal component analysis was applied to reduce the dimensionality of the input data. This approach was motivated by the limited dataset size, as it reduced the number of input features and corresponding model parameters, helping to mitigate overfitting. To perform PCA, each two-dimensional (2D) DXA image was flattened into a one-dimensional (1D) vector of pixel intensity values, transforming images of size 224×224 into vectors of length 50,176. These vectors were stacked to form a 2D array of shape (n_samples, n_pixels) for the training, validation, and testing sets. The PCA object from the scikit-learn library was used to perform dimensionality reduction on the training dataset. The PCA procedure involved

computing the mean vector across all samples in the training matrix (shaped $564 \times 50,176$) and subtracting this mean from each element to produce a mean-centred data matrix. Using this matrix, singular value decomposition (SVD) was performed with the full SVD solver to extract the principal components.

When performing PCA with scikit-learn, the number of components to retain must be specified. Choosing this value is a critical step, since retaining too few components risks losing important information, while retaining too many may offer little reduction in complexity and provide limited benefit for downstream model training [48]. To determine how many principal components should be retained, a plot of cumulative explained variance was calculated for the training set (Figure 3.2). This plot illustrates the number of components required to capture a specified proportion of the total variance, with thresholds at 90%, 95%, and 99% indicated. To visualize the principal components, images of the first eight principal components were reshaped into the original hip DXA image dimensions (Figure 3.3).

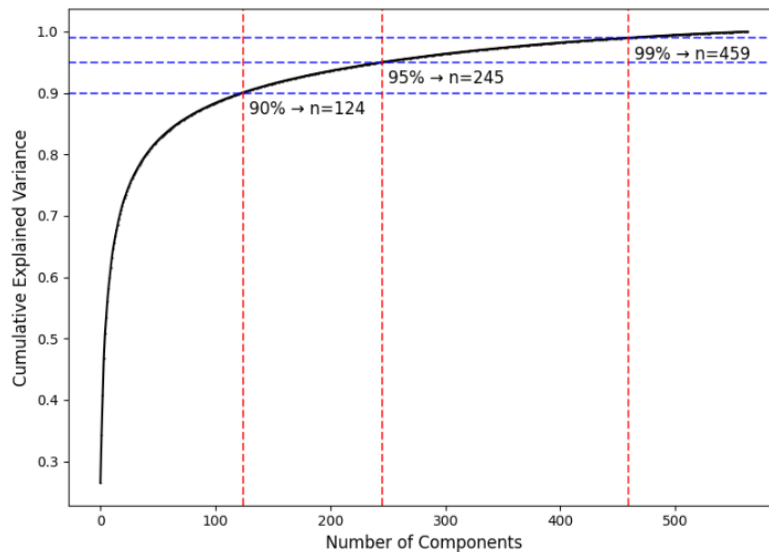


Figure 3.2: Cumulative Explained Variance Plot

Plot of cumulative explained variance vs. number of principal components, showing that 90, 95, and 99% of the variance is explained by 124, 245, and 459 principal components, respectively.

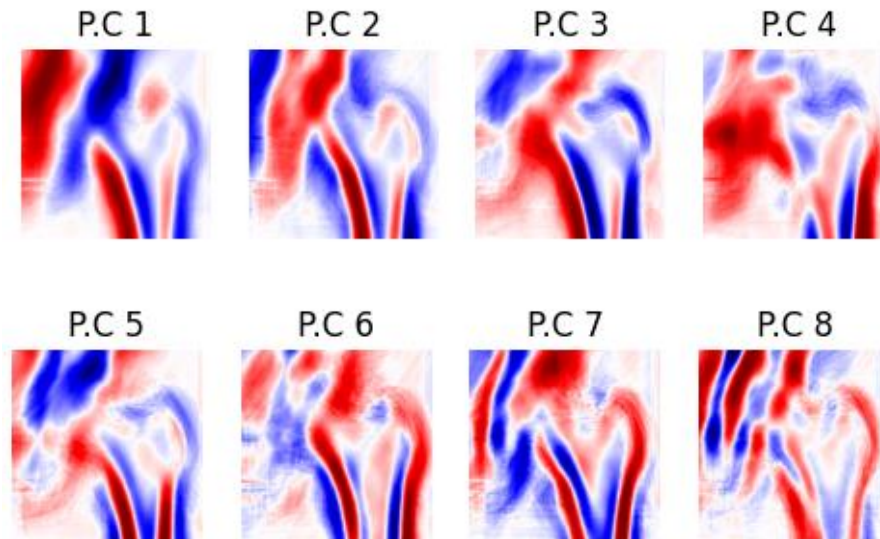


Figure 3.3: Principal Component Images

Principal component images derived from hip DXA scans, in which red and blue indicate groups of pixels that vary together in opposite directions across individuals, highlighting the hip regions that contribute most to differences between images. Earlier components capture the largest and most prominent patterns of variation, whereas later components represent finer or less common differences.

To examine the effect of the variance threshold on model performance, seven PCA cutoffs were evaluated, corresponding to retained variance levels ranging from 80% to 99%, as well as a 100% PCA baseline, using a logistic regression model (Table 3.2).

Table 3.2: Principal Component Analysis Groups to be Tested in Logistic Regression

Principal component analysis groups were defined using cumulative explained variance thresholds. Predefined groups retained 99%, 95%, and 90% of the variance, and additional groups at 88%, 85%, and 80% variance retention were included to evaluate performance across a range of dimensionalities. A no-PCA feature set was included as a baseline comparator.

Group Name	Percent Variance Retained (%)	Number of Components
PCA_100	100	564
PCA_99	99	459
PCA_95	95	245
PCA_90	90	124
PCA_88	88	100
PCA_85	85	68
PCA_80	80	41

One DXA image corresponding to each PCA variance threshold was reconstructed to qualitatively visualize the degree of anatomical detail preserved at different levels of dimensionality reduction. Reconstructions were generated to illustrate the visual effect of retaining fewer principal components, rather than to assess subject-specific differences (Figure 3.4). These reconstructions were intended solely as a visual aid to support interpretation of the PCA thresholds and were not used for quantitative analysis or model training.

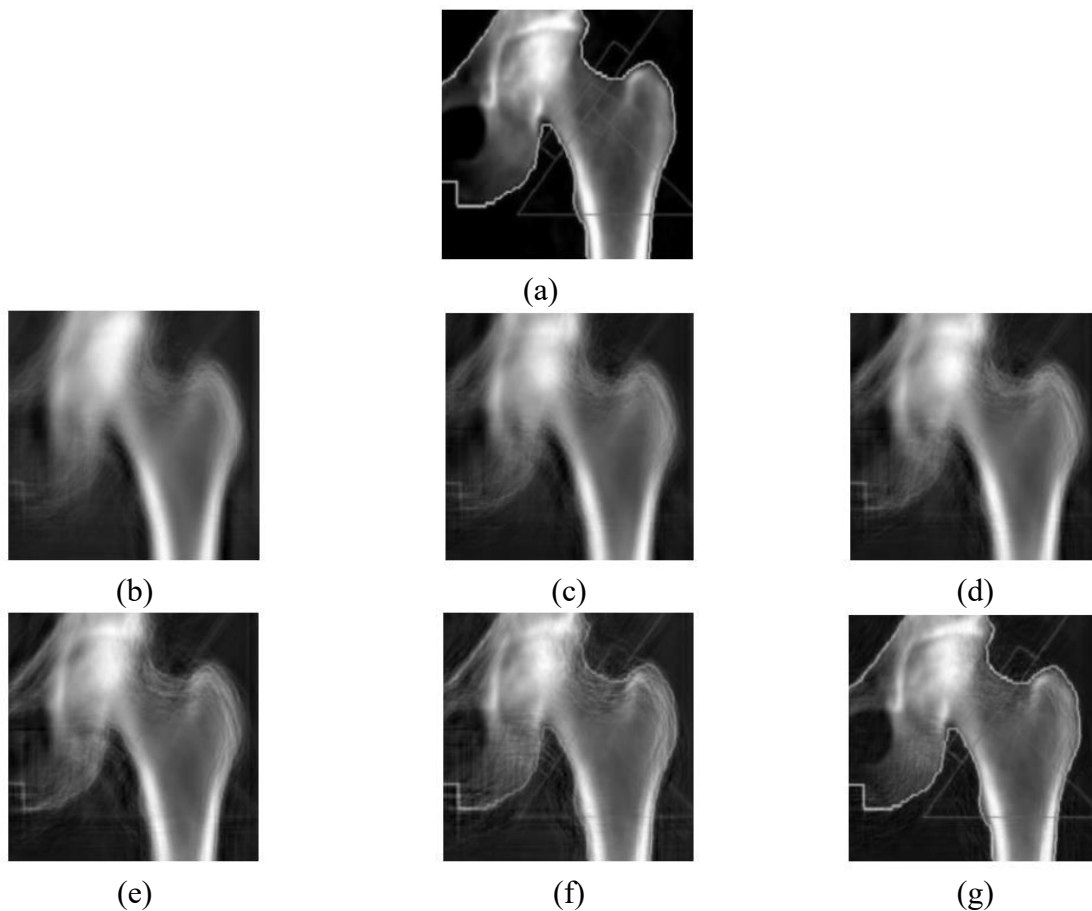


Figure 3.4: Anatomical Detail at Various PCA Reductions

Principal component analysis reconstruction showcasing anatomical detail at (a) Original, (b) 80%, (c) 85%, (d) 88%, (e) 90%, (f) 95%, and (g) 99% variance retention.

Dimensionality reduction was performed by applying the PCA methods to the training set to learn the principal components that captured the most variance for each specified group. The

learned components were then applied to the validation and test datasets using the scikit-learn `.transform()` method, without refitting. Logistic regression was used as a simple, interpretable baseline to quickly assess the effect of PCA-based dimensionality reduction. Its fast training allowed multiple PCA classes to be evaluated efficiently, helping to identify which reduced feature spaces captured meaningful patterns. Performance in logistic regression was entirely determined by area under the curve (AUC) (Table 3.3).

Table 3.3: Area Under the Curve Performance in Logistic Regression

Area under the curve for each PCA group, with PCA_88 achieving the highest AUC and PCA_100 achieving the lowest AUC.

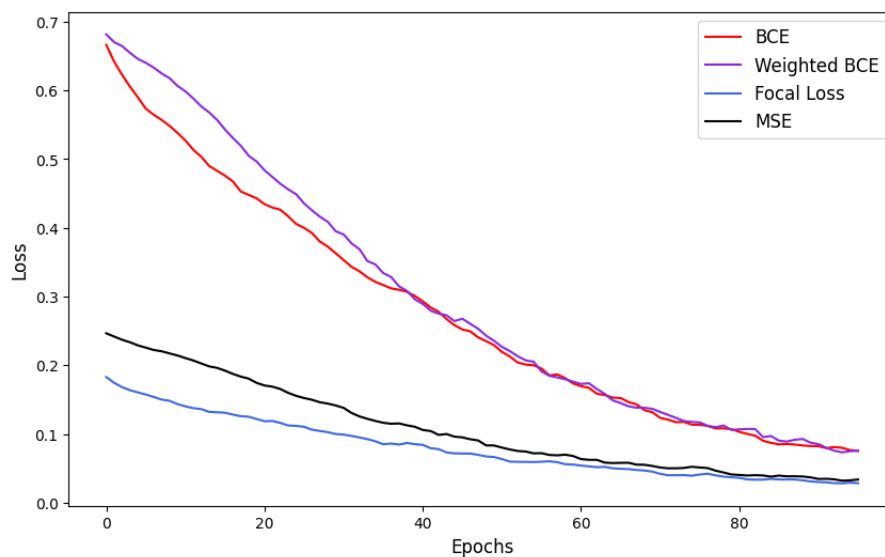
Group Name	AUC
PCA_100	0.54
PCA_99	0.56
PCA_95	0.56
PCA_90	0.60
PCA_88	0.66
PCA_85	0.64
PCA_80	0.63

The results from logistic regression confirmed that PCA_88 had the highest AUC; as such, it was used for model development in Chapters 3 and 4.

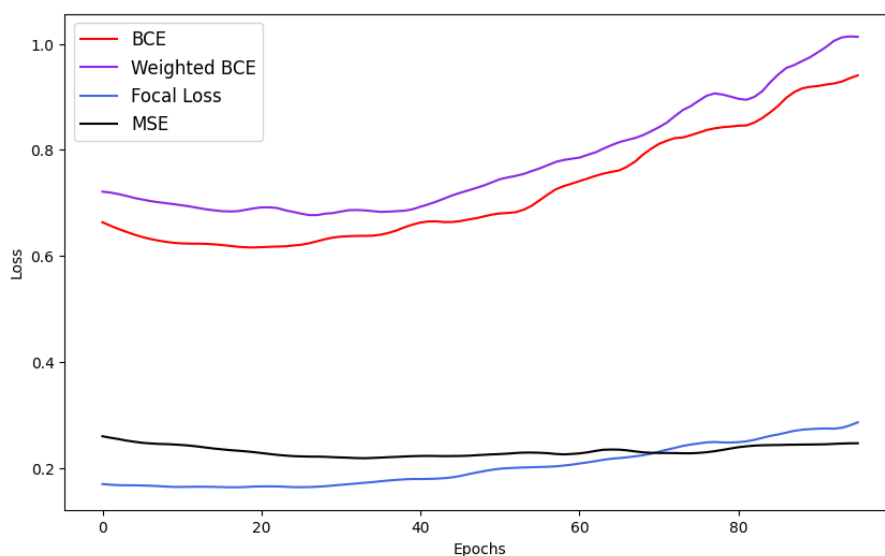
3.2.3 Selection of the Loss Function

Certain loss functions are better suited to specific tasks or data characteristics, such as classification versus regression or balanced versus imbalanced datasets [49], [55]. In this work, four different loss functions were tested using a simple fully connected FFNN trained on the PCA_88 reduced image features. The purpose of this test was not to define a learner with high classification accuracy, but rather to evaluate which loss function provided the most stable behavior during training on a simple model so that that loss function could be used for full model development.

The sample model consisted of two hidden layers with batch normalization, rectified linear unit (ReLU) activation, and a dropout layer after the first hidden layer. The final linear layer produced a single output value, as transformed by the sigmoid function, which was interpreted as a fracture probability between 0 (no risk of fracture) and 1 (certain risk of fracture). The evaluated loss functions were binary cross-entropy (BCE), weighted binary cross-entropy (wBCE), mean-squared error (MSE), and focal loss (Figure 3.5). Binary cross-entropy loss was included because it is the standard loss function for binary classification, penalizing incorrect predictions through the negative log-likelihood of the true label. Weighted BCE extends BCE by multiplying each sample's loss by a predefined weight, allowing the model to account for class imbalance. Mean-squared error, while traditionally used for regression, can also be applied to binary classification problems by comparing predicted probabilities to binary targets. Finally, focal loss was included because it further addresses class imbalance by down-weighting easy examples and placing greater emphasis on hard-to-classify samples. Training and validation loss plots were examined to assess stability, defined by smooth convergence to a plateau with minimal oscillation. It is important to note that hyperparameters were not tuned, as the goal of this assessment was not to build a high-performing classifier, but instead to gain an initial understanding of which loss function would be most suitable for further model development.



(a)



(b)

Figure 3.5: Training and Validation Loss Comparison

Plots highlighting (a) training and (b) validation loss calculated for BCE, wBCE, MSE, and focal loss. Training loss was similarly stable across all loss functions, whereas validation loss showed pronounced overfitting for BCE and weighted BCE. MSE and focal loss exhibited improved behavior; however, focal loss still showed a tendency toward overfitting. All validation loss plots indicated limited learned generalizability from training.

The focal loss function achieved the lowest training loss (0.025) and was used for the remainder of this thesis work. Focal loss modifies standard binary cross-entropy loss by adding two factors: 1) a weighting term α that balances positive and negative samples, and 2) a modulating factor $(1 - p_t)^\gamma$ that down-weights well-classified examples. The term p_t refers to the probability that the model assigns an example to the correct class. When an example is easy to classify, meaning the input features align well with the learned decision boundary, p_t will be close to 1, making $(1 - p_t)$ very small. In contrast, hard examples occur when the input is ambiguous or falls near or across the decision boundary, leading the model to assign a lower probability to the correct class; in these cases, p_t is small and $(1 - p_t)$ becomes large. The γ parameter controls how aggressively the model down-weights easy samples. When $\gamma = 0$, there is no modulation, so with $\alpha = 0$ the loss formula reduces to the standard BCE loss. With a larger γ , there is a much larger penalty for hard examples and almost no penalty for easy examples, forcing the model to focus its learning capacity on the most challenging cases. In the original formulation of focal loss, γ was defined for values greater than or equal to zero, and values between zero and five were empirically evaluated [62]. This evaluation was recreated in this work (Figure 3.6).

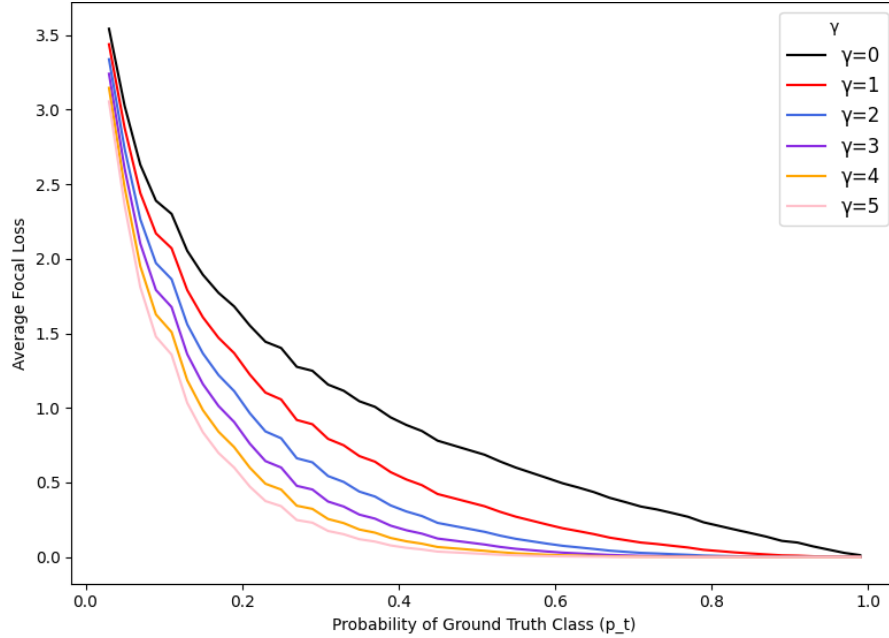


Figure 3.6: Focal Loss vs. Ground-Truth Probability

Visualization of how focal loss influences the outcome for $\alpha = 0$, $\gamma \in \{0, 1, 2, 3, 4, 5\}$. For hard examples (low p_t), the focal loss remains relatively large, so these samples continue to contribute strongly to the gradient updates. For easy examples (high p_t), the loss is heavily down-weighted, so their contribution to the gradients becomes very small and they have little influence on further parameter updates.

3.2.4 Image-Only Algorithm

3.2.4.1 Model Architecture

The FFNN (Figure 3.7) consisted of three fully connected layers. It had an input layer of 100 neurons, corresponding to the dimension of the PCA-reduced features. The input layer was mapped to the first hidden layer, which had a dimension of 32 units. After the first hidden layer, batch normalization was performed to stabilize the activation distributions during training. The ReLU activation function was used, allowing the model to capture non-linear relationships within the image data. A dropout rate of 0.5 was then applied, meaning that half of the neurons in this layer were randomly deactivated during training to reduce overfitting [49]. A second fully connected layer reduced the hidden representation from 32 to 16 units, followed again by batch normalization and ReLU activation. The network concluded with a single-output linear layer,

which produced a raw logit that could later be transformed into a prediction score using the sigmoid function. An overview of the model design used in this work is shown below.

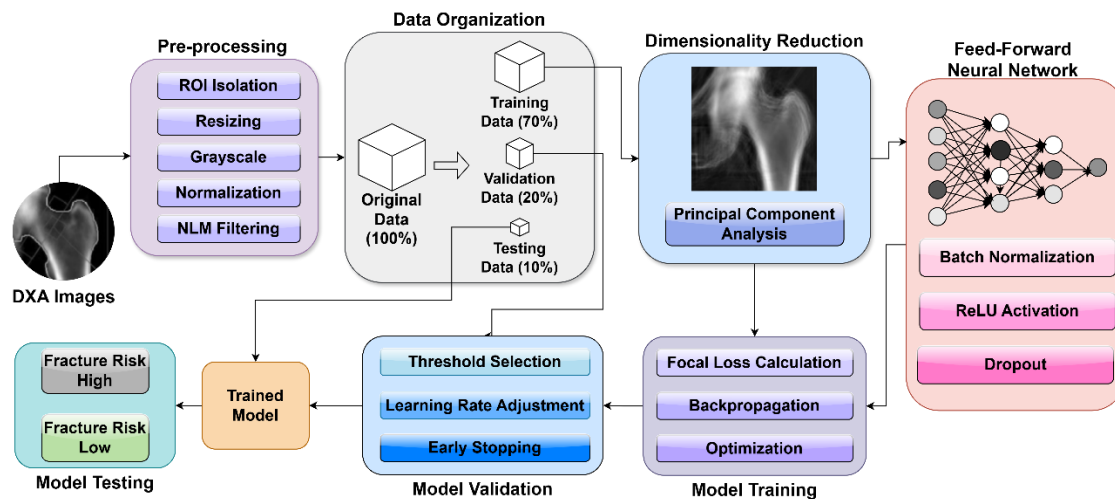


Figure 3.7: Architecture Design

Flowchart highlighting key steps in pre-processing, data organization, dimensionality reduction, network architecture, model training, model validation, and model testing.

3.2.4.2 Model Training

A neural network trains by making gradual adjustments to its learnable parameters (weights and biases) to make better predictions. Weights control how strongly one neuron influences the next, while biases allow neurons to shift when they activate. Together, these parameters determine the output of each neuron. During training, the network makes a prediction and compares it to the correct answer (the label), producing a value called the loss that measures how incorrect the prediction was. After computing the loss, the network propagates error signals backward through the layers to determine how each weight and bias influenced the final error. These error signals are then used to update the parameters in a way that reduces the loss, a process known as gradient descent. The learning rate controls how large each update is. If the learning rate is too high, the model may overshoot optimal values and fail to converge. If it is too low, parameter updates

become very small, slowing training and increasing the risk of getting stuck in a local minimum. By repeating this forward and backward process across many training examples, the network gradually improves its ability to capture patterns in the data.

In this study, the PCA-reduced images were fed directly into the neural network for training. The model's tunable parameters, referred to as hyperparameters (*e.g.*, learning rate, batch size, dropout rate, number of epochs, number of hidden layers), were first selected using scikit-learn's Grid Search. This tool evaluates different parameter combinations to identify those that best fit the model. After this initial search the parameters were further tuned manually. Manual tuning was performed because while Grid Search is effective at identifying strong hyperparameter combinations, it is restricted to the values provided and may miss more optimal settings that lie between or outside the provided range. It also does not account for the interrelationships that exist between different hyperparameters.

The Adaptive Moment Estimation with Weight Decay (AdamW) optimizer was used with an initial learning rate of 0.003 [63]. This value was selected based on standard values reported for Adam-based optimizers [45]. The AdamW optimizer was implemented over standard the Adam optimizer because decoupled weight decay has been proven to lead to more stable and well-regularized parameter updates [63]. The focal loss function ($\alpha = 0.25$, $\gamma = 4$) was used to address class imbalance by down-weighting highly confident, easy-to-classify samples. A batch size of 64 was used, allowing the model to process the training data in smaller subsets, meaning one complete pass through the dataset (an epoch) consisted of 9 batches [49]. Training was performed by minimizing the batch-averaged focal loss using gradient-based optimization, with each sample weighted to account for class imbalance.

3.2.4.3 Model Validation

At the end of each training epoch, the model was evaluated on the validation set. The entire validation set was passed through the model in evaluation mode, producing probability estimates for all 169 validation samples. These predictions were then used to compute several metrics, including mean validation loss and AUC.

To determine the most effective decision threshold, a sweep across 100 candidate thresholds between 0.01 and 0.99 was performed at each epoch. For every threshold, binary predictions were derived from the validation probabilities, and the corresponding F2 score was computed. The threshold that yielded the highest F2 score was recorded at each epoch as the optimal threshold. The optimal threshold used for testing was 0.45. This does not constitute data leakage, as the model weights remained fixed during validation and the test set remained completely unseen. Using the validation set to tune the threshold provided a justified threshold choice, as there was no predefined “correct” threshold (such as 0.5) for this application. Early stopping was implemented to prevent the model from overtraining [49], [64]. To ensure sufficient learning, a minimum of 90 epochs was required before early stopping could be implemented [43], [45]. After this point, training was stopped if the validation F2 failed to improve by at least 0.001 for five consecutive epochs. After early stopping, the model that corresponded to the highest validation F2 was retained for final evaluation. A learning rate scheduler was used to monitor the training progress. When the network stopped improving for a patience of five epochs, the learning rate was halved. The final parameters that were used in this model are summarized in Table 3.4 below:

Table 3.4: Tuned Model Parameters

Final parameters used in model development.

Parameter	Value
Learning Rate	1e-6
Batch Size	64
Weight Decay	1e-6
Dropout Rate	0.5
Alpha (Focal Loss)	0.25
Gamma (Focal Loss)	4
Minimum Epochs	90
Early Stopping Patience	5
Minimum Improvement (Delta) Required	0.001
Learning Rate Scheduling Factor	0.5
Learning Rate Scheduler Patience	5

To evaluate the model, training and validation loss as well as validation AUC were plotted. The data were smoothed using a moving average function with a window size of five to reduce noise and better highlight the underlying trends. A confusion matrix was generated to determine the classification accuracy of the model, and additional quantitative performance metrics relevant to binary classification problems were calculated. To assess the sensitivity of model performance to the chosen decision threshold, several alternative thresholds beyond the selected optimal value were also evaluated. This was done because binary classifiers output continuous probabilities rather than discrete labels, making decision threshold selection critical to performance. Given that the decision threshold was optimized on the validation set, it was important to assess how well this selected value generalized to the test set. Finally, a receiver operating characteristic (ROC) curve was used to evaluate the model's discriminative performance.

3.3 Results

The model's training loss (Figure 3.8) experienced an overall downward trend, and the validation loss (Figure 3.8) experienced a very subtle increase, mostly remaining stagnant. The validation AUC (Figure 3.9) experienced a steep incline to a plateau after about 40 epochs.

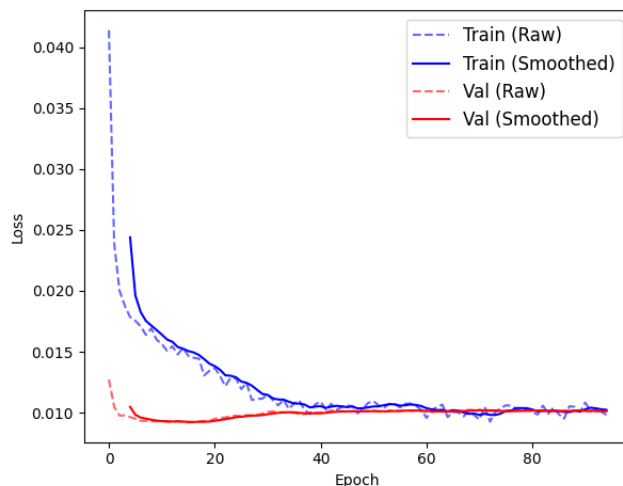


Figure 3.8: Training and Validation Loss

Training and validation loss plotted across 95 training epochs, with a moving average applied to highlight overall trends and reduce short-term variability.

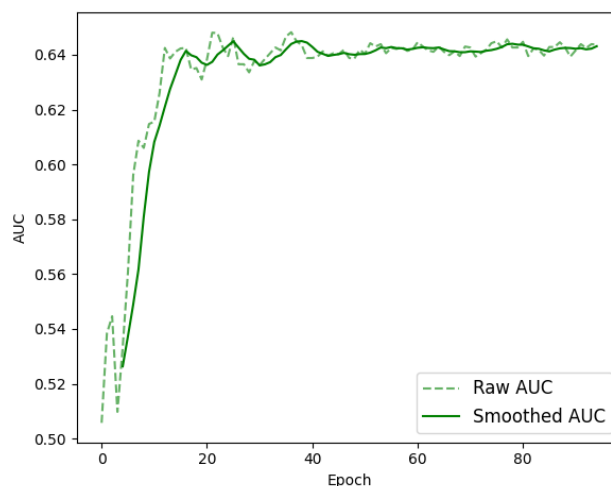


Figure 3.9: Validation AUC

Validation AUC plotted against 95 epochs of training, with a moving average applied to highlight overall trends and reduce short-term variability.

The confusion matrix (Figure 3.10) highlighted that the model achieved a classification accuracy of 60%, as it misclassified eight true positive cases and 21 true negative cases. Multiple thresholds were tested to ascertain how sensitive the classification performance was to threshold selection (Figure 3.11). Additional quantitative metrics indicated moderate classification performance (Table 3.5), and the ROC curve (Figure 3.12) achieved a test AUC of 0.66.

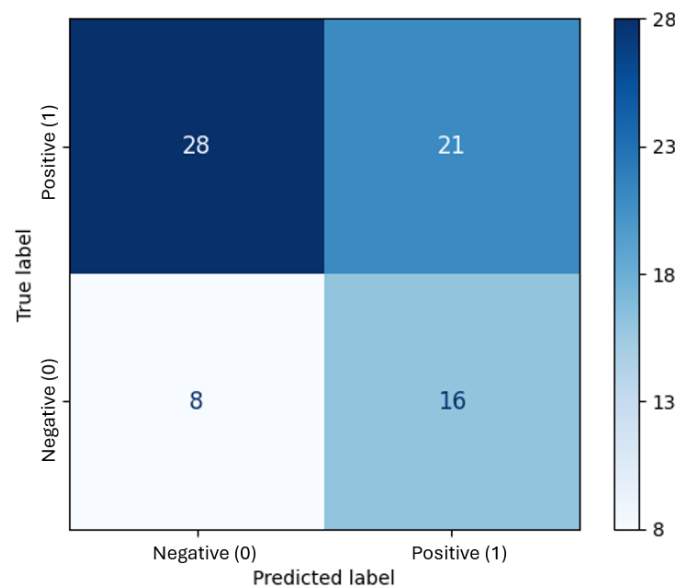


Figure 3.10: Confusion Matrix

Confusion matrix (classification threshold = 0.45) showcasing the model's ability to decipher true positive and true negative cases. The model was able to capture 16 true positive cases and 28 true negative cases, misclassifying 29 of 73 participant examples (40%).

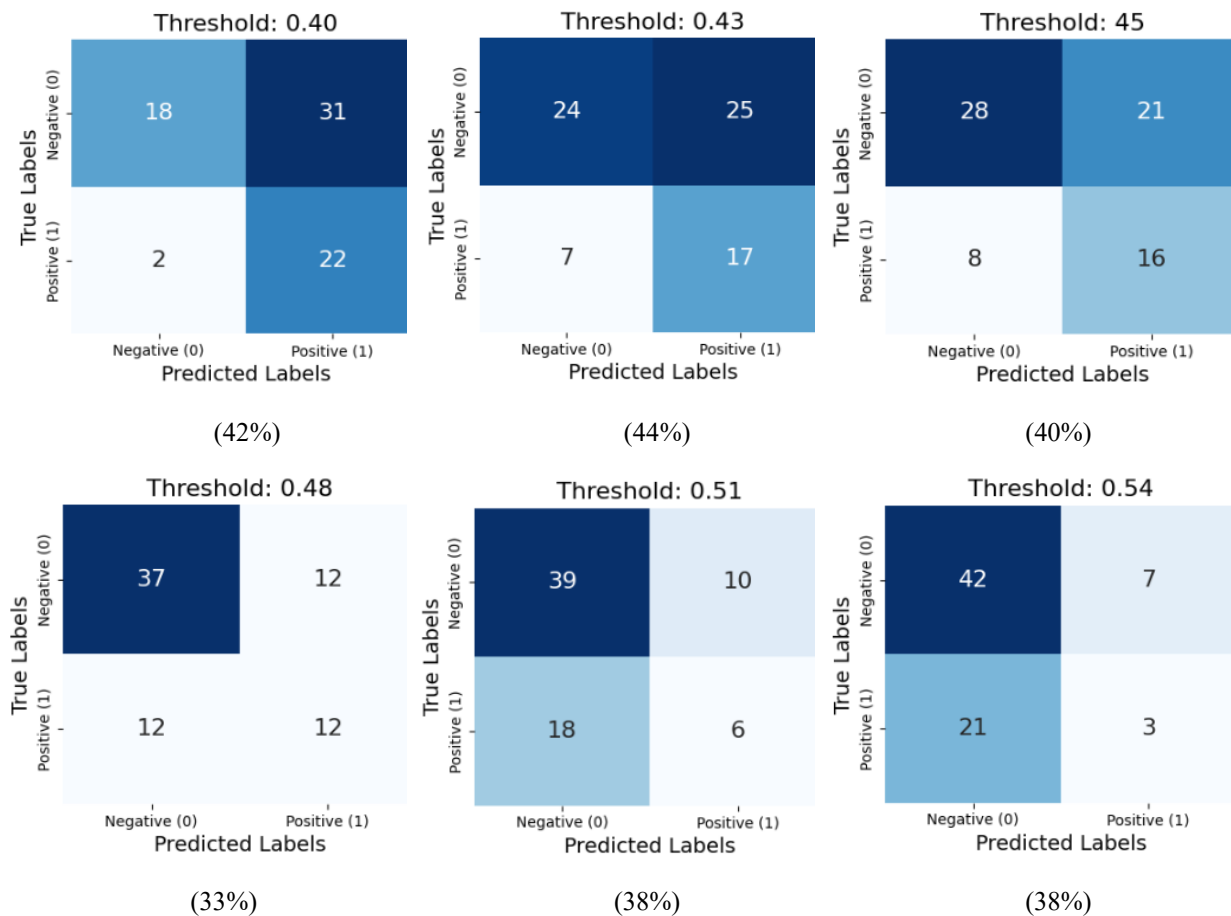


Figure 3.11: Confusion Matrices for Thresholds in Range 0.40-0.54 (Misclassification Percentage in Brackets)

Confusion matrices highlighting how classification outcomes changed when the selected threshold was changed.

Table 3.5: Model Performance Metrics

Performance metrics obtained from the final FFNN.

Performance Metric	Value
Sensitivity	0.67
Specificity	0.57
Precision	0.43
Accuracy	0.60
Training Loss	0.0098
Validation Loss	0.010
Validation AUC	0.65
Validation F2 Score	0.67
Test AUC	0.66
Test F2 Score	0.60

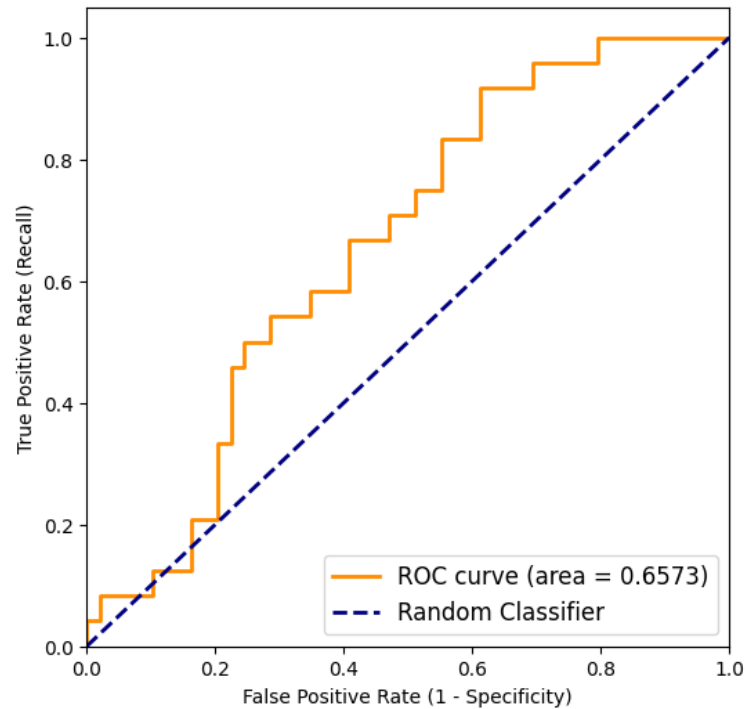


Figure 3.12: ROC Curve

ROC curve highlighting the trade-off between true positive rate (recall) and false positive rate ($1 - \text{specificity}$) as the classification threshold is swept from strict to permissive; the area under this curve summarizes how well the model separates positive from negative cases across all thresholds, with curves closer to the upper left indicating better discrimination than the diagonal random classifier line.

3.4 Discussion

This work presented a novel approach to fracture risk classification by leveraging neural network-based feature extraction combined with principal component analysis directly from hip DXA images, rather than relying solely on bone mineral density metrics. The findings of this study support the development of low-cost, scalable, image-based fracture screening tools for postmenopausal women, enabling earlier identification of individuals at high risk. The proposed model demonstrated moderate predictive performance prior to the inclusion of clinical risk factors, indicating that DXA images alone contain meaningful predictive information even in relatively

small datasets. This level of performance is comparable to that reported for BMD-based assessment (often represented as a T-score) alone, which has been shown to exhibit only modest discrimination for fracture risk (e.g., $AUC \approx 0.66$) [65]. Reported discrimination for FRAX is similarly moderate, despite its inclusion of clinical risk factors [36], [66].

The PCA group with 88% variance retention showed the best performance when evaluated using simple logistic regression. This may be due to the fact that this level of dimensionality reduction retained a large proportion of the underlying signal while still reducing the original feature space, which could help limit noise and overfitting. The focal loss function was selected for training because it showed a lower tendency toward overfitting compared with the BCE and weighted BCE loss functions. This behavior was shown in the validation loss curves, where both BCE-based losses exhibited increasing loss over successive epochs. While the MSE loss function showed indications of reduced overfitting, focal loss was selected for implementation because it provides class-balancing and a mechanism for emphasizing more challenging samples during training, which was particularly relevant given the imbalanced nature of the dataset.

The training loss curve had strong initial convergence, with loss dropping quickly within the first 15-20 epochs, demonstrating the learner's ability to quickly capture meaningful signals in the feature space. The validation loss plot remained fairly stagnant, indicating that the model showed limited improvement in generalization beyond the early stages of training and may have begun to overfit as training progressed. To prevent overfitting, batch regularization, early stopping, feature space reduction, and 50% dropout were implemented [49], [61]. Other work evaluating the predictive ability of DXA images in fracture risk prediction has reported training curves with more pronounced overfitting and a sharper divergence between training and validation loss plots [39].

The comparatively stable behavior observed in this work suggests improved control of overfitting despite similar sample size constraints.

The validation AUC reflected the model's ability to distinguish between positive and negative cases, rising sharply early in training before exhibiting some oscillations. This pattern indicated that the model may have learned to rank most samples correctly early on, but the feature space did not provide enough discriminative signal to raise the AUC much higher. Once the main signal was captured, further training only made small, noisy adjustments to predictions.

Despite using the validation-tuned threshold, the model struggled to achieve good recall and specificity. This indicated that, although the network detected a better proportion of true fractures, it did so at the cost of a relatively high false-positive rate, which has been seen in the literature [39]. The low accuracy of the model reflected this imbalance, and the corresponding F2 score on the test set confirmed that the recall-oriented training objective favoured sensitivity over precision. While prioritizing recall was appropriate given the clinical importance of detecting fractures, the model still failed to identify 33% of the fracture cases in the test set.

When the decision threshold was changed from 0.45 to 0.48, ten positive cases of the 24 total true positives were affected. This demonstrated that the model performance was extremely sensitive to the selection of the decision threshold, indicating challenges for reliable clinical deployment. Overall, the threshold optimized on the validation set translated reasonably well to the test set and offered a good balance in prediction behavior. Although other thresholds achieved lower overall misclassification rates, these cases struggled more with identifying true positives, which was important for the recall-focused task.

While selecting the threshold from the validation data does not introduce data leakage, it is important to note that tuning the threshold this way means that the final test results are no longer

a perfectly pure measure of the model's ability to generalize to brand new data. The test set still represents the same population and data-collection process, and with a relatively small sample size, the threshold that appeared optimal for this cohort may not transfer reliably to a truly external dataset. While it is important to acknowledge that tuning decision thresholds on validation data may influence reported performance, comparable approaches have been reported in prior studies that optimize classification cutoffs using validation metrics [67], [68].

The ROC curve demonstrated that the model performed better than random chance, but the confusion matrix challenged the clinical utility of this model in its current form for performing binary classification. Despite this result, the reported test AUC was within the range commonly reported for the FRAX tool (approximately 0.60–0.75) [36], [66]. Notably, studies incorporating more extensive predictor sets have achieved AUC values between 0.70-0.80, suggesting that predictive performance may improve with the addition of CRFs [36], [40], [44].

While the proposed model struggled to correctly classify non-fracture cases, in the context of fracture risk screening, identifying individuals at high risk of fracture is often more important than correctly excluding low risk cases. Although the model's current performance limits its standalone clinical utility, the results indicate that DXA images contain fracture-relevant information. Further improvements may be achievable through the inclusion of clinical risk factors.

3.4.1 Relevance, Strengths, and Limitations

This study served as an exploratory model to evaluate whether machine learning models could support early identification of 3-year fracture risk using routinely collected DXA imaging data. Some strengths of this study were that it was developed using a robust architecture designed to limit overfitting, and despite its limited predictive accuracy, it did achieve similar performance metrics to those reported for the widely used FRAX tool [36], [66]. Multiple evaluation metrics

were calculated to prevent overly optimistic results and to ensure balanced assessment across discrimination, sensitivity, and specificity. Additionally, the application of principal component analysis to the images improved AUC, as demonstrated by preliminary analysis using logistic regression. To our knowledge, applying principal component analysis to DXA images within an image-only feed-forward neural network has not previously been reported, highlighting the novelty of this approach.

Although the study demonstrated that ML-based prediction is feasible, there were some limitations to this work. The dataset was relatively small for image-based learning, with only 806 participants in total, which restricts the model's ability to learn subtle or complex patterns. Although similar limitations have been reported in previous work, the developed pipeline remains a strength of this study, as it can be readily adapted and tuned when more data become available [39], [45], [61]. Additionally, by using PCA to stabilize training, all spatial structure was removed from the images. This means the model could only use global variance patterns rather than local morphological patterns. In the context of machine learning for medical imaging, spatial relationships often carry meaningful value. As a result, the predictive ability of the model was limited, and more spatially aware architectures (*e.g.*, CNNs) may be better suited to this task and should be investigated in the future.

Finally, despite implementing stratified sampling, the test set contained a slightly higher proportion of fracture cases (7% higher than the training split, 5% higher than the validation split), which may have influenced the reported performance metrics. Although this may have led to worse performance in the model, it is a more realistic test of how well the model handles clinically meaningful cases. Overall, these limitations indicate that while the present model represents a useful early exploration, larger datasets, richer feature representations, and methods that retain

spatial information will be important for developing a tool that can perform reliably in a clinical setting.

3.5 Conclusions

This chapter presented the development, training, and evaluation of a machine learning model designed to predict fracture status using PCA-reduced imaging data. The model was able to extract meaningful patterns and achieve moderate predictive performance from image data alone. This work provided a solid foundation for the next stage of the project, the addition of clinical CRFs to the model.

Chapter 4 – Design and Implementation of a Feed-Forward Neural Network for Binary Fracture Classification from DXA Images and Clinical Risk Factors

***Overview:** This chapter builds on the previously introduced machine learning (ML) approach to classifying fracture risk from dual energy x-ray absorptiometry (DXA) images by incorporating clinical risk factors (CRFs). The proposed algorithm is a multi-modal neural network designed to learn patterns from image-derived features and clinical variables. This model builds on the algorithm developed in Chapter 3, as it includes additional health variables known to influence bone strength. This chapter describes the methodology used in designing this model, as well as the processes used to train, tune, and test the model. It also discusses the techniques used to evaluate the model's performance, and how the inclusion of CRFs impacted the final results.*

4.1 Introduction

Traditional machine learning models have been demonstrated to be effective in medical classification tasks [39], [44], [69], [70]. Shallow neural networks require less data than deep neural networks and have been applied with success in medical research studies [71], [72]. Their lower model complexity reduces the risk of overfitting, which is an important consideration in medical applications where datasets are often limited and case events are rare. The previously proposed model was a multi layer perceptron (MLP) that consisted of three fully connected layers trained on dual energy x-ray absorptiometry (DXA) images reduced by principal component

analysis (PCA). While this model demonstrated moderate predictive capability, incorporating additional features known to influence bone health has the potential to substantially improve its performance [40]. Expanding the input space to include a broader set of clinically meaningful variables has been shown to enhance model accuracy and generalizability, and is commonly performed in research [7 - 10]. There are many factors that contribute to bone health outside of bone mineral density (BMD) data, such as age, body mass index (BMI), and previous fracture history [5], [66]. By including these additional predictors in the feature space, the model will have more relevant information from which to learn, and thus a better chance of achieving a higher predictive ability.

Multi-modal networks are neural network architectures that are designed to combine information from multiple data sources within a single predictive framework. In the context of fracture-risk assessment, imaging and clinical variables capture different patient information. Image features represent structural and material characteristics of the bone, while clinical variables reflect the demographic and physiological factors that influence bone health and fracture risk. Combining the two types of information offers an opportunity for improved learning and motivates the use of a multi-modal algorithmic structure. This learner style has been reported in medical imaging studies with good success [75], [76]. This work aimed to evaluate whether a multi-modal neural network could improve fracture-risk classification performance when trained on DXA images and CRFs.

4.2 Methodology

The PCA reduced image features, target labels, train/validate/test split, and loss function from Chapter 3 formed the basis of the data used in this work. The selected CRFs for this study

were age, BMI, number of previous non-vertebral fractures, number of previous vertebral fractures, number of prevalent fractures at baseline, treatment status, and total hip DXA BMD. These variables were selected based on their well-established associations with fracture risk and bone health, as well as their demonstrated value as predictive features in prior fracture-risk models [5], [8]. The MLP developed in this work extended the single-branch FFNN described in Chapter 3 by introducing two separate branches: one for image-derived features and one for CRFs.

4.2.1 Model Architecture

The image branch had an input layer of 100 neurons, corresponding to the dimension of the PCA-reduced features. The first layer in this branch was a fully connected linear layer that projected the input to a hidden layer with 128 units. After the first hidden layer, batch normalization was performed to stabilize the activation distributions during training. A rectified linear unit (ReLU) activation function was employed, allowing the model to capture non-linear relationships within the image data [39]. A dropout rate of 0.5 was applied to reduce overfitting [49]. A second fully connected layer reduced the hidden representation from 128 to 64 units, again followed by batch normalization, ReLU activation, and a second dropout layer with a rate of 0.4 [77]. Early layers used the full dropout rate (0.5) to strongly reduce overfitting when feature representations were high-dimensional, while the deeper layer used a reduced rate (0.4) to avoid discarding too much information. These values were selected based on empirical evaluation of training and validation performance during model development.

The clinical branch of the multi-modal MLP followed the same two-layer structure but was tailored to the smaller clinical feature space. The input dimension consisted of seven clinical risk factors (CRFs), which were linearly combined into 96 learned features forming the first hidden layer, allowing the network to capture a richer internal representation of their interactions [55].

Batch normalization, ReLU activation, and dropout (0.5) were implemented. A second linear layer reduced the hidden representation from 96 to 48 units, again followed by batch normalization, ReLU activation, and dropout (0.4) [77].

The image and clinical branches produced 64 and 48 learned features, respectively, which were concatenated into a 112-dimensional fused feature vector, with feature dimensionality chosen to support representational capacity in each pathway. This step allowed the model to combine information derived from bone morphology with demographic and physiological risk factors. The network consisted of a fully connected layer that projected the fused 112-dimensional vector into a 32-unit hidden representation [77]. Rectified linear unit activation introduced nonlinearity, and a dropout rate of 0.5 was applied to regularize the fused modality space. The final layer was a single-unit linear layer that outputted a raw logit corresponding to the predicted fracture risk. During testing, this raw logit was transformed to a prediction score (between zero and one) using the sigmoid function. The model output remained in logit form because supplying raw logits to the loss function ensures more stable training and avoids issues like vanishing gradients that can arise from directly applying the sigmoid activation within the network [78]. A depiction of the model design is shown below (Figure 4.1).

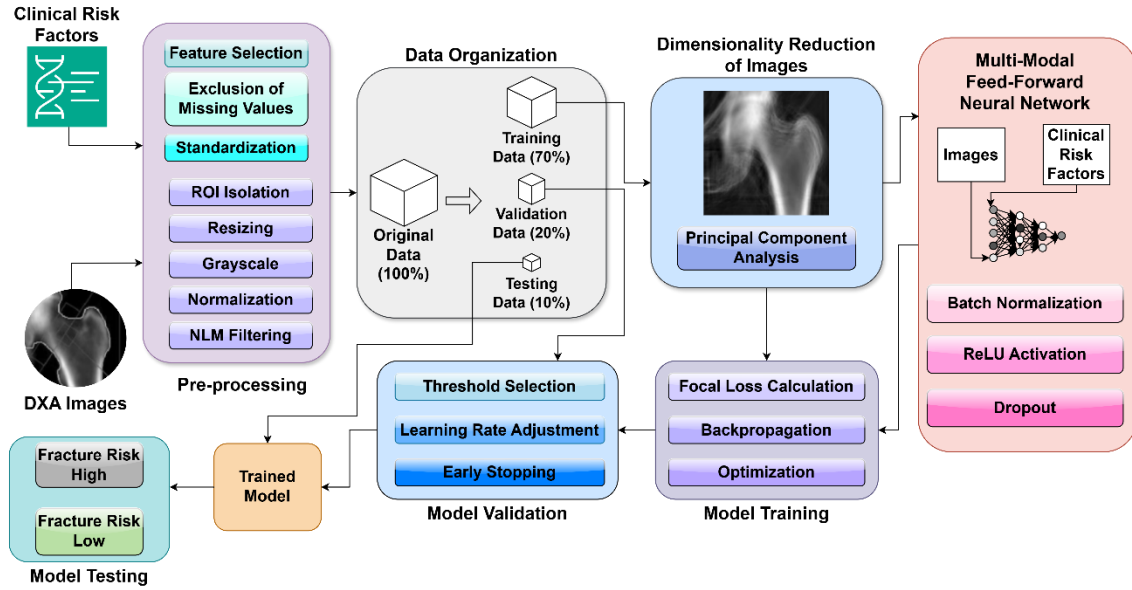


Figure 4.1: Architecture Design

Flowchart highlighting key steps in pre-processing, data organization, dimensionality reduction, network architecture, model training, model validation, and model testing.

4.2.2 Model Training

The Adaptive Momentum Estimation with Weight Decay (AdamW) optimizer was used with a weight decay of $1e-4$ and an initial learning rate of 0.001. The focal loss function was used ($\alpha = 0.25$, $\gamma = 4$) to emphasize the minority class during training. A batch size of 32 was used, and in each batch the image and CRF tensors were passed through their respective branches and concatenated into a single fused vector to produce one output logit per sample representing the estimated fracture risk prior to sigmoid transformation. Training was performed by minimizing the focal loss function averaged over each training batch using gradient-based optimization [49]. To prevent unstable weight updates, gradient norm clipping was applied with a maximum norm of 0.5, limiting the overall magnitude of the gradient vector during backpropagation. This ensured that gradients from both branches were scaled proportionally when large updates occurred, helping

to stabilize optimization in the multi-modal architecture [79]. An average training loss for the epoch was recorded to track optimization progress.

4.2.3 Model Validation

Model validation was conducted on a held-out validation set with the model in evaluation mode, disabling dropout, fixing batch normalization statistics, and preventing gradient updates to ensure true inference behavior. At the end of each training epoch, the model (with weights and biases fixed at their updated values from training) was evaluated on the full validation set to produce a vector of logits, which were used to compute the focal loss and were then transformed into probabilities using the sigmoid function. Validation performance was quantified using several metrics derived from the output probabilities. First, the mean focal loss for the epoch was computed to assess how well the model fit the validation data. Then, the area under the curve (AUC) was calculated to evaluate the model's discriminative ability.

To determine the most effective decision threshold, a sweep across 100 candidate thresholds between 0.01 and 0.99 was performed at each epoch. For each threshold, predicted probabilities were converted to binary class labels, and the F2 score was computed to emphasize recall. Early stopping was implemented after a minimum of 70 epochs to keep the model from overtraining. After 70 epochs, improvements were only considered to be meaningful if (i) validation loss decreased by at least $1e-4$, or (ii) validation AUC increased by at least $1e-4$, relative to the best observed results [80]. The same margin was applied to both metrics to suppress validation noise, in line with standard early stopping criteria [80]. If no improvement was observed for a patience of five consecutive epochs, training was stopped early to prevent the network from overfitting to the training data [80]. The best model was selected as the last model that achieved an improvement in validation performance before early stopping was triggered, and the threshold

that maximized the validation F2 score in this model was selected as the final threshold for testing. A learning rate scheduler was used to monitor the training progress. When the network stopped improving, the learning rate was halved. This was implemented with a patience of eight epochs to avoid unnecessarily frequent adjustments. The learning-rate patience was set higher than the early-stopping patience to avoid unnecessary learning rate reductions if training was likely to terminate shortly afterward. The final parameters that were used to develop this model are summarized in Table 4.1.

Table 4.1: Global Training Parameters
Final parameters used to train the model.

Parameter	Value
Learning Rate	1e-6
Batch Size	32
Weight Decay	1e-4
Global Dropout	0.5
Alpha (Focal Loss)	0.25
Gamma (Focal Loss)	4.0
Minimum Epochs	70
Early Stopping Patience	5
Minimum Improvement (Delta) Required	1e-4
Learning Rate Scheduling Factor	0.5
Learning Rate Scheduler Patience	8

To evaluate the model, training and validation loss curves along with validation AUC were plotted and smoothed using a moving average filter with a window size of five to reduce noise and better reveal underlying trends. Binary classification performance was then assessed using a confusion matrix derived from the test data. To assess how the chosen threshold influenced the confusion matrix, several alternative thresholds were applied to the test set after model evaluation. An ROC curve was plotted to evaluate the model's discriminative ability. To quantify the

contribution of individual clinical features and imaging inputs to the final prediction, SHAP analysis was performed.

4.3 Results

The training loss (Figure 4.2) demonstrated an overall declining trend with some oscillations. The validation loss (Figure 4.2) remained mostly stagnant but experienced a slow rise. The validation AUC (Figure 4.3) was oscillatory and hovered at around 0.68.

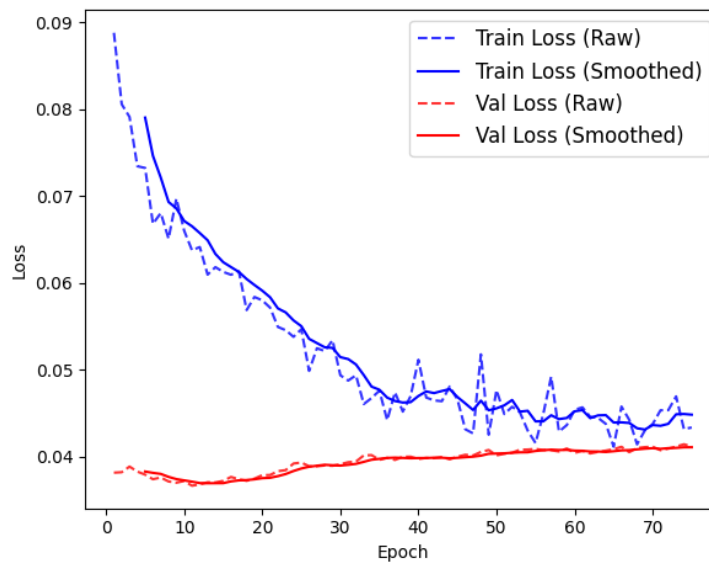


Figure 4.2: Training and Validation Loss

Training and validation loss plotted across 75 training epochs, with a moving average applied to highlight overall trends and reduce short-term variability.

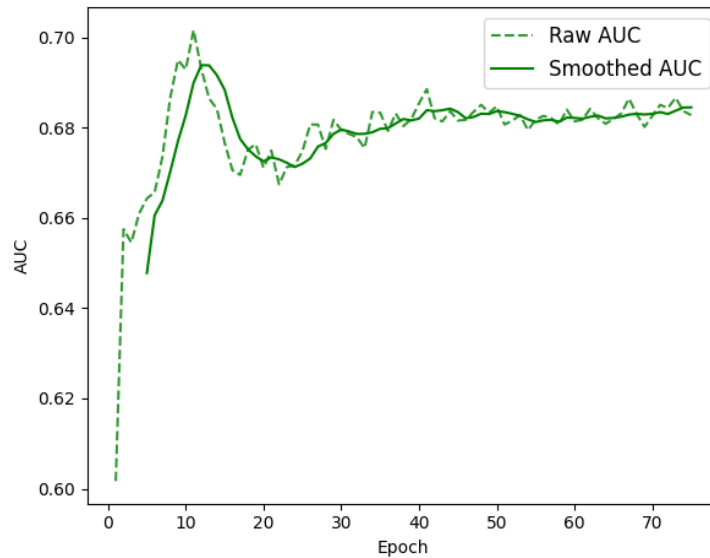


Figure 4.3: Validation AUC

Validation AUC plotted against 75 epochs of training, with a moving average applied to highlight overall trends and reduce short-term variability.

The confusion matrix (Figure 4.4) indicated that the model achieved an overall classification accuracy of 75%. Additional quantitative performance metrics (Table 4.2) were reported.

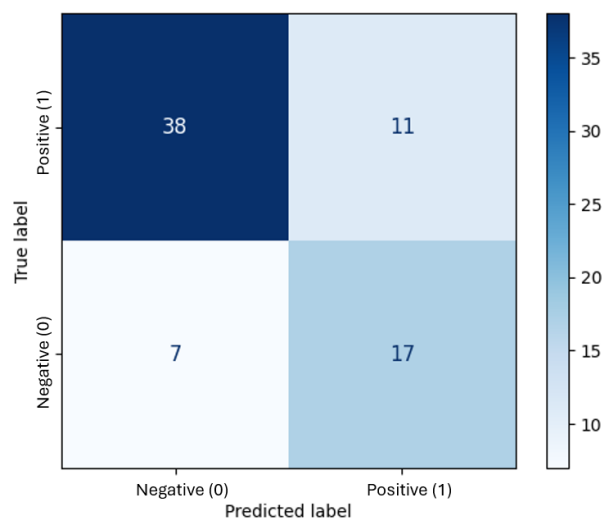


Figure 4.4: Confusion Matrix

Confusion matrix showcasing the model's ability to decipher true positive and true negative cases. The model was able to capture 17 true positive cases and 36 true negative cases, misclassifying 18 of 73 participants (25%).

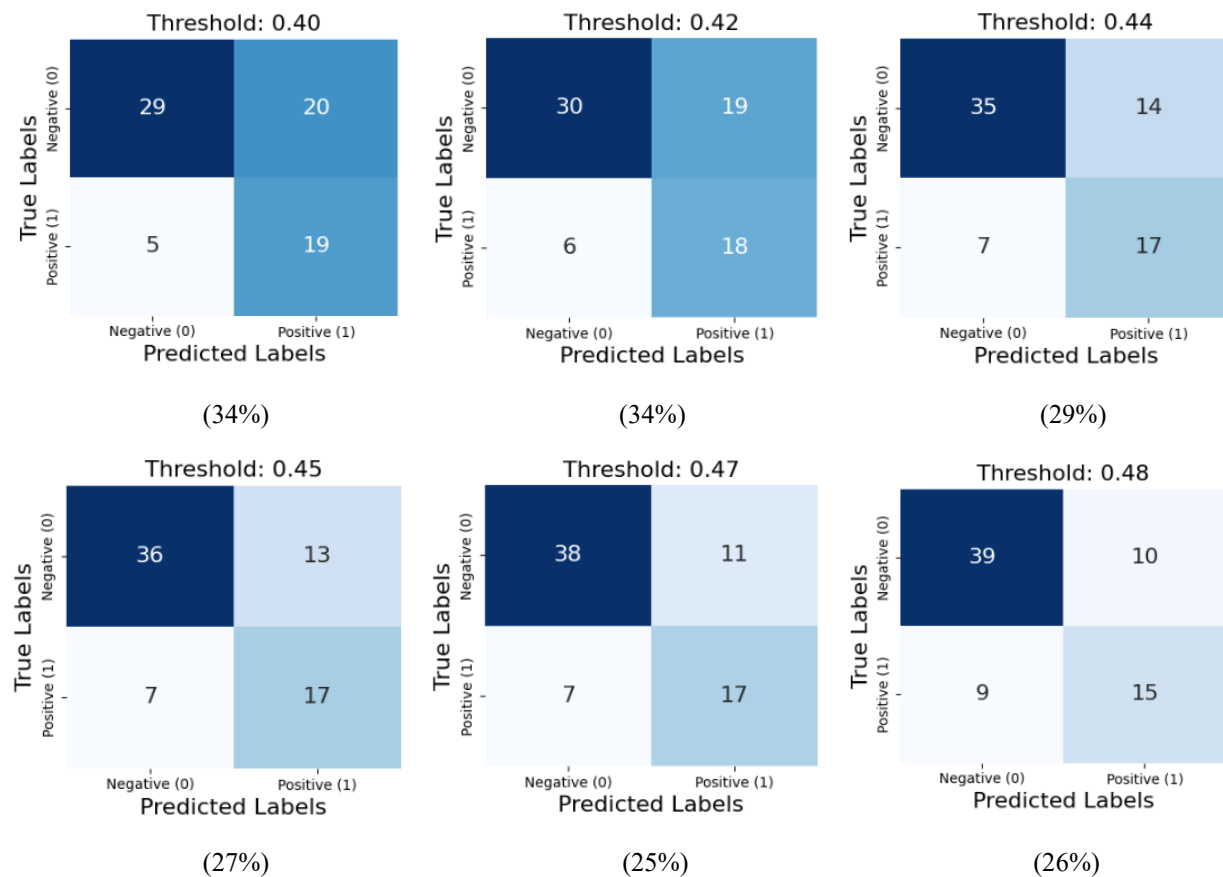


Figure 4.5: Confusion Matrices for Thresholds in Range 0.40-0.48 (Misclassification Percentage in Brackets)

Confusion matrices highlighting how classification outcomes change when the selected threshold is changed.

Table 4.2: Model Performance Metrics

Summary of the performance metrics obtained from the final model.

Performance Metric	Value
Sensitivity	0.71
Specificity	0.78
Precision	0.61
Accuracy	0.75
Training Loss	0.043
Validation Loss	0.037
Validation AUC	0.70
Validation F2 Score	0.68
Test AUC	0.78
Test F2 Score	0.69

The ROC curve (Figure 4.6) was consistently above the line of random chance, achieving an AUC of 0.78.

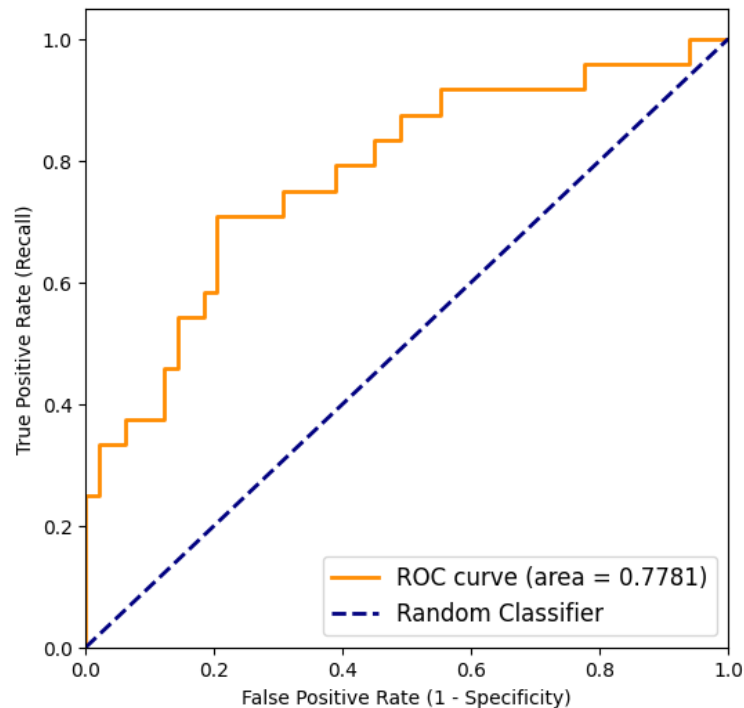
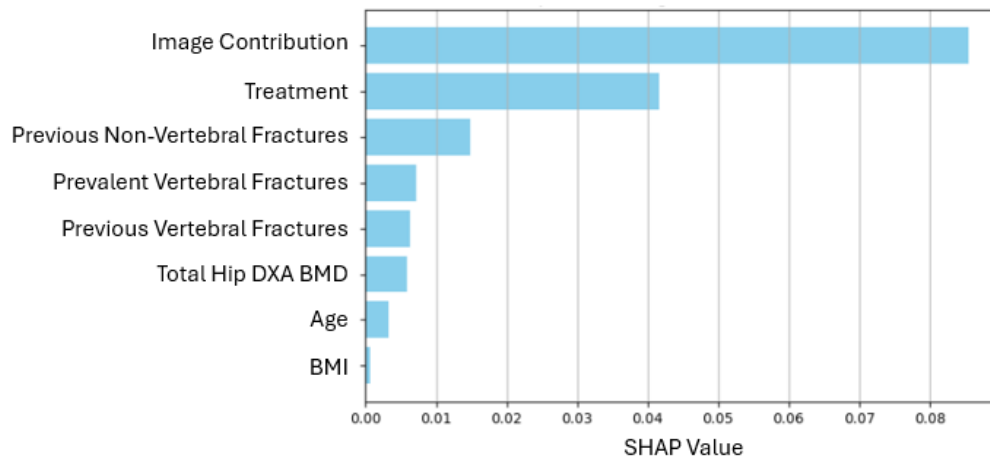


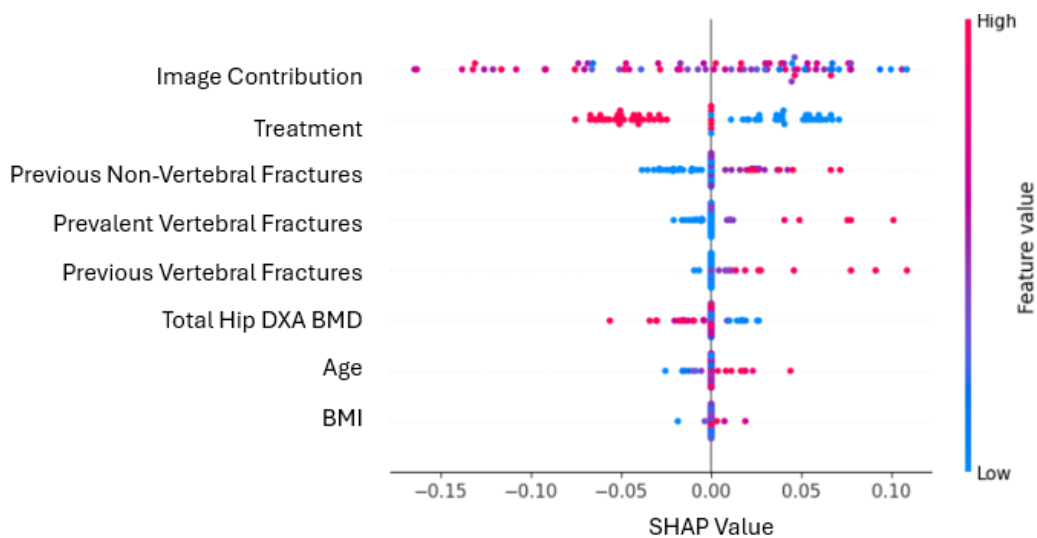
Figure 4.6: ROC Curve

ROC curve highlighting how the model performed compared to a random chance classifier.

The SHAP analysis (Figure 4.7) showed that image features contributed most strongly to the final prediction score, followed by medication status, previous fracture history, total hip BMD, and age. Body mass index (BMI) had the lowest impact on fracture risk prediction.



(a)



(b)

Figure 4.7: SHAP Analysis

(a) SHAP feature importance plot showing the relative contribution of imaging PCA features and clinical variables to the model's predictions. Values represent mean absolute SHAP scores, indicating each feature's overall impact on the output.

(b) SHAP summary plot showing the direction and magnitude of each feature's contribution. Each point represents a test sample, with color indicating the feature value and horizontal position indicating its positive or negative impact on the output. The plot highlights that absence of treatment (blue), low DXA BMD (blue), and a high number of previous fractures (red) is associated with higher fracture risk. The presence of treatment (red), high DXA BMD value (red), and a small number of previous fractures (blue) is associated with lower fracture risk. SHAP values clustered near zero indicate minimal influence on the model's decision compared to other features.

Branch-specific SHAP analysis showed that the two branches of the multi-modal architecture contributed nearly equally to the final prediction score (Figure 4.8).

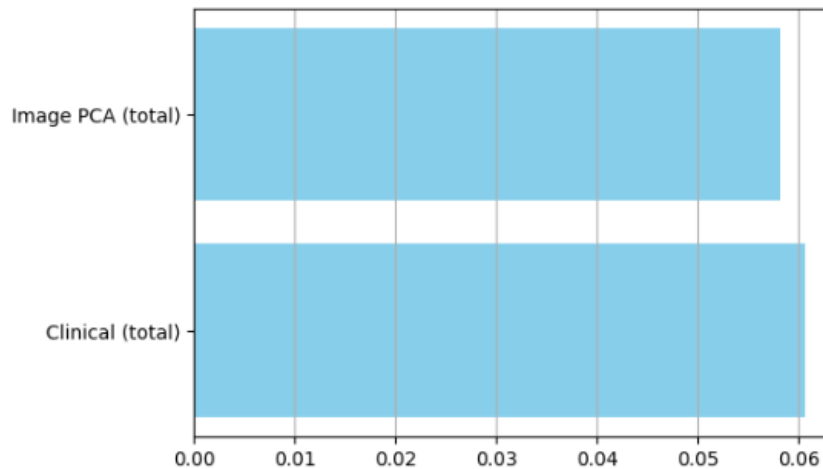


Figure 4.8: Branch Specific SHAP Analysis

Branch-level SHAP analysis highlighting branch contributions to the final prediction score.

4.4 Discussion

The goal of this work was to develop a machine learning model capable of predicting fracture risk in postmenopausal women with an osteoporosis diagnosis using clinical, demographic, and image data. This work demonstrated the benefits of using a multi-modal algorithm structure for fracture risk prediction and showed that imaging features provided meaningful predictive value beyond traditional clinical risk factors. This study also gave insight into which clinical risk factors were most meaningful for fracture risk prediction, highlighting the relative importance and directional effects of each predictor. This work demonstrated that machine learning models incorporating both clinical risk factors and imaging data can support more accurate and cost-effective fracture risk prediction in aging populations.

In this model, the training loss decreased rapidly after approximately the first 30 epochs and then continued to decrease gradually until the early stopping criteria were met. The validation

loss decreased mostly during the early stages of training before slightly climbing. The slight upwards trends in the validation loss is an indication that the model was beginning to overfit. Overfitting is a major issue in the field of predictive ML modeling, and regularization techniques such as dropout, early stopping, and dimensionality reduction were implemented to address this [80]. Despite this concern, the gap between the training loss and the validation loss remained small, indicating that the model was able to achieve moderate generalization, which was further confirmed by its test AUC (0.78). The validation AUC improved rapidly during the early stages of training, indicating that the model quickly learned informative patterns from the data. This initial improvement was followed by stabilization at a consistent performance level, suggesting successful convergence without evidence of progressive overfitting. These findings are reasonable for a neural network trained on 564 participants and demonstrate competitive performance relative to training and validation metrics reported in other ML-focused prediction studies [39], [49], [75], [81].

The confusion matrix showed that the model achieved an improved classification performance across both classes from the model developed in Chapter 3. The model achieved good sensitivity and specificity metrics that were comparable to other studies in the field [45], [75]. It also achieved a moderate F2 score, which is reasonable given the decision threshold was optimized on the validation data to prioritize recall. The selected threshold generalized well to the test set, which included a marginally higher proportion of fracture cases. A notable strength of this model, compared with the model developed in Chapter 3, was that its threshold exhibited greater stability for identifying true positives. Adjusting the threshold between 0.40 and 0.47 only affected two positive cases, whereas in Chapter 3, this change affected nine positive cases. This suggests that the model's sensitivity to threshold changes for identifying true positive cases was improved. This

also suggests consistency between the validation and test sets, as the optimal threshold from the validation set translated well to the test set.

Based on the AUC value, the model achieved reasonable discriminative ability beyond random chance. The curve remained consistently above the diagonal baseline of random chance, demonstrating that the model maintained a moderate balance between sensitivity and specificity across thresholds. Its AUC score was in line with the range commonly reported for the FRAX tool (approximately 0.60-0.80), indicating comparable predictive performance [36], [66].

The image features contributed most to the model's final prediction score, indicating that the image representation captured information relevant to fracture risk. This was expected, as it is consistent with the findings from Chapter 3, and the images represented a larger proportion of the features space. Each image was represented by 100 PCA-derived features, whereas each clinical risk factor was represented by a single feature, resulting in greater representation, and thus greater influence during training, for the image data. The importance of medication status, fracture history, and DXA BMD aligns with the established clinical understanding of osteoporosis, while the relatively low contribution of BMI reflects its less direct association with fracture risk [5]. The SHAP analysis supported established risk patterns by showing that lower bone mineral density, a history of prior fractures, and absence of treatment were associated with increased fracture risk. It was unexpected that BMD contributed relatively little to the model outcome, which may be due to the PCA-derived image features capturing overlapping skeletal information that reduced the independent influence of BMD in the final predictions.

4.4.1 Relevance, Strengths, and Limitations

The results demonstrated that the multi-modal model captured clinically meaningful patterns by combining imaging-derived PCA features with selected clinical risk factors. Compared

with the model developed in Chapter 3, this approach achieved a 12% increase in test AUC. The model performed well across diverse clinical subgroups, including medicated and unmedicated participants and those with prior fractures, in contrast to prior work that excluded individuals on medication or with previous fractures [45]. Despite sample size being a limitation of this study, the model still outperformed several fracture risk prediction models developed using larger cohorts, achieving higher AUC values than many reported approaches [36], [39], [44], [60]. In addition to performing within the expected range reported for the FRAX tool, the model demonstrated improved discrimination compared with BMD-based T-score assessment, achieving an approximately 12% increase in AUC relative to reported values [65]. Finally, the model substantially improved its ability to classify true negatives when compared to the approach developed in Chapter 3, with a 20% increase in correctly identified non-fracture cases. The stability of the selected decision threshold further supported the robustness of the model, as small threshold variations had minimal impact on positive case classification.

Despite the improved performance from the model developed in Chapter 3, there were still limitations to this work. The model still had a low F2 score (0.69), which although higher than the F2 score attained in Chapter 3 (0.60), still proves that the model struggles to consistently identify high risk individuals, likely due to the limited training sample size. Additionally, it was unexpected that BMD appeared so low in the SHAP ranking, which may reflect redundancy with the image-derived features or may indicate potential concerns about the model's validity. While earlier non-imaging studies ranked BMD above fracture history in SHAP analyses, their models also achieved AUC values up to 12% lower [44]. Additionally, linear interrelationships among clinical risk factors and DXA measures should be considered, as variables such as age are inherently correlated with lower BMD, leading to overlapping information captured across predictors. The training

curves also demonstrated a slight tendency towards overfitting, indicating that future work should focus on acquiring larger datasets to learn more generalizable feature patterns. Existing studies in research have also trained their learners on multiple anatomical images per participant (*e.g.*, full body DXA, DXA images of the forearm and spine) and additional CRFs (*e.g.*, genomic features, glucocorticoid exposure) [8], [40], [58], [64]. While increasing the number of predictors may have caused this learner to overfit, when larger datasets become available in the future it would be of interest to increase the feature space accordingly [55].

Finally, all results were dependent on a single train-validation-test split rather than k-fold cross-validation (KFCV). In practice, KFCV is often used because it can help to provide a reliable estimate of model performance while making efficient use of limited data [55], [58], [82]. Despite its advantages, this technique was not employed in this study because its use would have conflicted with several key components of the training pipeline. In standard KFCV, the held-out fold functions as the test set, leaving no separate validation data available. This was problematic because the model was trained using early stopping to prevent overfitting, which required a dedicated validation set to monitor improvements and identify the stopping epoch. Implementing early stopping within this structure would have required an additional inner split inside each training fold, converting the process into full nested cross-validation. Nested KFCV is computationally demanding and rarely used in practice, as it multiplies training cost and complexity across folds [83]. This also would have reduced the amount of data available for training within each fold, and it has been shown that reductions in training data can lead to less stable parameter estimation and suboptimal model performance within cross-validation runs [83]. Additionally, because each fold would have been assigned a different validation subset, each KFCV training run could have reached a different early stopping point, producing multiple

incomparable models with inconsistent training depth and learning dynamics [83]. For these reasons, a fixed train-validation-test design was implemented.

4.5 Conclusions

The multi-modal network presented in this work demonstrated a clear improvement over the model trained without CRFs, indicating that combining image-derived features with the selected CRFs provided additional predictive value. Compared to the model developed in Chapter 3, this work demonstrated a 12% improvement in test AUC, a 15% increase in classification accuracy, a 9% gain in F2 score, and a 5% improvement in validation AUC, along with higher reported metrics for sensitivity, specificity, and precision. The inclusion of treatment status and fracture history strengthened the model's ability to identify higher-risk individuals, and the SHAP analyses confirmed that these variables contributed meaningfully alongside the imaging representation. While overall performance remained moderate, with room for improvement in recall-focused metrics and the independent contribution of clinically important variables such as BMD, the model outperformed existing research models and was on the upper end of the AUC range typically reported for FRAX [23], [36], [44], [60]. These findings have provided a proof of concept for developing predictive ML models capable of identifying fracture risk from DXA images and CRFs in postmenopausal women with osteoporosis.

Chapter 5 – Deep Learning and Transfer Learning for Binary Fracture Classification from DXA Images and Clinical Risk Factors

***Overview:** This chapter introduces deep learning and transfer learning approaches for predicting fracture risk from dual energy x-ray absorptiometry (DXA) images. A custom deep learning model and a ResNet-50–based transfer learning (TL) framework were developed for binary classification, with the goal of classifying study participants as either high- or low-risk for fracture. This work builds on the machine learning methods presented in earlier chapters and explores the potential of more complex model architectures for fracture risk prediction. This chapter describes the processes used to train, tune, and evaluate both models, compares their performance with the Fracture Risk Assessment Tool (FRAX), and examines model explainability using SHapley Additive exPlanations (SHAP)-based image analysis.*

5.1 Introduction

Deep learning (DL) models are becoming powerful tools in medical image analysis because of their ability to easily learn complex, hierarchical feature representations from raw imaging data [55]. While shallower neural networks offered a good balance between model complexity and generalization, deeper architectures that operate directly on image data can better preserve spatial information that is lost when images are reduced using PCA. Transfer learning (TL) builds on this approach by using models that have been pre-trained on large datasets, allowing

previously learned features to be adapted to new tasks and making it possible to train effective models even when medical data are limited [49].

Both deep learning and transfer learning have demonstrated strong performance across a range of medical imaging applications, including disease detection and risk prediction [40], [43], [49], [84], [85], [86]. One of the primary challenges in developing deep learning models is their reliance on large, well-labeled, and balanced datasets [55]. Despite this limitation, several studies have reported strong performance even when training on relatively small datasets [36], [45]. This performance was achieved through regularization techniques and optimization strategies to reduce overfitting. Additional to DL models, transfer learning has emerged as a promising tool to mitigate data constraints by providing a pre-trained starting point for model development. In transfer learning approaches, models are initialized using existing weights and then tuned for the target task in later layers [49]. This is particularly advantageous for image-based tasks, as many transfer learning models are pre-trained to learn generic visual features, such as edges and textures, reducing the need for large datasets to learn these low-level representations from scratch [41], [42], [85]. Previous medical imaging studies using transfer learning have demonstrated high predictive performance, with reported accuracies of up to 98% using moderately sized datasets ($n = 1200$) [49]. Furthermore, the models developed in Chapter 3 and 4 of this thesis work relied on dimensionally reduced image inputs that did not reserve qualitative anatomical or spatial orientation. For these reasons, deep learning and transfer learning approaches were investigated in this work to assess their suitability for fracture risk prediction and to inform future model development.

5.2 Methodology

5.2.1 Deep Learning Model

This chapter includes the same target labels, clinical risk factors (CRFs), train/validate/test split, and loss function as described in Chapter 4. The images used in this algorithm were not reduced by principal component analysis, preserving their spatial resolution.

The model developed herein was a neural network (NN) that combined image-based features learned by a deep convolutional neural network (CNN) with tabular clinical risk factor (CRF) features learned by a multi-layer perceptron (MLP) [40]. This introduced a deeper architecture that extracted spatial features by operating directly on convolutional feature maps, rather than relying on dimensionality reduction techniques [45].

5.2.1.1 Model Architecture

The image branch of this MLP was a deep convolutional network with four blocks [39]. Each block had two convolutional layers with 3×3 kernels, stride and padding of 1, rectified linear unit (ReLU) activations, and a 2×2 max-pooling layer [55], [87]. The number of kernels increased progressively across blocks (32, 64, 128, and 256) to learn more complex features from the images. After the fourth block, global average pooling was performed to produce a 256-dimensional feature vector summarizing the learned visual representations [39]. The CRF branch consisted of a fully connected layer that mapped the seven input CRF features to a 16-dimensional feature vector, followed by ReLU activation and dropout (0.2) [87]. The CRFs were mapped to a higher-dimensional space because this allowed the network to combine the input variables in different ways, helping it capture relationships between risk factors before applying nonlinear transformations. The image and CRF feature vectors were then concatenated to form a joint feature vector that integrated image and CRF-derived features prior to classification [45]. The fused

feature vector was passed through a fully connected classifier consisting of two hidden layers with 64 and 32 units respectively, each followed by ReLU activation and dropout with a rate of 0.5 [49].

The final linear layer produced a single output logit for binary classification (Figure 5.1).

5.2.1.2 Model Training

The Focal Loss function was implemented ($\alpha = 0.25$, $\gamma = 4.0$) and the Adaptive Momentum Estimation with Weight Decay (AdamW) optimizer was used with a learning rate of $1e-4$ and a weight decay coefficient of $1e-2$ [62], [63]. Training was performed in mini batches (batch size = 32) over multiple epochs (minimum epochs = 50) with early stopping implemented (patience = 5) [64]. During each training epoch, the model was set to training mode to ensure that dropout layers were active and gradients could be updated.

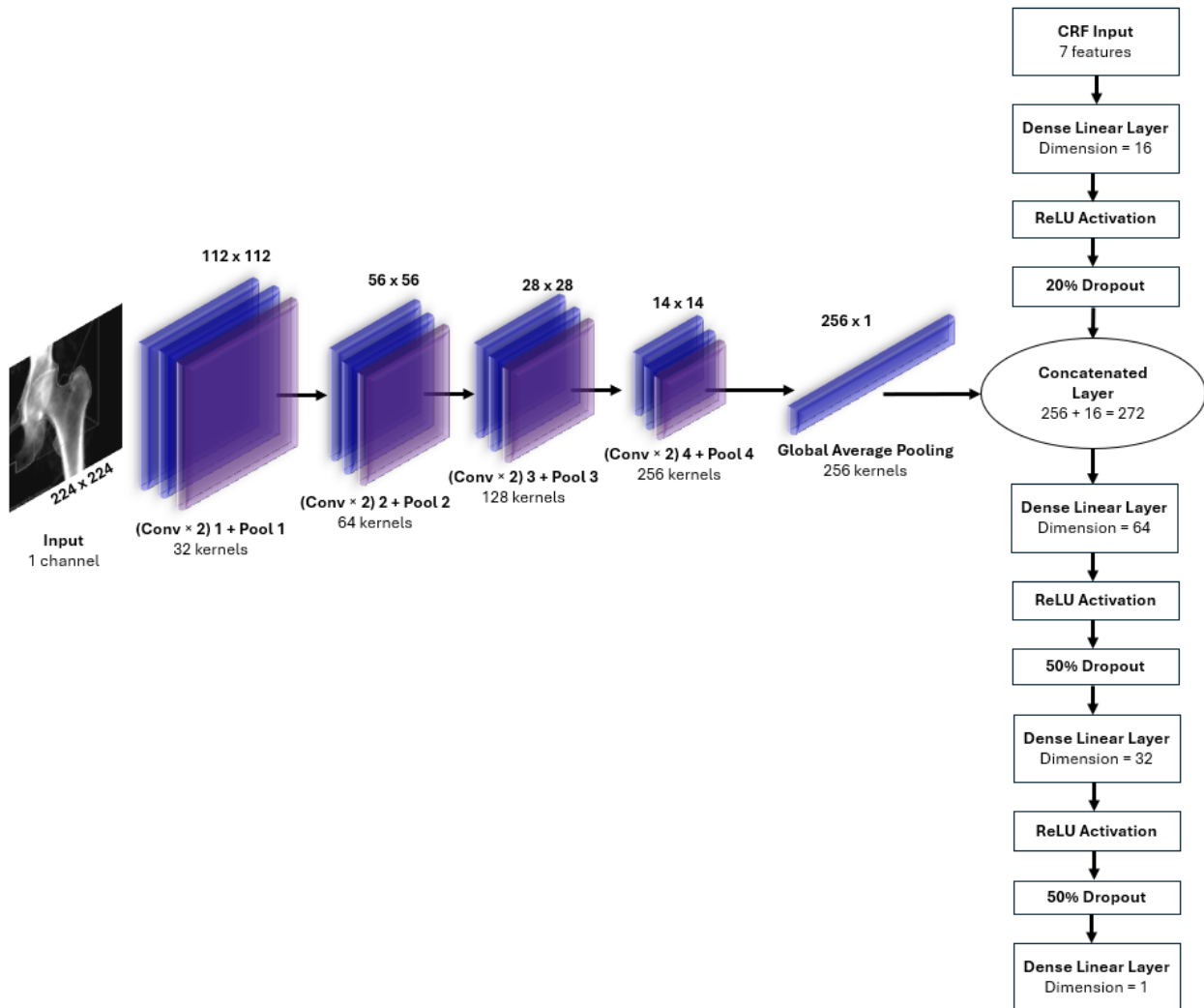


Figure 5.1: Deep Learning Architecture Overview

The grayscale images were pre-processed and cropped to 224 x 224 pixels before being fed as input into the CNN. Two convolutional layers with pooling layers extracted features and gradually downsampled the size of the feature maps. The 256 extracted feature images were concatenated with CRF features from a simple MLP with fully connected layers.

5.2.1.3 Model Validation

Images and CRF features in the validation set were passed through the model and focal loss was calculated to obtain the mean validation loss for each epoch. The logits were transformed into probability scores using the sigmoid function [84]. These probabilities were aggregated across all validation batches to compute the validation accuracy and the area under the curve (AUC) using

the full probability vector. The training loop recorded validation loss, validation accuracy, and validation AUC at each epoch [64]. Early stopping was driven by validation performance [64], where after a minimum of 50 epochs, an epoch was classified as meaningfully improved if either: (i) validation loss decreased by at least $1e-4$, or (ii) validation AUC increased by at least $1e-4$, relative to the best observed results. If no improvement occurred for five consecutive epochs, training was stopped, and the best model parameters were saved [88], [89]. The classification threshold was optimized on the validation set by maximizing the F1 score over probability cutoffs from 0.01 to 0.99, and the resulting threshold was applied to the test set [67], [68], [90]. Validation was performed to tune the model parameters before evaluating the final predictive ability on the test set (Table 5.1).

Table 5.1: Global Training Parameters

Final training parameters used after hyperparameter tuning.

Parameter	Value
Learning Rate	$1e-4$
Batch Size	32
Weight Decay	$1e-4$
Alpha (Focal Loss)	0.25
Gamma (Focal Loss)	4.0
Minimum Epochs	50
Early Stopping Patience	5
Minimum Improvement (Delta) Required	$1e-4$

To evaluate the model, plots were generated for training and validation loss as well as validation AUC. The data were smoothed using a moving average function with a window size of five to reduce noise and better highlight the underlying trends. To assess its performance in binary classification, a confusion matrix was generated from the test data. The classification threshold was selected from F1 optimization on the validation set and the confusion matrix was used to calculate model sensitivity, specificity, precision, and accuracy. A receiver operating characteristic

(ROC) curve was generated as an assessment of the model’s predictive ability independent of the decision threshold. To quantify how each clinical feature and the images impacted the final prediction score, Shapley Additive exPlanations (SHAP) analysis was performed.

5.2.2 Transfer Learning Model

Additional to the DL model, a preliminary investigation into the feasibility of transfer learning for fracture risk prediction from dual-energy x-ray absorptiometry (DXA) images was conducted. The goal of this experiment was not to establish an optimized clinical model, but to develop a simple, reproducible baseline using a widely adopted CNN architecture. ResNet-50 is a CNN pre-trained on the ImageNet database, and it was selected for this application because it provides a strong, well-validated architecture capable of learning rich visual features while remaining computationally efficient and stable to train [40], [41], [49], [64]. This model was implemented using the PyTorch torchvision library. By leveraging a pre-trained ResNet-50 model, this study explored whether features learned from large-scale natural image datasets could transfer to DXA images for fracture risk prediction [49], [91]. This served as an initial benchmark to guide future model development and identify domain-specific challenges. For this application, only images were used and clinical risk factors were excluded to specifically assess ResNet-50 as a TL model for DXA-based fracture risk prediction. Through its residual learning framework, this network can learn robust hierarchical visual features that have been shown to transfer effectively to medical imaging problems [40], [49], [91].

The same DXA image dataset used in previous chapters was adapted for use with the pre-trained ResNet-50 architecture. Single-channel images were resized to $224 \times 224 \times 3$ and normalized using ImageNet mean (0.485) and standard deviation (0.229) values [49]. The final fully connected layer was replaced with a single-logit output for binary classification. Model

optimization used binary focal loss with logits ($\alpha_{\text{positive}} = 0.75$, $\alpha_{\text{negative}} = 0.15$, $\gamma = 2.0$), which was intended to increase the penalty for false negatives. This was done because early iterations showed that the model tended to miss true positive cases.

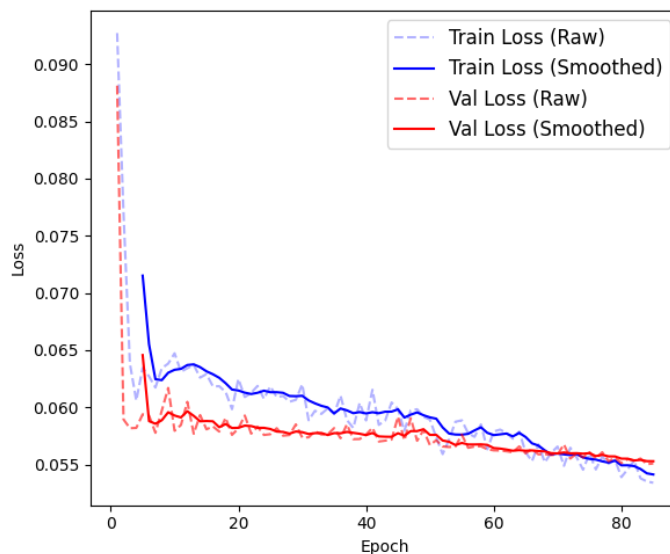
Training was performed in two phases. In Phase 1, all backbone parameters were frozen and only the final classification layer was trained for 5 epochs using AdamW [49]. In Phase 2, the final residual block and the classifier were unfrozen and fine-tuned for 10 epochs using AdamW. Performance was monitored at each epoch using validation loss and area under the ROC curve (AUC). After training, predicted probabilities were obtained by applying a sigmoid transform to the logits. Decision threshold selection was performed on the validation set using a grid search over thresholds (0.01–0.99) [67], [68], [90]. The primary threshold was chosen to maximize the F2 score. Final performance on the held-out test set was reported using accuracy, precision, recall, AUC, and confusion matrices.

To evaluate the model, training and validation loss as well as validation AUC were plotted. A confusion matrix was generated to determine the classification accuracy of the model, and additional quantitative performance metrics relevant to binary classification problems were calculated.

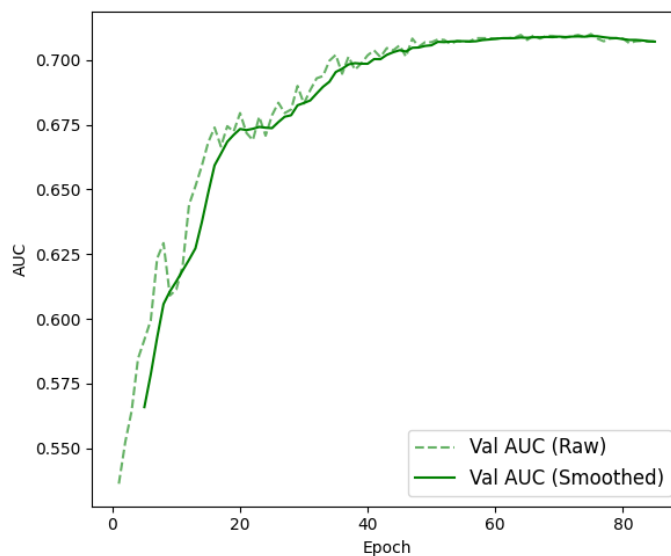
5.3 Results

5.3.1 Deep Learning Model

The model's training and validation loss (Figure 5.2) experienced an overall downward trend, with training loss decreasing more quickly across epochs than validation loss. The validation AUC (Figure 5.3) experienced a steep incline to a plateau at 0.71 after approximately 60 epochs.

**Figure 5.2: Training and Validation Loss**

Training and validation loss plotted across 85 training epochs, with a moving average applied to highlight overall trends and reduce short-term variability.

**Figure 5.3: Validation AUC**

Validation AUC plotted against 85 epochs of training, with a moving average applied to highlight overall trends and reduce short-term variability.

The confusion matrix (Figure 5.4) highlighted that the model achieved a classification accuracy of 70%.

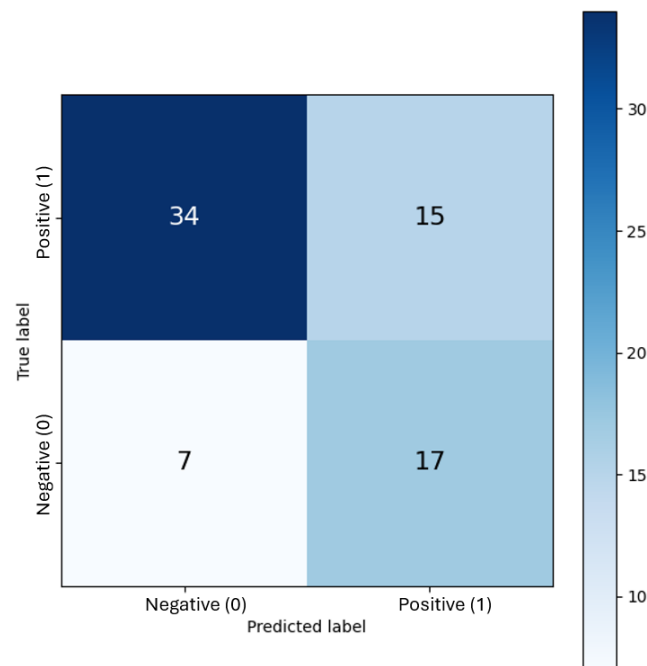


Figure 5.4: Confusion Matrix

Confusion matrix showcasing the model’s ability to decipher true positive and true negative cases. The model was able to capture 17 true positive cases and 34 true negative cases, misclassifying 22 of 73 participant examples (30%) (threshold = 0.31).

Table 5.2: Model Performance Metrics

Performance metrics obtained from final model evaluation on the test set, as well as training loss, validation loss, and validation AUC.

Performance Metric	Value
Sensitivity	0.71
Specificity	0.69
Precision	0.53
Accuracy	0.70
Training Loss	0.053
Validation Loss	0.055
Validation AUC	0.71
Test AUC	0.72
Test F2	0.66

The ROC curve (Figure 5.5) demonstrated that the model stayed consistently above the line of random chance and achieved an AUC of 0.72.

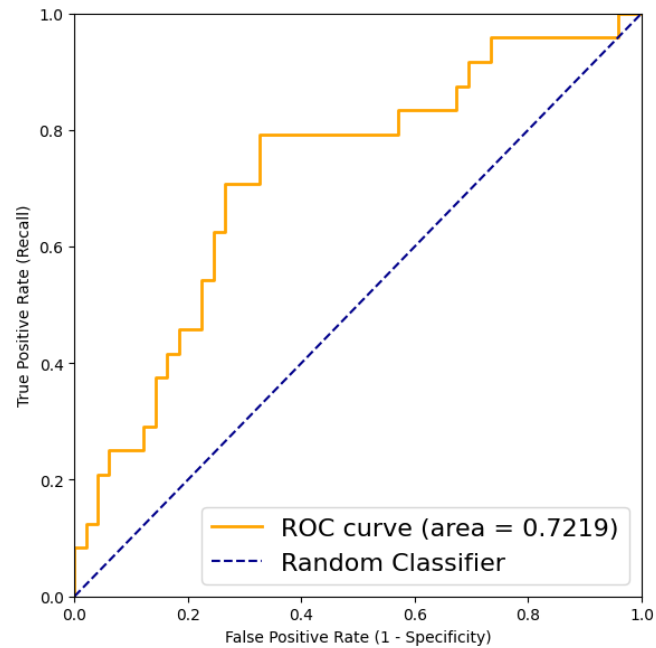
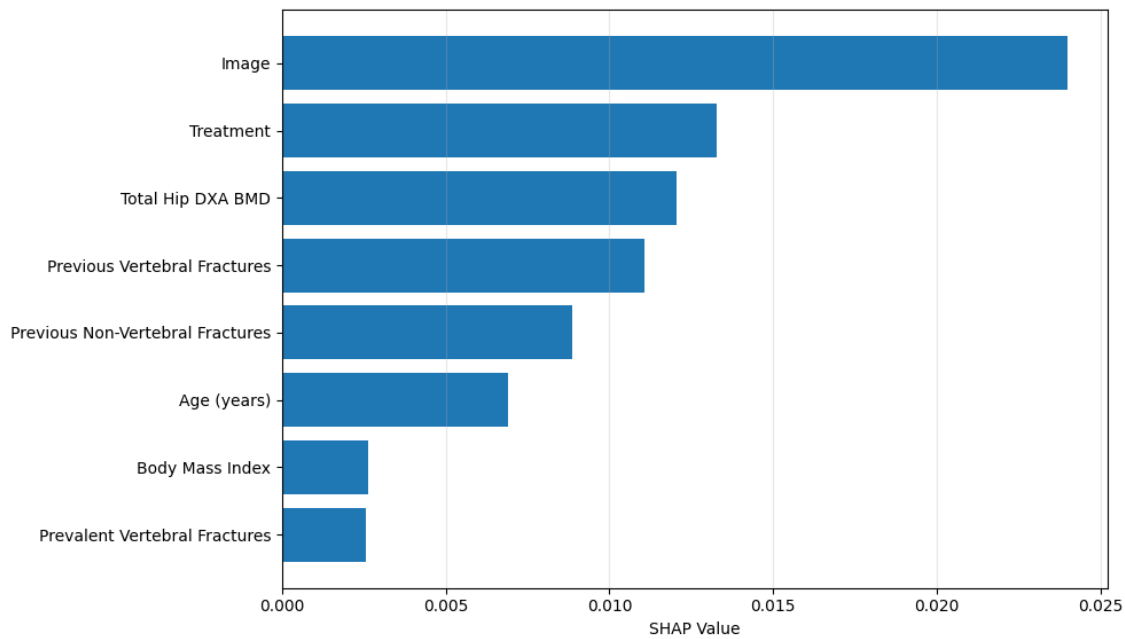


Figure 5.5: ROC Curve

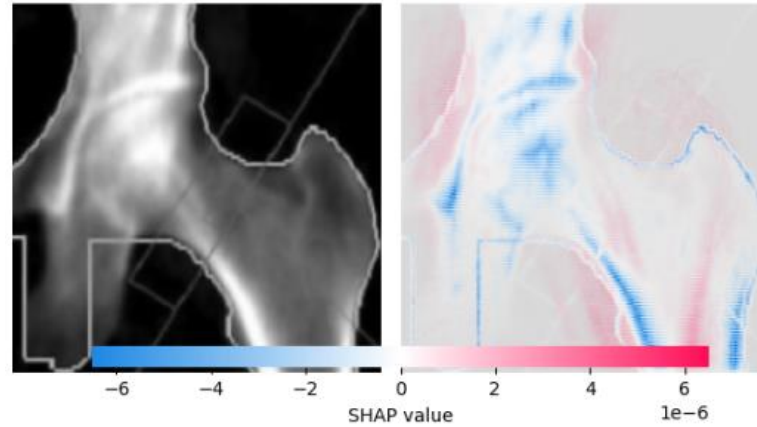
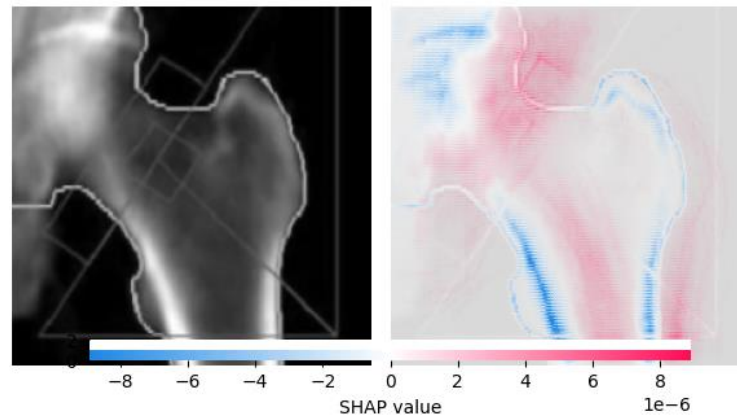
ROC curve highlighting how the model performs compared to a random chance classifier. The model achieved an AUC of 0.72.

The SHAP analysis (Figure 5.6) highlighted that the images contributed most to the final prediction score, followed by treatment and total hip DXA bone mineral density (BMD). The factor that contributed the least to the model's output was the number of prevalent vertebral fractures.

**Figure 5.6: SHAP Analysis**

SHAP feature importance plot showing the relative contribution of image features and clinical variables to the model's predictions. Values represent mean absolute SHAP scores, indicating each feature's overall impact on the output.

To interpret the image component of the model, gradient SHAP analysis was applied to all the test images [92]. Pixel-level SHAP values were computed to visualize regions of the image that contributed positively or negatively to the predicted fracture probability. Two correctly identified test cases are shown in Figure 5.7.

(a) True Negative Case | $P(\text{fracture}) = 0.28$ (b) True Positive Case | $P(\text{fracture}) = 0.32$ **Figure 5.7: Image-Level SHAP Analysis**

Examples of gradient SHAP visualizations overlaid on hip DXA images for one true positive and one true negative participant. For each participant, the original image (left) and the corresponding SHAP map (right) are shown. Red regions indicate image areas that contribute positively to the predicted fracture probability, while blue regions indicate areas that contribute negatively. Image (a) highlights a higher density of blue anatomical regions and image (b) highlights a higher density of red anatomical regions, illustrating how localized image features influence the model's decision making. $P(\text{fracture})$ indicates the model's prediction score from 0 (no fracture risk) to 1 (certain fracture risk), with a classification threshold of 0.31.

5.3.2 Transfer Learning Model

The model's training loss (Figure 5.8) experienced an overall downward trend, and the validation loss (Figure 5.8) plateaued after approximately one epoch. The validation AUC (Figure 5.9) experienced some oscillation before flattening after about six epochs.

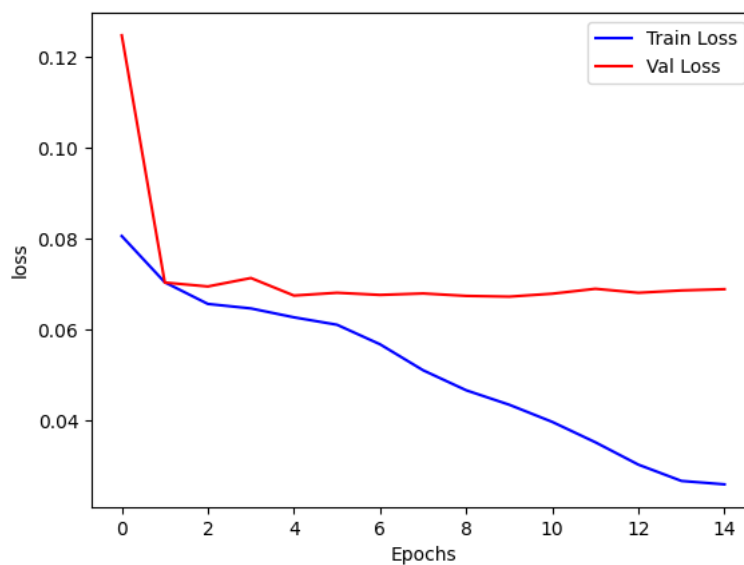


Figure 5.8: Training and Validation Loss

Training and validation loss plotted against 14 training epochs.

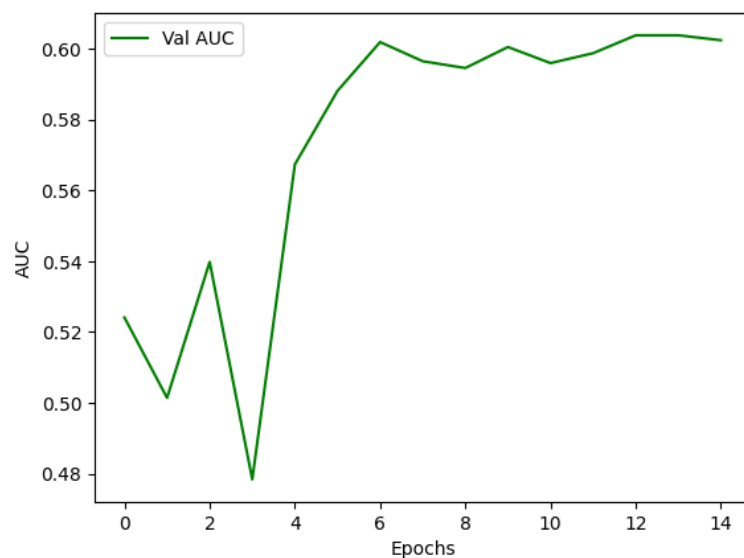


Figure 5.9: Validation AUC

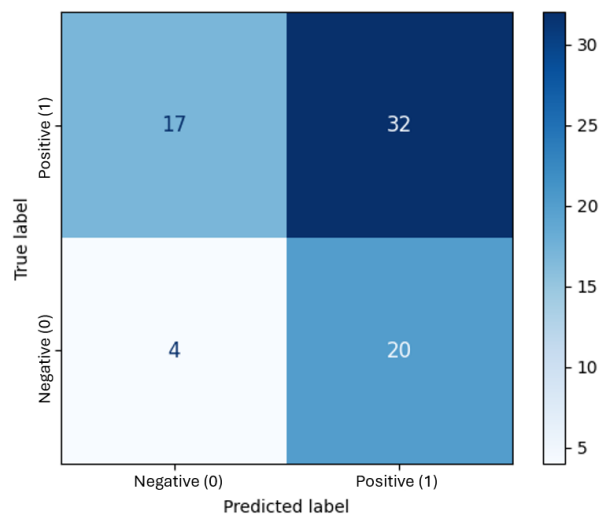
Validation AUC plotted against 14 training epochs.

Table 5.3: Model Performance Metrics

Performance metrics obtained for TL model.

Performance Metric	Value
Precision	0.39
Recall	0.83
Accuracy	0.50
Test AUC	0.65

The confusion matrix (Figure 5.10) highlighted a classification accuracy of 51%.

**Figure 5.10: Confusion Matrix**

Confusion matrix showcasing the model's ability to decipher true positive and true negative cases (misclassification = 49%, threshold = 0.42).

5.4 Discussion

5.4.1 Deep Learning Model

This work served as an investigation into predicting fracture risk in postmenopausal women using deep learning techniques, introducing a multi-modal framework that models imaging features alongside clinical risk factors to predict fracture risk. Despite the limited sample size, the model achieved moderate predictive performance, attaining AUC values comparable to the established (and widely used) FRAX tool and exceeding AUC values for BMD T-score assessment

[65], [66]. These findings demonstrated that deep learning applied to DXA imaging can extract clinically relevant information beyond traditional risk models and highlighted the potential of multi-modal approaches to enhance fracture risk assessment in postmenopausal women.

During training, the model exhibited stable learning behavior, as evidenced by the parallel decrease in training and validation loss across successive epochs. Validation AUC increased rapidly during early epochs and plateaued around 0.71, indicating moderate discriminative ability. When evaluated on the test set, the model achieved a comparable AUC of 0.72, indicating repeatable performance. At the selected threshold (0.31), the model's sensitivity and specificity reflected a balanced tradeoff between detecting true positives and limiting false positives. The model incorrectly classified 22 out of 73 cases, including seven instances in which true positive cases were missed. These metrics reflected moderate predictive ability that is comparable to FRAX and other research tools [36], [44], [60], [66]. It is noteworthy that this model converged on a lower optimal decision threshold (0.31) than earlier models in this thesis, which converged at thresholds of 0.45 and 0.47. The CNN produced a narrower range of predicted risk probabilities (0.25–0.41) compared with the image-only model (0.20–0.60) and the image/CRF model (0.17–0.61), suggesting lower confidence in its predictions. In contrast, the image/CRF model achieved the highest classification accuracy and the widest range of predicted probabilities, indicating stronger discriminative ability and making it the most robust model developed in this work.

The low decision threshold in the DL model may be explained by the CNN's predicted probabilities being tightly clustered. In addition, unlike the other models, the deep learning model showed better predictive performance in validation when the threshold was optimized for the F1 score rather than the F2 score. Given F1 balances precision and recall, it is reasonable that the decision threshold would be lower. The model achieved good recall, which is particularly

important in clinical settings where failure to identify fracture events may result in serious adverse outcomes. Given that this deep learning study was conducted on a very small population, these results are encouraging and suggest that, with the availability of larger datasets, a more robust and clinically useful decision boundary could be learned.

SHAP analysis showed that images dominated model predictions, indicating that DXA images captured fracture-relevant information beyond clinical variables. While clinical risk factors provided strong low-dimensional signals, the image branch learned additional features not fully represented by clinical measurements, highlighting the value of incorporating imaging data into clinical risk assessment. The SHAP analysis also confirmed that BMD had a high influence on the model outcome, which differed from the model developed in Chapter 4, where BMD exhibited a lower SHAP ranking. Given that BMD is a well-established and clinically used determinant of fracture risk [5], this difference is clinically relevant and warrants further investigation to understand how model design and feature representation may influence the alignment between model explanations and established clinical knowledge. Gradient SHAP visualizations revealed consistent qualitative differences between true positive and true negative cases. True positive predictions were associated with broader and predominantly positive SHAP attributions across the proximal femur, whereas true negative predictions exhibited more negative attributions. This supports the hypothesis that imaging data can enhance fracture risk prediction when effectively integrated with clinical variables.

5.4.2 Transfer Learning Model

The training loss decreased steadily across epochs, while the validation loss dropped rapidly in the early epochs and then plateaued. This divergence indicates that the model continued to fit the training data but achieved limited additional generalization on unseen data. Although no

sharp increase in validation loss was observed, the widening gap between training and validation loss suggests mild overfitting [39]. The validation AUC improved from approximately 0.52 to around 0.61 before stabilizing. While this trend demonstrates that the model learned some class-separating signal, the overall AUC remained modest, indicating limited discriminative performance. The test AUC was still within the range of AUCs reported by FRAX [66]. It is important to note that the number of training epochs is much smaller in this application than in previous work because in transfer learning the model starts from pre-trained weights that capture useful features, so training focuses only on adapting these features to the new task rather than learning them from scratch. Existing work has reported similar performance curves when training over a small number of epochs, while other studies have used substantially more epochs and achieved comparable predictive accuracy [42], [49], [85], [91]. This suggests that the optimal number of training epochs is highly context-dependent and influenced by factors such as the dataset, task complexity, and training strategy [49], [85], [91].

The model achieved a high recall (0.83), correctly identifying most positive cases, which is desirable in a clinical context. However; this came at the cost of very low precision (0.39), with a large number of false positives. In a clinical setting, this would lead to over-treatment of patients, exposing many individuals to unnecessary risks and side effects without clear benefit, while also increasing healthcare costs. These results indicate that while the recall-leaning focal loss and threshold selection successfully reduced false negatives, the model struggled to distinguish negative cases reliably.

Although the predictive performance of the proposed model was limited, TL approaches still hold value for fracture risk classification. Prior studies have demonstrated improved predictive ability using TL in medical imaging, including applications in stroke risk prediction and DXA-

based assessment [40], [49], [91]. Several factors may have contributed to the poor performance of the present model. Although ResNet-50 has performed well in other medical imaging studies, these typically used larger, more diverse datasets and incorporated additional predictors. This investigation excluded CRFs to isolate the image contribution for an image-based TL model; however, incorporating CRFs in future work is likely to improve model performance [40], [49]. Exploring alternative TL architectures and multi-modal designs may lead to more robust predictive ability.

5.5 Conclusions

This chapter demonstrated the feasibility of applying deep learning methods to fracture risk prediction using DXA imaging. The proposed deep learning model achieved performance outcomes comparable to FRAX outcomes reported in literature, while also providing meaningful model explainability through SHAP-based image analysis [66]. These findings suggested that the model captured clinically relevant imaging features. In contrast, the transfer learning approach showed weaker performance, which was likely attributed to a mismatch between the complexity of the ResNet-50 architecture and the relatively limited input data used in this proof-of-concept study. However, this result does not preclude continued investigation into the use of TL for this application. Rather, it suggested that larger and more diverse imaging datasets, inclusion of clinical risk factors, and alternative TL architectures may enable more effective TL applications. Overall, these findings support the potential utility of deep learning for fracture risk assessment and motivate further evaluation of machine learning approaches in medical imaging.

Chapter 6 – General Discussion and Conclusions

***Overview:** This chapter provides a summary of the main findings and reflects on their significance and relevance within the broader field of research. In addition, the overall strengths and limitations of the study are discussed and potential directions for future research that build on the results presented in this thesis are outlined.*

6.1 Summary

Osteoporosis is a critical public health challenge that affects over 2.3 million Canadians [93]. This disease is characterized by progressive loss of bone mass, and its first symptom is often a fragility fracture, making it challenging to diagnose before serious consequences occur [1]. Hip fractures are especially detrimental to patient quality of life, with one-year mortality rates post-fracture reaching as high as 22% in women and 33% in men [3]. Current diagnostic approaches for osteoporosis rely on a bone mineral density (BMD) measure (obtained at the femoral neck, total hip, or lumbar spine) obtained through dual-energy x-ray absorptiometry (DXA) imaging [12], [94]. Although BMD is correlated to fracture risk, this assessment method provides an incomplete representation of an individual's overall fracture risk [5]. Although the Fracture Risk Assessment Tool (FRAX) is widely used for fracture risk assessment, it has recognized limitations and has demonstrated challenges in achieving strong predictive accuracy for fracture risk [24], [66]. Four novel machine learning (ML) models were developed in this work, incorporating hip DXA images and known clinical risk factors to predict fracture risk in postmenopausal women. These developments represent meaningful progress in the field of machine learning for biomedical research and highlight the potential for further improvement in fracture risk assessment. The

overall purpose of this work was to evaluate whether machine learning approaches could utilize routinely collected DXA images with CRFs to enhance fracture risk prediction in older adults diagnosed with osteoporosis. This was achieved through acquiring relevant, longitudinal clinical data, performing proper pre-processing techniques for ML integration, assessing different modeling approaches, and performing a final, unbiased evaluation of each of the developed models.

The first phase of this work was to perform data pre-processing. This is a critical step in any ML problem, as it reduces noise, addresses missing or inconsistent values, and improves model performance [55]. The data used in this study were received from the Amgen (Amgen, Thousand Oaks, CA, USA) Fracture REduction Evaluation of Denosumab in Osteoporosis Every 6 Months (FREEDOM) trials [28]. To pre-process the images, the anatomical region of interest was isolated from the complete DXA report, followed by standard pre-processing methods for images [45]. Clinical risk factors were pre-processed using collinearity assessment and additional standard techniques to ensure suitability for model training.

Following pre-processing, the data were split into training, validation, and test sets to ensure reliable training, hyperparameter tuning, and independent evaluation of model performance. Principal component analysis (PCA) was performed on the images to reduce the feature space. A feed-forward neural network (FFNN) was trained to classify participants as high or low risk for fracture based on image data independently (*i.e.*, Objective 1, Chapter 3). The model correctly classified a majority of the test cases but failed to identify a substantial portion of fractured cases, limiting its suitability for clinical use where identifying true positive cases is critical. It achieved a moderate area under the curve (AUC), indicating it had better predictive ability than random chance, which is in line with the metrics of other reported fracture risk

prediction tools including FRAX and BMD T-score [23], [36], [44], [65]. The performance metrics were shown to be highly dependent on the threshold used for binary classification. This work provided a strong foundation for the subsequent inclusion of clinical risk factors in the predictive modeling algorithm.

Factors known to influence skeletal strength and fracture risk were included through a multi-modal architecture (*i.e.*, Objective 2, Chapter 4). The classification accuracy of this algorithm improved markedly compared to the version without CRFs, along with a notable enhancement in its ability to correctly identify true positive cases. This model was similarly sensitive to changes in the decision threshold; however, its ability to identify true positive cases remained relatively stable, with threshold adjustments resulting in only minimal variation in true positive classification. SHapley Additive exPlanations (SHAP) analysis was employed to enhance model interpretability by identifying the factors that most strongly influenced the final prediction score, revealing that the image and CRF branches contributed approximately equal information to the final outcome. To evaluate a model capable of capturing more spatial patterns, a simple convolutional neural network (CNN) and a transfer learning (TL) model were developed (*i.e.*, Objective 3, Chapter 5). The CNN achieved good predictive accuracy, an AUC value comparable to that of FRAX, and a superior AUC to BMD T-score [23], [36], [44], [65]. The TL model struggled to learn a meaningful decision boundary, as proven by its high rate of misclassification. After considering all four models developed in this work, each had their own strengths and limitations. The TL and image-only PCA models performed the poorest in terms of incorrect classification and test AUC. Although the image-only model demonstrated low classification accuracy, it produced a relatively wide range of predicted risk scores, suggesting greater confidence in its predictions compared with the deep learning model. In contrast, the deep learning

model achieved a higher test AUC (0.72) but generated a narrow range of predicted probabilities (0.25–0.41), indicating limited confidence in individual predictions, potentially due to the complexity of applying convolutional and pooling layers to a limited sample size.

The image/CRF model achieved the strongest overall performance. It produced the highest test AUC (0.78), a broad range of predicted risk scores (0.17–0.61), and moderate sensitivity to decision threshold selection. In addition, this model showed no strong tendency toward overfitting as it demonstrated stable behavior in both training and validation. These findings suggest that PCA-based dimensionality reduction, despite sacrificing some spatial information compared with deep learning approaches, effectively reduced the feature space and limited overfitting while preserving meaningful predictive signal. As a result, the image/CRF model represents the most balanced and robust approach for this dataset and would be the most appropriate model to pursue in future work, particularly given the limited sample size. This model demonstrated improved discriminatory performance relative to AUC values reported in the literature for fracture risk assessment based solely on BMD T-score and achieved performance within the range commonly reported for FRAX [23], [36], [44]. To further contextualize these findings, Table 6.1 summarizes the principal tools currently used for fracture risk prediction, the clinical risk factors they incorporate, and the performance metrics reported in the literature, alongside the results obtained in this thesis.

Table 6.1: Comparison of Risk Prediction Tools

Comparison of various fracture risk prediction tools including BMD T-score, FRAX/FRAXplus, Canadian Association of Radiologists and Osteoporosis Canada (CAROC), and the models developed in this work.

Tool	Categorical / Binary CRFs Included	Continuous / Dose- Dependent CRFs Included	Model Outcome	Reported Area Under the Curve
BMD T-score	N/A	BMD T-score (total hip, femoral neck, or lumbar spine) [94]	Fracture risk	~ 0.66 [65]
Fracture Risk Assessment Tool (FRAX)	Continent, country, previous fracture, parental hip fracture, current smoking, glucocorticoids, rheumatoid arthritis, secondary osteoporosis, alcohol (3 or more units/day) [8]	Age, sex, weight, height, femoral neck BMD [8]	10-year fracture probability [8]	0.60-0.80 [36], [66]
FRAX+	Available in beta form. FRAX with additional adjustment for fracture recency, high-dose glucocorticoid exposure, duration of type 2 diabetes, recent falls, and supplementary skeletal measures (lumbar spine BMD, TBS, and hip axis length) [8]			
Canadian Association of Radiologists and Osteoporosis Canada (CAROC) Tool	Sex, fracture history, glucocorticoid use [26]	Age, femoral neck BMD [26]	Low / moderate / high 10-year major osteoporotic fracture risk [25], [26]	Not widely reported
Image-Only PCA Model	Osteoporosis diagnosis (T-score $\leq$ - 2.5)	DXA images reduced by PCA	Low / high 3-year hip, vertebral, or major osteoporotic fracture risk	0.66

Tool	Categorical / Binary CRFs Included	Continuous / Dose- Dependent CRFs Included	Model Outcome	Reported Area Under the Curve
Image-CRF PCA Model	Osteoporosis diagnosis (T-score $\leq$ - 2.5), treatment status (placebo vs. denosuamb)	DXA images reduced by PCA, age, body mass index, number of previous vertebral fractures, number of previous non- vertebral fractures, number of prevalent vertebral fractures, femoral neck DXA BMD	Low / high 3- year hip, vertebral, or major osteoporotic fracture risk	0.78
Deep Learning Model	Osteoporosis diagnosis (T-score $\leq$ - 2.5), treatment status (placebo vs. denosuamb)	DXA images reduced by PCA, age, body mass index, number of previous vertebral fractures, number of previous non- vertebral fractures, number of prevalent vertebral fractures, femoral neck DXA BMD	Low / high 3- year hip, vertebral, or major osteoporotic fracture risk	0.72
Transfer Learning Model	Osteoporosis diagnosis (T-score $\leq$ - 2.5)	DXA images reduced by PCA	Low / high 3- year hip, vertebral, or major osteoporotic fracture risk	0.65

6.2 Limitations and Strengths

Limitations and strengths specific to each phase of this thesis work were discussed in detail in each chapter; however, there are certain themes that apply to the entire thesis work.

The greatest limitation of this work was that all models were trained, tested, and evaluated on only one cohort consisting of 806 participants from the Amgen FREEDOM trials. To develop a robust learner, a model should be trained on large amounts of diverse data, and to get a fair assessment of its generalizability, it should be tested on independent, external datasets. While the Amgen cohort provided detailed longitudinal data that made this work possible, several characteristics limit its generalizability to the broader population. The study population consisted exclusively of postmenopausal women with a diagnosed case of osteoporosis, meaning the findings are applicable only to this specific subgroup [28]. Because all participants had osteoporosis, the “low risk” classification in this cohort does not correspond to the baseline risk level of older adults without a clinical osteoporosis diagnosis. Despite this limitation, postmenopausal women represent a population at elevated risk for osteoporotic fracture, making them an appropriate and clinically relevant group for this application [5]. Additionally, all medicated participants were receiving denosumab; as osteoporosis treatments act through different mechanisms, the findings may not be generalizable to individuals treated with other medications.

Only 33 hip fractures were observed during the follow-up period. To better balance the classes, fractures at other anatomical sites were included in the analysis, despite the model using only hip images. As a hip-focused prediction tool, this reduces anatomical specificity. Conversely, if the goal were to predict general fracture risk, incorporating images from multiple skeletal sites would have been more appropriate [40]; however, this would have required a larger cohort. Separately, no weighting scheme was applied to differentiate between participants who

experienced forearm fractures (classified as major osteoporotic fractures) and those who sustained more severe outcomes, such as both hip and vertebral fractures during the three-year follow-up period. Treating these events as equivalent may have obscured meaningful differences in fracture severity and clinical impact, potentially limiting the model's ability to learn nuanced risk patterns and to prioritize individuals at highest risk of serious fracture outcomes.

Another limitation is that many factors known to influence fracture risk (*e.g.*, smoking history, alcohol intake, ethnicity, trabecular bone score) were not included in this model. Some variables were excluded because they were not collected during the clinical trial, while others were omitted to limit feature space complexity, as overly complex feature sets can lead to overfitting in machine learning models [55]. While additional predictors may have improved predictive performance, the limited sample size required careful control of feature space complexity to avoid overfitting. This work also did not account for fall risk, which is a major contributor to fracture outcomes [95]. Factors related to balance and stability were not considered and may capture aspects of fracture risk not reflected in bone- or image-based predictors. Additionally, the model was trained only on Lunar (GE HealthCare, Chicago, IL, USA) machine type images, and no information was provided regarding the specific machine generations used, which may limit the model's generalizability and performance when applied to data acquired from other manufacturers or even different versions of the same system. All models were somewhat sensitive to the decision threshold used in binary classification, which was obtained from tuning the validation set. Selecting the decision threshold this way inherently proved that it would not be generalizable to other cohorts, and even across different models it experienced some degree of variation, most notably in the deep learning model. Finally, while PCA is an effective technique for reducing the dimensionality of a feature space, when applied to images it concatenates all pixels into a single

vector, thereby discarding spatial relationships and local structural information that may be important for accurate image-based predictions.

Despite its limitations, there were also many strengths to this work. By using a longitudinal dataset, the model was able to capture disease progression over time, enabling more robust risk prediction than what would be possible with cross-sectional data alone. The period between baseline and follow-up was three years, meaning the tool predicts fracture risk within a three-year window. Although this is shorter than the ten-year risk horizon provided by FRAX, it offers a more immediate assessment of fracture risk [66]. This timeframe aligns well with clinical practice, as DXA scans for low-risk individuals are typically recommended at intervals of one to three years [96]. The cohort included both medicated and unmedicated individuals, allowing the models to learn from a broader range of fracture risk profiles that more closely reflect real-world clinical populations. In contrast, previous work in this field has often excluded individuals taking anti-osteoporotic medications [45], which can limit the applicability of results to routine clinical practice. Additionally, the models developed in this work accounted for the number of previous fractures rather than discretizing previous fractures into a binary yes/no variable, representing an improvement over the standard FRAX approach [8]. Another strength of this work was the use of multiple performance metrics (e.g., AUC, sensitivity, specificity, accuracy, recall, F-scores, training/validation loss plots, etc.) to provide a comprehensive assessment of model behavior. When metrics conflicted, interpretation was guided by the recall-focused nature of fracture screening (prioritizing correct classification of true positive cases), and by AUC as a threshold-independent measure of overall discriminatory ability.

Finally, and all models excluding the achieved AUCs comparable to FRAX and higher than those reported for BMD-based T-score assessment (which remains the clinical standard),

supporting the value of incorporating image-derived information into fracture risk prediction [23], [36], [44]. Despite working with a limited, imbalanced dataset, good performance metrics were achieved across multiple models. Although the selection of the classification threshold introduced limitations in terms of generalizability, a key strength of this study was that threshold tuning was performed deliberately rather than relying on an arbitrary default value (such as 0.5), which could have artificially inflated performance. This approach, along with explicit design choices aimed at reducing overfitting and the use of SHAP analysis to enhance model interpretability, reflected a well-structured and robust experimental framework. Finally, few studies have explored the integration of hip DXA images with clinical risk factors. This multi-modal approach strengthened this work by capturing complementary information, rather than relying on a single data source.

6.3 Future Directions

With end-to-end data pipelines now established, this framework could be readily applied to other datasets, enabling more reliable evaluation and improved generalizability in future studies. When more data becomes available, it would be interesting to implement survival analysis techniques that can predict time-to-event outcomes. Future work should focus on training and evaluating the models using images acquired from multiple DXA manufacturers and machine versions to reduce device-specific bias and improve robustness in real-world clinical settings. Additionally, implementing weighting schemes or hierarchical outcome definitions would allow the model to better distinguish between fracture types and severities, resulting in more clinically meaningful and nuanced risk predictions.

A natural extension of this work would be to extend the models to work with Hologic (Hologic Inc., Marlborough, MA, USA) scanner types, or to develop a robust model that can accept scan

types from different manufacturers. It would also be of interest to access data from the Amgen follow-up studies, which provide continued longitudinal monitoring of participants from the original osteoporosis trials [28]. These follow-up cohorts offer longer observation periods and additional fracture outcomes, presenting an opportunity to evaluate model performance over extended time horizons and to better capture disease progression and long-term fracture risk. Leveraging the Amgen follow-up data could strengthen internal validation by assessing the stability of model predictions across additional time points and clinical events. However, because the follow-up data largely consists of the same cohort, this would not fully address limitations related to population diversity or generalizability. Additionally, variations in treatment exposure and imaging availability during the follow-up period would need to be carefully accounted for to avoid confounding effects.

6.4 Significance

This work represents a meaningful advancement toward the development of a clinically relevant fracture risk prediction framework that integrates imaging and longitudinal clinical data. The proposed approach was designed to move beyond traditional fracture risk assessment methods that rely solely on clinical risk factors or bone mineral density metrics. Through comparative evaluation of multiple modeling strategies, this work demonstrated that incorporating clinical risk factors alongside imaging data improved classification performance and the identification of true positive fracture cases.

The results of this work will contribute to the development of a clinical tool that could be directly implemented into DXA machines, enabling routine scans to be automatically assessed for fracture risk, allowing early intervention of at-risk individuals. By providing a personalized

approach that evaluates each patient independently based on patterns learned from population-level data, this work has the potential to alleviate the burden of fracture-related hospitalizations, healthcare costs, and long-term care needs in older adults.

References

- [1] M. Van Oostwaard, “Osteoporosis and the Nature of Fragility Fracture: An Overview,” in *Fragility Fracture Nursing*, K. Hertz and J. Santy-Tomlinson, Eds., in Perspectives in Nursing Management and Care for Older Adults. , Cham: Springer International Publishing, 2018, pp. 1–13. doi: 10.1007/978-3-319-76681-2_1.
- [2] Osteoporosis Canada, 2022 Osteoporosis Canada Facts and Stats, Osteoporosis Canada, Canada, 2022. [Online]. Available: https://osteoporosis.ca/wp-content/uploads/2022-Osteoporosis-Canada-Facts-and-Stats_Web.pdf
- [3] M. Nikitovic, W. P. Wodchis, M. D. Krahm, and S. M. Cadarette, “Direct health-care costs attributed to hip fractures among seniors: a matched cohort study,” *Osteoporos. Int.*, vol. 24, no. 2, pp. 659–669, Feb. 2013, doi: 10.1007/s00198-012-2034-6.
- [4] R. B. Hopkins *et al.*, “The current economic burden of illness of osteoporosis in Canada,” *Osteoporos. Int.*, vol. 27, no. 10, pp. 3023–3032, Oct. 2016, doi: 10.1007/s00198-016-3631-6.
- [5] D. C. Azevedo, L. S. Hoff, S. C. Kowalski, C. A. F. De Andrade, V. F. M. Trevisani, and A. K. G. De Melo, “Risk factors for osteoporotic hip fracture among community-dwelling older adults: a real-world evidence study,” *Adv. Rheumatol.*, vol. 64, no. 1, p. 8, Jan. 2024, doi: 10.1186/s42358-024-00350-6.
- [6] E. M. Curtis, R. J. Moon, N. C. Harvey, and C. Cooper, “The impact of fragility fracture and approaches to osteoporosis risk assessment worldwide,” *Bone*, vol. 104, pp. 29–38, Nov. 2017, doi: 10.1016/j.bone.2017.01.024.

- [7] R. Zerikly and E. W. Demetriou, “Use of Fracture Risk Assessment Tool in clinical practice and Fracture Risk Assessment Tool future directions,” *Womens Health*, vol. 20, p. 17455057241231387, Jan. 2024, doi: 10.1177/17455057241231387.
- [8] A. T. H. Tan *et al.*, “Assessment of fracture risk with FRAX and FRAXplus,” *Gac. Médica México*, vol. 160, no. 4, p. 14160, Jan. 2025, doi: 10.24875/GMM.24000107.
- [9] A. Ammar, F. Jazinizadeh, J. D. Adachi, and C. E. Quenneville, “The effect of femur positioning on dual-energy X-ray absorptiometry (DXA) measures and statistical shape and appearance modeling (SSAM) fracture risk assessments,” *Proc. Inst. Mech. Eng. [H]*, vol. 238, no. 1, pp. 90–98, Jan. 2024, doi: 10.1177/09544119231214651.
- [10] F. Jazinizadeh and C. E. Quenneville, “3D Analysis of the Proximal Femur Compared to 2D Analysis for Hip Fracture Risk Prediction in a Clinical Population,” *Ann. Biomed. Eng.*, vol. 49, no. 4, pp. 1222–1232, Apr. 2021, doi: 10.1007/s10439-020-02670-2.
- [11] B. H. Derrickson and G. J. Tortora, *Tortora's Principles of anatomy & physiology*, [15th ed.]. Danvers MA: Wiley, 2017.
- [12] M. Lorentzon and S. R. Cummings, “Osteoporosis: the evolution of a diagnosis,” *J. Intern. Med.*, vol. 277, no. 6, pp. 650–661, Jun. 2015, doi: 10.1111/joim.12369.
- [13] D. L. Kendler *et al.*, “A scorecard for osteoporosis in Canada and seven Canadian provinces,” *Osteoporos. Int.*, vol. 32, no. 1, pp. 123–132, Jan. 2021, doi: 10.1007/s00198-020-05554-2.
- [14] B. L. Riggs, “Overview of Osteoporosis,” *N. Engl. J. Med.*, vol. 154, no. 1, pp. 63–67, Jan. 1991. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC1002677/>
- [15] S. M. Ott, “Cortical or Trabecular Bone: What’s the Difference?,” *Am. J. Nephrol.*, vol. 47, no. 6, pp. 373–375, 2018, doi: 10.1159/000489672.

- [16] E. Seeman, “Structural basis of growth-related gain and age-related loss of bone strength Proceedings of a satellite symposium held on the occasion of the EULAR Congress, Paris, France, June 13, 2008,” *Rheumatology*, vol. 47, no. Supplement 4, pp. iv2–iv8, Jul. 2008, doi: 10.1093/rheumatology/ken177.
- [17] J. A. Cauley *et al.*, “Serum 25 HydroxyVitamin D Concentrations and the Risk of Hip Fractures: The Women’s Health Initiative,” *Ann. Intern. Med.*, vol. 149, no. 4, pp. 242–250, Aug. 2008, doi: 10.7326/0003-4819-149-4-200808190-00005..
- [18] R. Coronado-Zarco, A. Olascoaga-Gómez De León, A. García-Lara, J. Quinzaños-Fresnedo, T. I. Nava-Bringas, and S. I. Macías-Hernández, “Nonpharmacological interventions for osteoporosis treatment: Systematic review of clinical practice guidelines,” *Osteoporos. Sarcopenia*, vol. 5, no. 3, pp. 69–77, Sep. 2019, doi: 10.1016/j.afos.2019.09.005.
- [19] G. Meyer, A. Warnke, R. Bender, and I. Mühlhauser, “Effect on hip fractures of increased use of hip protectors in nursing homes: cluster randomised controlled trial,” *BMJ*, vol. 326, no. 7380, p. 76.1, Jan. 2003, doi: 10.1136/bmj.326.7380.76.
- [20] Y. Lu and H. S. Uppal, “Hip Fractures: Relevant Anatomy, Classification, and Biomechanics of Fracture and Fixation,” *Geriatr. Orthop. Surg. Rehabil.*, vol. 10, p. 2151459319859139, Jan. 2019, doi: 10.1177/2151459319859139.
- [21] N. Walter *et al.*, “Epidemiology and treatment of proximal femoral fractures in the elderly U.S. population,” *Sci. Rep.*, vol. 13, no. 1, p. 12734, Aug. 2023, doi: 10.1038/s41598-023-40087-8.
- [22] Office of the Surgeon General (US), *Bone Health and Osteoporosis: A Report of the Surgeon General*. Rockville, Maryland: Office of the Surgeon General, 2004. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/20945569/>

- [23] X. Jiang *et al.*, “Diagnostic accuracy of FRAX in predicting the 10-year risk of osteoporotic fractures using the USA treatment thresholds: A systematic review and meta-analysis,” *Bone*, vol. 99, pp. 20–25, Jun. 2017, doi: 10.1016/j.bone.2017.02.008.
- [24] J. E. Compston and M. T. Drake, “Defining Very High Fracture Risk: Is FRAX Fit for Purpose?,” *J. Bone Miner. Res.*, vol. 35, no. 8, pp. 1399–1403, Dec. 2020, doi: 10.1002/jbmr.4134.
- [25] K. Beattie *et al.*, “Estimating osteoporotic fracture risk following a wrist fracture: a tale of two systems,” *Arch. Osteoporos.*, vol. 10, no. 1, p. 13, Dec. 2015, doi: 10.1007/s11657-015-0218-3.
- [26] Canadian Multicentre Osteoporosis Study (CaMos) Research Group *et al.*, “Construction and validation of a simplified fracture risk assessment tool for Canadian women and men: results from the CaMos and Manitoba cohorts,” *Osteoporos. Int.*, vol. 22, no. 6, pp. 1873–1883, Jun. 2011, doi: 10.1007/s00198-010-1445-5.
- [27] S. Riar *et al.*, “Comparison of 2 fracture risk estimation processes in Alberta: a cross-sectional chart review,” *CMAJ Open*, vol. 9, no. 2, pp. E711–E717, Apr. 2021, doi: 10.9778/cmajo.20200207.
- [28] H. G. Bone *et al.*, “10 years of denosumab treatment in postmenopausal women with osteoporosis: results from the phase 3 randomised FREEDOM trial and open-label extension,” *Lancet Diabetes Endocrinol.*, vol. 5, no. 7, pp. 513–523, Jul. 2017, doi: 10.1016/S2213-8587(17)30138-9.
- [29] A. Tenenhouse, N. Kreiger, and D. Hanley, “Canadian multicentre osteoporosis study (CaMos),” *Drug Dev. Res.*, vol. 49, no. 3, pp. 201–205, Mar. 2000, doi: 10.1002/(SICI)1098-2299(200003)49:3%3C201::AID-DDR10%3E3.0.CO;2-O.

- [30] M. A. Bolognese *et al.*, “Denosumab Significantly Increases DXA BMD at Both Trabecular and Cortical Sites: Results From the FREEDOM Study,” *J. Clin. Densitom.*, vol. 16, no. 2, pp. 147–153, Apr. 2013, doi: 10.1016/j.jocd.2012.02.006.
- [31] N. Hassanabadi *et al.*, “Variation in bone mineral density and fractures over 20 years among Canadians: a comparison of the Canadian Multicenter Osteoporosis Study and the Canadian Longitudinal Study on Aging,” *Osteoporos. Int.*, vol. 34, no. 2, pp. 357–367, Feb. 2023, doi: 10.1007/s00198-022-06623-4.
- [32] S. R. Cummings *et al.*, “Denosumab for Prevention of Fractures in Postmenopausal Women with Osteoporosis,” *N. Engl. J. Med.*, vol. 361, no. 8, pp. 756–765, Aug. 2009, doi: 10.1056/NEJMoa0809493.
- [33] J. A. Shepherd *et al.*, “A multinational study to develop universal standardization of whole-body bone density and composition using GE Healthcare Lunar and Hologic DXA systems,” *J. Bone Miner. Res.*, vol. 27, no. 10, pp. 2208–2216, Oct. 2012, doi: 10.1002/jbmr.1654.
- [34] C. J. Mertens, “Overview of the Radiation Dosimetry Experiment (RaD-X) flight mission: RAD-X OVERVIEW,” *Space Weather*, vol. 14, no. 11, pp. 921–934, Nov. 2016, doi: 10.1002/2016SW001399.
- [35] N. K. Chauhan and K. Singh, “A Review on Conventional Machine Learning vs Deep Learning,” in *2018 International Conference on Computing, Power and Communication Technologies (GUCON)*, Greater Noida, Uttar Pradesh, India: IEEE, Sep. 2018, pp. 347–352. doi: 10.1109/GUCON.2018.8675097.
- [36] S. J. Kang, M. J. Kim, Y.-I. Hur, J.-H. Haam, and Y.-S. Kim, “Application of Machine Learning Algorithms to Predict Osteoporotic Fractures in Women,” *Korean J. Fam. Med.*, vol. 45, no. 3, pp. 144–148, May 2024, doi: 10.4082/kjfm.23.0186.

- [37] C. Kruse, P. Eiken, and P. Vestergaard, “Machine Learning Principles Can Improve Hip Fracture Prediction,” *Calcif. Tissue Int.*, vol. 100, no. 4, pp. 348–360, Apr. 2017, doi: 10.1007/s00223-017-0238-7.
- [38] D. J. Hand, “Measuring classifier performance: a coherent alternative to the area under the ROC curve,” *Mach. Learn.*, vol. 77, no. 1, pp. 103–123, Oct. 2009, doi: 10.1007/s10994-009-5119-5.
- [39] T. Nissinen *et al.*, “Detecting pathological features and predicting fracture risk from dual-energy X-ray absorptiometry images using deep learning,” *Bone Rep.*, vol. 14, p. 101070, Jun. 2021, doi: 10.1016/j.bonr.2021.101070.
- [40] D. Senanayake *et al.*, “Classification of Fracture Risk in Fallers Using Dual-Energy X-Ray Absorptiometry (DXA) Images and Deep Learning-Based Feature Extraction,” *JBMR Plus*, vol. 7, no. 12, p. e10828, Dec. 2023, doi: 10.1002/jbm4.10828.
- [41] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” Dec. 10, 2015, *arXiv*: arXiv:1512.03385. doi: 10.48550/arXiv.1512.03385.
- [42] M. A. Kassem, S. M. Naguib, H. M. Hamza, M. M. Fouda, M. K. Saleh, and K. M. Hosny, “Explainable Transfer Learning-Based Deep Learning Model for Pelvis Fracture Detection,” *Int. J. Intell. Syst.*, vol. 2023, no. 1, p. 3281998, Jan. 2023, doi: 10.1155/2023/3281998.
- [43] C.-T. Cheng *et al.*, “Application of a deep learning algorithm for detection and visualization of hip fractures on plain pelvic radiographs,” *Eur. Radiol.*, vol. 29, no. 10, pp. 5469–5477, Oct. 2019, doi: 10.1007/s00330-019-06167-y.
- [44] S. H. Kong *et al.*, “A Novel Fracture Prediction Model Using Machine Learning in a Community-Based Cohort,” *JBMR Plus*, vol. 4, no. 3, p. e10337, Mar. 2020, doi: 10.1002/jbm4.10337.

- [45] S. H. Kong *et al.*, “Development of a Spine X-Ray-Based Fracture Prediction Model Using a Deep Learning Algorithm,” *Endocrinol. Metab.*, vol. 37, no. 4, pp. 674–683, Aug. 2022, doi: 10.3803/EnM.2022.1461.
- [46] S. H. Kong *et al.*, “A Computed Tomography–Based Fracture Prediction Model With Images of Vertebral Bones and Muscles by Employing Deep Learning: Development and Validation Study,” *J. Med. Internet Res.*, vol. 26, p. e48535, Jul. 2024, doi: 10.2196/48535.
- [47] I. T. Jolliffe and J. Cadima, “Principal component analysis: a review and recent developments,” *Philos. Trans. R. Soc. Math. Phys. Eng. Sci.*, vol. 374, no. 2065, p. 20150202, Apr. 2016, doi: 10.1098/rsta.2015.0202.
- [48] U. Ferizi *et al.*, “Artificial Intelligence Applied to Osteoporosis: A Performance Comparison of Machine Learning Algorithms in Predicting Fragility Fractures From MRI Data,” *J. Magn. Reson. Imaging*, vol. 49, no. 4, pp. 1029–1038, Apr. 2019, doi: 10.1002/jmri.26280.
- [49] M. A. Saleem *et al.*, “Enhancing stroke risk prediction through class balancing and data augmentation with CBDA-ResNet50,” *Sci. Rep.*, vol. 15, no. 1, p. 24553, Jul. 2025, doi: 10.1038/s41598-025-07350-6.
- [50] M. J. Willemink *et al.*, “Preparing Medical Imaging Data for Machine Learning,” *Radiology*, vol. 295, no. 1, pp. 4–15, Apr. 2020, doi: 10.1148/radiol.2020192224.
- [51] A. Ammar, F. Jazinizadeh, J. D. Adachi, and C. E. Quenneville, “The effect of file type and DXA protocol on an image processing fracture risk prediction tool,” *Proc. Inst. Mech. Eng. [H]*, vol. 239, no. 3, pp. 308–320, Mar. 2025, doi: 10.1177/09544119251327649.
- [52] T. Schlett, S. Schachner, C. Rathgeb, J. Tapia, and C. Busch, “Effect of Lossy Compression Algorithms on Face Image Quality and Recognition,” Feb. 24, 2023, *arXiv*: arXiv:2302.12593. doi: 10.48550/arXiv.2302.12593.

- [53] C. F. Dormann *et al.*, “Collinearity: a review of methods to deal with it and a simulation study evaluating their performance,” *Ecography*, vol. 36, no. 1, pp. 27–46, Jan. 2013, doi: 10.1111/j.1600-0587.2012.07348.x.
- [54] S. Papapoulos *et al.*, “Five years of denosumab exposure in women with postmenopausal osteoporosis: Results from the first two years of the FREEDOM extension,” *J. Bone Miner. Res.*, vol. 27, no. 3, pp. 694–701, Mar. 2012, doi: 10.1002/jbmr.1479.
- [55] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*, 1st ed. Cambridge University Press, 2014. doi: 10.1017/CBO9781107298019.
- [56] E. Tasci, Y. Zhuge, K. Camphausen, and A. V. Krauze, “Bias and Class Imbalance in Oncologic Data—Towards Inclusive and Transferrable AI in Large Scale Oncology Data Sets,” *Cancers*, vol. 14, no. 12, p. 2897, Jun. 2022, doi: 10.3390/cancers14122897.
- [57] E. Siris *et al.*, “Estimation of Long-Term Efficacy of Denosumab Treatment in Postmenopausal Women With Osteoporosis: A FRAX- and Virtual Twin-Based Post Hoc Analysis From the FREEDOM and FREEDOM Extension Trials,” *JBMR Plus*, vol. 4, no. 4, p. e10348, Apr. 2020, doi: 10.1002/jbm4.10348.
- [58] Q. Wu, F. Nasoz, J. Jung, B. Bhattarai, and M. V. Han, “Machine Learning Approaches for Fracture Risk Assessment: A Comparative Analysis of Genomic and Phenotypic Data in 5130 Older Men,” *Calcif. Tissue Int.*, vol. 107, no. 4, pp. 353–361, Oct. 2020, doi: 10.1007/s00223-020-00734-y.
- [59] M. Liu *et al.*, “Predicting fracture risk for elderly osteoporosis patients by hybrid machine learning model,” *Digit. Health*, vol. 10, p. 20552076241257456, Jan. 2024, doi: 10.1177/20552076241257456.

- [60] A. Engels *et al.*, “Osteoporotic hip fracture prediction from risk factors available in administrative claims data – A machine learning approach,” *PLOS ONE*, vol. 15, no. 5, p. e0232969, May 2020, doi: 10.1371/journal.pone.0232969.
- [61] A. Galassi, J. D. Martín-Guerrero, E. Villamor, C. Monserrat, and M. J. Rupérez, “Risk Assessment of Hip Fracture Based on Machine Learning,” *Appl. Bionics Biomech.*, vol. 2020, pp. 1–13, Dec. 2020, doi: 10.1155/2020/8880786.
- [62] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal Loss for Dense Object Detection,” Feb. 07, 2018, *arXiv*: arXiv:1708.02002. doi: 10.48550/arXiv.1708.02002.
- [63] Z. Zhuang, M. Liu, A. Cutkosky, and F. Orabona, “Understanding AdamW through Proximal Methods and Scale-Freeness,” Jan. 31, 2022, *arXiv*: arXiv:2202.00089. doi: 10.48550/arXiv.2202.00089.
- [64] J. Schilcher, A. Nilsson, O. Andlid, and A. Eklund, “Fusion of electronic health records and radiographic images for a multimodal deep learning prediction model of atypical femur fractures,” *Comput. Biol. Med.*, vol. 168, p. 107704, Jan. 2024, doi: 10.1016/j.compbimed.2023.107704.
- [65] K. E. Naylor, E. V. McCloskey, R. Eastell, and L. Yang, “Use of DXA-based finite element analysis of the proximal femur in a longitudinal study of hip fracture,” *J. Bone Miner. Res.*, vol. 28, no. 5, pp. 1014–1021, May 2013, doi: 10.1002/jbmr.1856.
- [66] M. G. Donaldson, L. Palermo, J. T. Schousboe, K. E. Ensrud, M. C. Hochberg, and S. R. Cummings, “FRAX and Risk of Vertebral Fractures: The Fracture Intervention Trial,” *J. Bone Miner. Res.*, vol. 24, no. 11, pp. 1793–1799, Nov. 2009, doi: 10.1359/jbmr.090511.
- [67] J. Bae *et al.*, “External Validation of Deep Learning Algorithm for Detecting and Visualizing Femoral Neck Fracture Including Displaced and Non-displaced Fracture on Plain X-

ray,” *J. Digit. Imaging*, vol. 34, no. 5, pp. 1099–1109, Oct. 2021, doi: 10.1007/s10278-021-00499-2.

[68] D. Guo, X. Liu, D. Wang, X. Tang, and Y. Qin, “Development and clinical validation of deep learning for auto-diagnosis of supraspinatus tears,” *J. Orthop. Surg.*, vol. 18, no. 1, p. 426, Jun. 2023, doi: 10.1186/s13018-023-03909-z.

[69] Y. Zhang, H. T. N. Le, N. Lopez, and K. Phan, “Comparative Study of Machine Learning Models for Textual Medical Note Classification,” *Computers*, vol. 15, no. 1, p. 7, Dec. 2025, doi: 10.3390/computers15010007.

[70] Q. An, S. Rahman, J. Zhou, and J. J. Kang, “A Comprehensive Review on Machine Learning in Healthcare Industry: Classification, Restrictions, Opportunities and Challenges,” *Sensors*, vol. 23, no. 9, p. 4178, Apr. 2023, doi: 10.3390/s23094178.

[71] H. Yu, D. C. Samuels, Y. Zhao, and Y. Guo, “Architectures and accuracy of artificial neural network for disease classification from omics data,” *BMC Genomics*, vol. 20, no. 1, p. 167, Dec. 2019, doi: 10.1186/s12864-019-5546-z.

[72] P. Mukherjee *et al.*, “A shallow convolutional neural network predicts prognosis of lung cancer patients in multi-institutional computed tomography image datasets,” *Nat. Mach. Intell.*, vol. 2, no. 5, pp. 274–282, May 2020, doi: 10.1038/s42256-020-0173-6.

[73] E. Villamor, C. Monserrat, L. Del Río, J. A. Romero-Martín, and M. J. Rupérez, “Prediction of osteoporotic hip fracture in postmenopausal women through patient-specific FE analyses and machine learning,” *Comput. Methods Programs Biomed.*, vol. 193, p. 105484, Sep. 2020, doi: 10.1016/j.cmpb.2020.105484.

- [74] E. Stenwig, P. S. Rossi, G. Salvi, and N. K. Skjærvold, “The impact of feature combinations on machine learning models for in-hospital mortality prediction,” *Sci. Rep.*, vol. 15, no. 1, p. 39119, Nov. 2025, doi: 10.1038/s41598-025-26611-y.
- [75] F. Khader *et al.*, “Multimodal Deep Learning for Integrating Chest Radiographs and Clinical Parameters: A Case for Transformers,” *Radiology*, vol. 309, no. 1, p. e230806, Oct. 2023, doi: 10.1148/radiol.230806.
- [76] F. Khader *et al.*, “Medical transformer for multimodal survival prediction in intensive care: integration of imaging and non-imaging data,” *Sci. Rep.*, vol. 13, no. 1, p. 10666, Jul. 2023, doi: 10.1038/s41598-023-37835-1.
- [77] S. K. D. C. Mesquita *et al.*, “Multilayer Perceptron Artificial Neural Network to Support Nurses’ Decision-Making on Topical Therapies for Venous Ulcers: Construction, Validation, and Evaluation,” *BioMedInformatics*, vol. 5, no. 4, p. 72, Dec. 2025, doi: 10.3390/biomedinformatics5040072.
- [78] P. Bruno, M. Macrì, and C. Dodaro, “A Dual-stage Deep Learning Framework for Breast Ultrasound Image Segmentation and Classification,” *J. Med. Syst.*, vol. 49, no. 1, p. 162, Nov. 2025, doi: 10.1007/s10916-025-02298-6.
- [79] A. Ramaswamy, “Gradient Clipping in Deep Learning: A Dynamical Systems Perspective:,” in *Proceedings of the 12th International Conference on Pattern Recognition Applications and Methods*, Lisbon, Portugal: SCITEPRESS - Science and Technology Publications, 2023, pp. 107–114. doi: 10.5220/0011678000003411.
- [80] L. Prechelt, “Early stopping — but when?” in *Neural Networks: Tricks of the Trade*, G. B. Orr and K.-R. Müller, Eds. Berlin, Germany: Springer, 1998, pp. 55–69.

- [81] M. A. Badgeley *et al.*, “Deep learning predicts hip fracture using confounding patient and healthcare variables,” *Npj Digit. Med.*, vol. 2, no. 1, p. 31, Apr. 2019, doi: 10.1038/s41746-019-0105-1.
- [82] A. Geroldinger, L. Lusa, M. Nold, and G. Heinze, “Leave-one-out cross-validation, penalization, and differential bias of some prediction model performance measures—a simulation study,” *Diagn. Progn. Res.*, vol. 7, no. 1, p. 9, May 2023, doi: 10.1186/s41512-023-00146-0.
- [83] G. C. Cawley and N. L. C. Talbot, “On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation,” *J. Mach. Learn. Res.*, vol. 11, pp. 2079–2107, Jul. 2010.
- [84] T. Chauhan, H. Palivela, and S. Tiwari, “Optimization and fine-tuning of DenseNet model for classification of COVID-19 cases in medical imaging,” *Int. J. Inf. Manag. Data Insights*, vol. 1, no. 2, p. 100020, Nov. 2021, doi: 10.1016/j.jjime.2021.100020.
- [85] A. Alam *et al.*, “Novel transfer learning based bone fracture detection using radiographic images,” *BMC Med. Imaging*, vol. 25, no. 1, p. 5, Jan. 2025, doi: 10.1186/s12880-024-01546-4.
- [86] N. Hong *et al.*, “Deep-Learning-Based Detection of Vertebral Fracture and Osteoporosis Using Lateral Spine X-Ray Radiography,” *J. Bone Miner. Res.*, vol. 38, no. 6, pp. 887–895, Dec. 2020, doi: 10.1002/jbmr.4814.
- [87] Y. Huang *et al.*, “Two-stage CNNs for computerized BI-RADS categorization in breast ultrasound images,” *Biomed. Eng. OnLine*, vol. 18, no. 1, p. 8, Dec. 2019, doi: 10.1186/s12938-019-0626-5.
- [88] U. A. Hasib, R. Md Abu, J. Yang, U. A. Bhatti, C. S. Ku, and L. Y. Por, “YOLOv8 framework for COVID-19 and pneumonia detection using synthetic image augmentation,” *Digit. Health*, vol. 11, p. 20552076251341092, May 2025, doi: 10.1177/20552076251341092.

- [89] M. A. Mahmoud, Y. Wen, Y. Liufu, X. Pan, R. Su, and Y. Guan, “A Lightweight Dual-Output Vision Transformer for Enhanced Lung Nodule Classification Using CT Images,” *Technol. Cancer Res. Treat.*, vol. 24, p. 15330338251370439, Jun. 2025, doi: 10.1177/15330338251370439.
- [90] C. H. Suh *et al.*, “Development and Validation of a Deep Learning–Based Automatic Brain Segmentation and Classification Algorithm for Alzheimer Disease Using 3D T1-Weighted Volumetric Images,” *Am. J. Neuroradiol.*, vol. 41, no. 12, pp. 2227–2234, Dec. 2020, doi: 10.3174/ajnr.A6848.
- [91] H. Alshahrani *et al.*, “An Intelligent Attention-Based Transfer Learning Model for Accurate Differentiation of Bone Marrow Stains to Diagnose Hematological Disorder,” *Life*, vol. 13, no. 10, p. 2091, Oct. 2023, doi: 10.3390/life13102091.
- [92] M. Watson and N. A. Moubayed, “Attack-agnostic Adversarial Detection on Medical Data Using Explainable Machine Learning,” May 05, 2021, *arXiv*: arXiv:2105.01959. doi: 10.48550/arXiv.2105.01959.
- [93] Osteoporosis Canada, “2022 Osteoporosis Canada Facts and Stats.” [Online]. Available: <https://osteoporosis.ca/facts-and-stats/>
- [94] M. S. LeBoff *et al.*, “The clinician’s guide to prevention and treatment of osteoporosis,” *Osteoporos. Int.*, vol. 33, no. 10, pp. 2049–2102, Oct. 2022, doi: 10.1007/s00198-021-05900-y.
- [95] K. M. Kim, L.-Y. Lui, and S. R. Cummings, “Recent fall and high imminent risk of fracture in older men and women,” *Age Ageing*, vol. 51, no. 6, p. afac141, Jun. 2022, doi: 10.1093/ageing/afac141.

- [96] A. Papaioannou *et al.*, “2010 clinical practice guidelines for the diagnosis and management of osteoporosis in Canada: summary,” *Can. Med. Assoc. J.*, vol. 182, no. 17, pp. 1864–1873, Nov. 2010, doi: 10.1503/cmaj.100771.

Appendix A – Glossary of Anatomical Terms

This appendix provides definitions of the medical terms used throughout this thesis. This appendix is intended for readers with less familiarity in medical terminology.

Acetabulum	The cup-shaped socket of the pelvis that holds the head of the femur to form the hip joint.
Alendronate	A bisphosphonate drug used to treat osteoporosis by reducing bone loss.
Anterior	Referring to the front side of the body.
Articular Capsule	The fibrous structure that encloses and stabilizes a joint.
Articular Cartilage	Smooth tissue that covers bone ends in joints to allow smooth movement.
Bisphosphonates	A class of drugs that slow bone loss by reducing bone resorption, commonly used to treat osteoporosis.
Bone Resorption	The process by which bone is broken down and minerals are released into the blood.
Cortical Bone	The dense, hard outer layer of bone that provides strength and protection.
Denosumab	A monoclonal antibody drug that reduces bone resorption by inhibiting osteoclast activity.
Diaphysis	The long central shaft of a long bone.

Epiphysis	The rounded end of a long bone, involved in joint formation.
Estrogen	A hormone that helps maintain bone density; reduced levels increase osteoporosis risk.
Extracapsular	Located outside the joint capsule.
Femoral Head	The ball-shaped top of the femur that fits into the acetabulum.
Femoral Neck	The narrow region connecting the femoral head to the femoral shaft.
Femoral Shaft	The long, straight portion of the femur.
Femur	The thigh bone and the longest, strongest bone in the human body.
Genant Semi-Quantitative Method	A standardized imaging-based method used to assess vertebral fractures by visually grading vertebral shape deformities (mild, moderate, or severe) on spinal X-rays or DXA images.
Greater Trochanter	A large bony projection on the femur where muscles attach.
Intertrochanteric	The region between the greater and lesser trochanters of the femur.
Intracapsular	Located within the joint capsule.
Lateral	Referring to the side of the body away from the midline.
Lesser Trochanter	A smaller bony projection on the femur for muscle attachment.
Major Osteoporotic Fracture	A fracture occurring at common osteoporosis sites such as hip, spine, humerus, or forearm.
Monoclonal	Derived from a single type of immune cell, used to describe targeted antibody-based drugs.
Musculoskeletal System	The system of bones, muscles, tendons, and ligaments that support movement.
Osteoclast	A cell responsible for breaking down bone tissue.

Osteoporotic	Describing bone that has reduced density and increased fracture risk.
Pelvis	The bony structure that connects the spine to the lower limbs and supports the hip joint.
Periosteal Expansion	The outward growth of bone that increases bone diameter and strength.
Porous	Having small spaces or holes; osteoporotic bone is more porous than healthy bone.
Posterior	Referring to the back side of the body.
Proximal	Closer to the center of the body.
Risedronate	A bisphosphonate medication that slows bone breakdown.
Subtrochanteric	The region just below the lesser trochanter of the femur.
Synovial Cavity	The space within a joint that contains synovial fluid.
Synovial Fluid	A lubricating fluid that reduces friction within joints.
Tendon	A strong connective tissue that attaches muscle to bone.
Tissue	A group of similar cells working together to perform a specific function.
Trabecular Bone	The spongy inner bone that absorbs shock and is common in vertebrae and femoral head.
T-Score	A measurement comparing bone density to that of a healthy young adult.
Vertebral	Related to the vertebrae or spine.
Zoledronic Acid	A potent bisphosphonate given intravenously to treat osteoporosis.